

Current MoE Reasoning Experiments

1 Scope

This report describes annotation, representation probing, observational routing analysis, and held-out routing interventions. The active packages are `gepaLLMAsJudge`, `probeTest`, `correlation_pipeline`, `moe_identity_guiding`, and `moe_margin_guiding`. Numerical comparisons use complete saved artifacts and retain their model, population, and deployment limitations.

2 Experimental Setup

Current experimental scope. The generative judge assigns Schoenfeld episode labels. Boundary probes test prospective label decodability in Qwen3.5-35B-A3B, GPT-OSS-20B, and Gemma-4-26B-A4B. Correlation analysis relates each model’s routing summaries to correctness and labelled reasoning episodes. Identity and margin guiding then evaluate frozen policies on held-out Qwen problems.

GEPA judge design. The GEPA workflow starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen predictions. Each request contains the SAT problem and the previous, current, and next response units. The original context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The Qwen3.8 alignment instead imports the 40-example prompt selected by the earlier final fit as its external seed, so it does not dynamically sample new demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the final-fit runs use a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Each model receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output. Qwen, GPT-OSS, and Gemma provide 41, 25, and 31 hidden-state indices, respectively. An intermediate index is the input to the equally numbered decoder block; the final output has no corresponding router.

At each index, seven separate target-versus-rest logistic regressions use raw activations (2,048 dimensions for Qwen, 2,880 for GPT-OSS, and 2,816 for Gemma), balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected Qwen probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a

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multiclass probability or new gold annotation. Only boundary vectors are stored; the models do not regenerate the released or benchmark solutions.

Correlation-pipeline design. Each of the three models has 4,647 attempts from 1,547 source problems: 32 attempts per AIME24, AIME25, and AMC23 problem, and one per MATH-500, OlympiadBench, and Minerva problem. There are 100 complete avg@32 groups. Generation uses temperature 0.6, top- p 0.95, a 32,768-token completion budget, and a 49,152-token context. Generation and replay quantizations are not identical: Qwen uses GPTQ generation and Unsloth runtime 4-bit replay; Gemma uses NVFP4 generation and NF4 replay; GPT-OSS uses its MXFP4 checkpoint. Native top- k is 8 of 256, 8 of 128, and 4 of 32, respectively. Probe-selected router indices are Qwen 27/34/35/36/38/39, Gemma 16/19/21/22/25/27/29, and GPT-OSS 15/19/21/23.

All token-limit completions remain included. Known-gold attempts with no extracted answer count as incorrect in correlation analysis. Whole-continuation features include final-answer tokens; full-reasoning and absolute-position views use all reasoning tokens independently of tagging. Class views use available sentence labels and omit missing features pairwise. Qwen uses a frozen Qwen3.8-NVFP4 judge at temperature zero, low effort, and a 4,096-token output budget, with a 100,000-unit sample-zero cap. Imported Gemma and GPT-OSS labels use temperature 1, top- p 0.95, top- k 20, low effort, and a 16,384-token budget. Gemma labels cover all six benchmarks; GPT-OSS labels cover MATH-500 and part of AIME24. These annotation protocols and populations are not matched across models. Source-integrity checks do not establish label accuracy.

Guiding design. Identity and margin policies use a dataset-stratified 70/30 source-problem split (seed 42), problem-weighted calibration, and at least four supporting problems per target. Their completed stochastic evaluations pair 1,395 attempts on 465 held-out problems with condition-specific baselines. Sampling settings and prompts match within each pair; original attempt seeds are retained. Hooks act during prefill and decoding with native expert dispatch, eager execution, and no speculative decoding or prefix cache. Section 7 gives the calibration exclusions, policy definitions, and numerical comparisons.

Table model and metric conventions. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen’s κ is chance-corrected nominal agreement, Kendall’s τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight [0, 1] rescalings. Routing entropy is Shannon entropy of the softmax router distribution; the correlation pipeline’s switch rate is the fraction of adjacent tokens whose top-1 expert changes; its top- k overlap is the adjacent-set intersection divided by k (not Jaccard overlap); and router margin is the top-1 minus top-2 router probability. Top- k mass sums the selected probabilities, while the boundary margin is $p_{(k)} - p_{(k+1)}$. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance. The five active packages are under `src/moe_exp`; their saved outputs are under the corresponding results subdirectories. Source hashes, comparison checks, and regeneration commands are listed in the reproducibility appendix. Correlation tables and guiding comparisons have different populations and must not be pooled.

3 gepaLLMAsJudge

The gepaLLMAsJudge workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was

Table 1. Preliminary gepaLLMAsJudge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, Cohen’s κ , Kendall’s τ_b , their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables 2 and 3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Completed LLM-judge final fit. The Qwen3.6 final fit uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336

Table 4. Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , Kendall’s τ_b , absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	0.7543	0.7769	0.7270	0.6968	0.6832
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Table 5. Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

units). Seed and optimized prompts both return a valid label for all 407 validation units. Table 4 reports exact gold-label metrics rather than the LLM judge’s diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt’s overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13 points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within

Table 6. Final Qwen3.8 model-alignment results on the 407-unit development validation set. Both prompts have 100% valid-output coverage; the 336-unit locked test is not evaluated. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL, used for both classification and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , and Kendall’s τ_b .

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
Imported 40-example seed	0.7174	0.7400	0.6938	0.6523	0.6122
GEPA optimized	0.7125	0.7288	0.6743	0.6475	0.6009
Absolute change	−0.0049	−0.0112	−0.0196	−0.0049	−0.0113

Table 7. Per-class recall in the final Qwen3.8 alignment. The selection gate rejects any candidate whose recall drops by more than 0.10 for any class. *Model(s)*: Qwen3.8-27B UD-Q4_K_XL. *Metrics*: class support and seed versus GEPA-optimized recall.

Class	Support	Seed recall	Opt. recall
Read	45	0.778	0.756
Analyze	122	0.549	0.533
Plan	32	0.750	0.563
Implement	103	0.874	0.874
Explore	17	0.529	0.647
Verify	70	0.700	0.786
Monitor	18	1.000	0.944

responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

Final Qwen3.8 model-alignment pass. The final alignment uses Qwen3.8-27B UD-Q4_K_XL, seed 42, the heavy GEPA budget, and medium reasoning effort for the same-model judge. It keeps the same response-grouped 2,382/407/336 train/validation/locked-test split. Rather than drawing new few-shot examples, it imports the previous 40-example Qwen3.6 optimized prompt as its external seed. Gold-label agreement supplies 40% of the internal judge reward; functional fit, contextual coherence, and boundary precision supply the remaining 60%. Final selection uses exact balanced accuracy with a maximum allowed per-class recall drop of 0.10.

The optimized candidate changes 49 labels: it fixes 17 seed errors, breaks 19 seed-correct cases, and changes 13 errors to other errors. Its largest failure is Plan recall, which falls by 0.188 and therefore breaches the recall gate; the selected prompt is byte-identical to the imported seed. An unclustered exact McNemar test on the 17 versus 19 discordant correctness outcomes gives $p = 0.868$, offering no evidence of an aggregate improvement, although this unit-level calculation understates response clustering.

This negative exact-label result contrasts with the internal GEPA objective, whose best candidate reward rises from 0.7684 to 0.7762. Because only 40% of that reward is gold agreement, the discrepancy is consistent with optimizing the judge’s broader functional criteria without improving corpus-label classification. The optimized instruction is shorter (2,591 versus 5,201 words) but internally inconsistent: an early rule treats “Let me compute that” as Monitor while a later rule treats “Let me calculate” as Plan. Plan-to-Monitor errors rise from 6 to 10, consistent with that boundary confusion. The appropriate conclusion is that the final alignment completed

correctly and the safety gate worked: GEPA found no safe improvement over the existing seed. The complete artifact is stored at `results/gepaLLMAsJudge/qwen3.8-27b-medium-final-s42-v3`.

Frozen-prompt deployment and reasoning effort. The NVFP4 deployment comparison evaluates the frozen 40-example prompt on the same 407 development-validation units, with temperature zero, a 4,096-token budget, seed 42, one worker, and one uncached pass. Table 8 reports the saved outputs. Both settings have 100% valid-output coverage and no request errors. They disagree on 41/407 labels (10.1%): identical aggregate accuracy does not imply interchangeable annotations. This is exploratory evidence on reused validation responses, not independent validation of labels on generated benchmark traces.

Table 8. Frozen Qwen3.8-27B NVFP4 judge on 407 reused validation units. Metrics are exact accuracy, balanced accuracy, and macro-F1.

Effort	Accuracy	Balanced accuracy	Macro-F1
Low	0.7101	0.7415	0.6863
Medium	0.7101	0.7444	0.6832

The 336-unit partition is excluded from these final fits. Because its responses have been inspected in preliminary work, it should not be described as genuinely untouched; an independent generalization estimate requires unseen responses.

4 probeTest

The probeTest workflow tests a different annotation mechanism from the generative judge in gepaLLMAsJudge. It asks whether each model’s representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Three-model comparison. All three encoders use the same 3,087 units and target-stratified sentence split: 2,469 training and 618 test units. Table 9 reports each target’s accuracy-selected hidden-state index, F1, and AUROC. All 38 response IDs occur in both partitions, and layer selection uses the evaluation split. These are descriptive within-corpus comparisons, not response-held-out generalization estimates. There are 287 Qwen, 175 GPT-OSS, and 217 Gemma binary fits. The detailed external inspection below applies only to Qwen.

Table 9. Accuracy-selected boundary probes: Qwen3.5 GPTQ, GPT-OSS-20B, and Gemma-4 NF4. Each model reports hidden-state index, F1, and AUROC; selection and evaluation share the sentence-level test split.

Target	Index	Qwen		Index	GPT-OSS		Index	Gemma	
		F1	AUROC		F1	AUROC		F1	AUROC
Read	34	0.704	0.934	19	0.686	0.906	21	0.675	0.898
Analyze	36	0.647	0.835	21	0.540	0.767	27	0.611	0.816
Plan	39	0.485	0.855	15	0.434	0.852	25	0.500	0.907
Implement	38	0.771	0.930	21	0.745	0.905	29	0.735	0.922
Explore	40	0.489	0.888	23	0.405	0.842	16	0.447	0.835
Verify	27	0.726	0.912	21	0.667	0.880	19	0.603	0.872
Monitor	35	0.482	0.903	15	0.344	0.774	22	0.382	0.831

Table 10. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics*: target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

Qwen within-corpus results. Table 10 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38 distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 benchmark examples complete without omitted units, yielding 581 sentence boundaries. Table 11 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a

Table 11. Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table 12 compares them with the subsequent provisional pragmatic review. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

Table 12. Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table 12 reports the complete dataset-level results.

The class-level comparison in Table 13 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference

Table 13. Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution (Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

5 Correlation Pipeline

5.1 Statistical measures and literature basis

The analysis selects the association measure from the outcome scale and unit of analysis, as summarized in Table 14. The detailed Qwen results below use the six-benchmark run: 4,647 attempts, including token-limit completions, with full-reasoning, class, and position views. The 100 complete avg@32 groups support repeated-attempt analysis. Classes use sampled labelled attempts; full and position views use all reasoning tokens. Section 6 adds the completed Gemma and GPT-OSS runs, with model-specific class coverage and annotation protocols.

Table 14. Prespecified measures for the expanded correlation analysis.

Analysis	Unit	Measure	Reason
Continuous feature versus correctness or a binary reasoning-event indicator	Generation trace	Point-biserial r_{pb}	The predictor is continuous and the outcome is binary; the signed coefficient is exactly Pearson's r after coding the outcome as 0/1.
Mean feature versus avg@ N correctness	Source problem	Spearman's ρ	Avg@ N is bounded, often tied, and not assumed to have a linear interval relationship with a feature. Ranking tests monotone association and one row per problem avoids treating repeated attempts as independent.
Continuous feature–feature and continuous expert-rate associations	Trace or source problem	Spearman's ρ	These exploratory scans ask whether quantities vary monotonically without assuming comparable scales or a linear relationship.
Correct-versus-incorrect feature difference within a repeated problem	Mixed-outcome source problem	Mean paired contrast	This directly controls for the problem by forming one correct-minus-incorrect contrast per problem; uncertainty is obtained by resampling problems.

For a continuous feature X and binary target $Y \in \{0, 1\}$, the first coefficient is

$$r_{pb} = \text{Corr}(X, Y). \quad (1)$$

Thus positive values mean that the feature tends to be higher for correct (or event-present) traces, and negative values mean that it tends to be lower. Liu et al. directly use point-biserial correlation for continuous language-model quantities against binary answer correctness, making it the closest precedent for this case [3].

For repeated-attempt datasets, correctness and each predictor are first aggregated once per source problem. Only groups containing exactly the declared avg@ N unique sample identifiers, no duplicate generations detected by the saved fingerprints, and a complete set of binary outcomes enter the primary Spearman analysis. Spearman correlation is appropriate here because its target is monotone rank association rather than a specifically linear change in avg@ N . Hong et al. correlate a scalar projection of residual-stream states with graded reasoning scores using Spearman correlation [2]; Xu et al. use Spearman correlation between language-model-derived conceptual quantities and human ratings [4]; and Hewitt and Manning use it to compare geometry derived from contextual representations with gold syntactic structure [1]. These studies are analogous precedents for rank correlation with representation-derived quantities; they do not by themselves establish Spearman as a unique choice for avg@32. Zhang et al. additionally report Pearson and Spearman correlations together when relating model uncertainty, including entropy-based measures, to graded human hallucination judgments [5].

Uncertainty is reported with 95% percentile bootstrap intervals that resample whole source problems, keeping attempts from the same problem together. Xu et al. provide relevant precedent for bootstrap intervals and use 10,000 resamples in one category–feature analysis [4]; the resample count in this pipeline remains explicit and configurable rather than being justified by that number alone. Benjamini–Hochberg q -values are computed separately within each declared dataset/scope, target, aggregation, or cross-feature family. Naive trace-level p -values are retained only as diagnostics when a dataset contains repeated attempts, and layerwise and expert-identity screens remain exploratory.

5.2 Completed six-benchmark run and analysis population

The Qwen numerical results use the run under results/correlation_pipeline/reasoning-vllm-v1. Forward replay and the saved aggregate analyses are complete for 4,647 attempts from 1,547 source problems: 500 MATH-500, 960 AIME24, 960 AIME25, 675 OlympiadBench, 1,280 AMC23, and 272 Minerva attempts. AIME and AMC retain 32 attempts per problem; the other benchmarks retain one. GPQA-Diamond and MMLU-Pro are absent.

Generation, replay, and annotation. Generation serves Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 through vLLM with temperature 0.6 and top- p 0.95. Replay uses unsloth/Qwen3.5-35B-A3B with runtime 4-bit quantization at the probe-selected decoder indices 27, 34, 35, 36, 38, and 39. Hidden-state features describe block inputs, not exact post-attention router inputs. Router confidence is $1 - H(p)/\log 256$. The continuation analysis has 83 scalar features; across-layer means are emphasized and layerwise screens remain exploratory. Saved bootstrap intervals use 500 replicates and seed 42, resampling whole source problems.

The sentence judge is unsloth/Qwen3.8-27B-NVFP4, temperature zero, thinking enabled with low reasoning effort, and a 4,096-token completion budget. Inputs contain the question and previous/current/next sentences. The frozen program SHA-256 is 69706a02c0484f345cf57775729ce7551feb5932954b9ad0b8f5bb63585c6fd9. Its accuracy on these generated math traces has not been independently measured; the earlier GGUF alignment evaluation uses a different deployment.

Population and token scopes. All supplied generations enter whole-continuation, full-reasoning, position, and expert analyses, including token-limit completions. Whole-continuation metrics include final-answer tokens. The full/reasoning view uses every reasoning token, independently of annotation coverage. Position views also use all reasoning tokens within their absolute windows. Class views alone use available validated sentence labels: the sampling protocol selects sample zero, with error-rate-weighted benchmark quotas, seed 42, and a 100,000-sentence cap. Missing labels never exclude a generation from the other views. Missing class/window features are pairwise missing, not zero.

Table 15. Completed-run coverage and accuracy, Qwen3.5-35B-A3B. All supplied attempts are retained, including limit hits. AIME/AMC use 32 attempts per problem; other benchmarks use one. Unparseable answers with known gold are incorrect. Pooled accuracy is descriptive.

Dataset	Input	Limit hits	Analyzed	Correct	Errors	Input acc.	Analyzed acc.
MATH-500	500	17	500	345	155	0.690	0.690
AIME24	960	105	960	817	143	0.851	0.851
AIME25	960	165	960	773	187	0.805	0.805
Olympiad	675	89	675	322	353	0.477	0.477
AMC23	1280	17	1280	1238	42	0.967	0.967
Minerva	272	19	272	73	199	0.268	0.268
Pooled	4647	412	4647	3568	1079	0.768	0.768

Table 16. Whole-continuation aggregate features versus correctness: point-biserial r (finite traces), Qwen3.5-35B-A3B. Token-limit completions are included. A dash denotes an undefined coefficient.

Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	-0.291 (500)	-0.726 (960)	-0.727 (960)	-0.387 (675)	-0.238 (1,280)	-0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	-0.035 (960)	0.222 (960)	0.118 (675)	-0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)

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Feature	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Hidden-state norm	0.203 (500)	−0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	−0.152 (500)	−0.058 (960)	−0.077 (960)	−0.102 (675)	−0.229 (1,280)	−0.096 (272)
Top-1 switch rate	−0.071 (500)	−0.297 (960)	0.141 (960)	0.005 (675)	−0.160 (1,280)	−0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	−0.155 (960)	0.016 (675)	−0.062 (1,280)	−0.010 (272)

Termination and accuracy denominators. There are 3,568 correct answers among 4,647 attempts (0.768). All 412 completion-limit hits (8.9%) remain in the analysis; only two are correct. Among the 4,235 non-limited attempts, 3,566 are correct (0.842). There are 388 unparseable answers counted as incorrect under the known-gold scoring contract. These denominators differ from accuracy computed only among parseable answers in generation summaries. Termination and length are strong potential confounders: these results do not estimate natural-length performance or a length-adjusted routing effect.

5.3 Correctness associations and uncertainty

Whole-continuation router confidence has positive point-biserial coefficients on all six benchmarks: MATH-500 $r = 0.199$, AIME24 0.533, AIME25 0.493, Olympiad 0.243, AMC23 0.175, and Minerva 0.112. Saved pointwise 95% problem-bootstrap intervals exclude zero except for AMC23 ($[-0.116, 0.678]$). MATH-500's interval is $[0.099, 0.276]$; AIME24's is $[0.382, 0.678]$ and AIME25's is $[0.302, 0.608]$. These are marginal associations, without length or dataset adjustment.

Leading-expert margin ($M = p_{(1)} - p_{(2)}$) is positive throughout; Minerva's interval includes zero. Hidden-state step distance is also positive throughout ($r = 0.100$ – 0.557), while token count is negative (-0.727 – -0.238). Attempts within AIME/AMC problems are dependent; naive trace-level p -values and their BH adjustments are diagnostic only.

Full-reasoning confidence gives similar coefficients: 0.191, 0.527, 0.486, 0.238, 0.164, and 0.124 in the same dataset order. All 4,647 traces have reasoning tokens. Similarity between these two scopes is descriptive; it does not remove length or termination effects.

5.4 Class-conditioned routing and hidden states

Table 17. Class coverage in the completed run. Qwen3.8-27B labels on Qwen3.5-35B-A3B generations. Units are tagged sentences; dataset columns count traces with selected class tokens. Classes overlap across traces; missing classes are not zero-valued features.

Class	Units	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva	Total traces
Read	6,358	378	30	26	514	27	270	1245
Analyze	22,378	424	30	30	619	38	271	1412
Plan	5,654	407	28	28	507	28	271	1269
Implement	23,844	477	30	30	667	39	271	1514
Explore	7,151	260	28	27	474	22	251	1062
Verify	10,745	400	29	30	617	38	270	1384
Monitor	23,870	451	30	29	583	29	272	1394

The 100,000 sampled sentences have complete labels across 1,546 nonempty attempt samples; one of the 1,547 sample-zero attempts has no labelled sentence. All seven class analyses are complete, with coverage reported separately from the full population. Explore has tokens in 1,062 traces; Verify in 1,384. AIME/AMC class views contain at most the 30/30/40 sample-zero attempts rather than all 32 attempts per problem. Annotation counts

describe sampled sentences, not class prevalence in the complete corpus. Class transition metrics require adjacent original tokens within the same sentence.

Class conditioning remains heterogeneous. Verify confidence is 0.066 on MATH-500, 0.123 on Olympiad, and -0.028 on Minerva; Plan confidence is -0.067 , 0.039, and -0.121 , respectively. These separately selected cohorts do not test a difference between classes. The smaller AIME/AMC class cohorts remain especially sensitive to individual errors. Annotation errors, length, difficulty, and coverage are uncontrolled, so these screens do not identify beneficial or harmful reasoning states.

5.5 Class counts, transitions, and outcome associations

Class-conditioned router features and the proportion of reasoning assigned to each class describe different properties. The following plots use *sampled sentence shares*: the number of labelled sentences of a class divided by the number of labelled sentences in that attempt. An observed zero share is retained; unlabelled sentences are never assigned a class. These are estimates from the sampling protocol, not complete-sequence class fractions. Each source problem contributes only its sample-zero attempt; benchmark comparisons retain their separate denominators. Attempts with no labelled sentences are excluded.

Table 18. Mean sampled sentence share (percent), with equal weight per labelled attempt. Empty-label attempts are excluded; absent classes in an observed sample have zero share. These are not full-corpus class proportions.

Dataset	Attempts	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
MATH-500	499	8.9	19.5	9.3	29.3	4.1	11.8	17.1
AIME24	30	6.7	33.9	2.9	28.3	7.0	14.4	6.8
AIME25	30	5.7	33.2	4.8	27.9	6.7	13.4	8.3
Olympiad	675	6.9	26.8	6.0	29.7	6.4	13.2	11.0
AMC23	40	6.6	25.9	5.8	32.0	3.8	15.2	10.6
Minerva	272	9.4	19.8	8.4	21.8	7.7	10.3	22.6

Class-state dynamics remain descriptive. Relating complete transition paths to correctness and interventions conditioned on those paths is future work. The currently sampled labels do not justify reconstructing missing transitions.

5.6 Position-dependent estimates and coverage

The position reference uses all 4,647 supplied generations and mean reasoning length 12,498.062 tokens. Ten non-cumulative windows have boundaries $[i\bar{L}/10]$, approximately 1,249.8 tokens apart; overflow begins after the final boundary at 12,499. Every reasoning token contributes to its window regardless of tagging. Token count is omitted as a position predictor. Coverage falls from 4,647 traces in the first window to 1,789 in overflow, so later-window coefficients describe different cohorts.

Table 19. Reasoning length in tokens, all attempts including limit hits. This replaces the position-coverage count plot; lengths are observed under the generation budget.

Dataset	Attempts	Mean	Median	90th percentile
MATH-500	500	6,226.5	4,250	12,772
AIME24	960	15,908.4	14,011	32,766
AIME25	960	17,874.9	16,620	32,768
Olympiad	675	13,402.5	9,150	32,768
AMC23	1280	8,452.8	7,822	14,929
Minerva	272	9,805.0	7,465	19,533

5.7 Metric-pair dependence

Token count and confidence are negatively associated on all six datasets: $\rho = -0.681, -0.627, -0.836, -0.870, -0.845$, and -0.280 in MATH/AIME24/AIME25/Olympiad/AMC/Minerva order. This dependence is one reason confidence–correctness correlations need length-aware follow-up. Top-eight mass S and the selected/unselected mean probability gap G are algebraically redundant for $k = 8$ and $E = 256$:

$$G = \frac{S}{k} - \frac{1-S}{E-k} = \frac{E}{k(E-k)}S - \frac{1}{E-k}. \quad (2)$$

The affine relation survives token/layer averaging. These metrics should not count as independent corroboration or enter an unregularized model twice. The appendix presents correlation heatmaps for all 20 scopes, including class-conditioned metrics versus correctness and token count. The numeric tables remain available as a separate generated artifact.

The legacy regex flags (backtracking, contradiction, and self-correction) are removed from the report’s outcome panels, to keep the outcome panels focused on model-assigned classes. They are not the seven predicted reasoning classes. Detector outputs remain in the source artifacts.

5.8 Repeated-attempt findings

There are 100 eligible complete avg@32 groups: 30 AIME24, 30 AIME25, and 40 AMC23 problems. Problem means use all declared attempts, including limit hits, and binary outcomes are complete. The run supports both between-problem rank associations and within-problem correct/incorrect contrasts.

Table 20. Whole-continuation mean feature versus avg@32 correctness, Spearman ρ (source problems). All 32 attempts, including token-limit completions, enter each eligible group.

Feature	AIME24	AIME25	AMC23
Token count	−0.861 (30)	−0.909 (30)	−0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	−0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	−0.112 (30)	0.388 (30)	−0.215 (40)
Selection-boundary margin	−0.367 (30)	0.135 (30)	−0.145 (40)
Selected/rest mean gap	−0.112 (30)	0.388 (30)	−0.215 (40)
Hidden-state norm	−0.384 (30)	−0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	−0.331 (40)
Top-1 switch rate	−0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	−0.047 (30)	−0.196 (30)	−0.327 (40)

Table 21. Within-problem confidence differences, Qwen3.5: mean correct minus incorrect, equally weighted over mixed-outcome problems; saved 95% whole-problem bootstrap intervals.

Dataset	Mixed problems	Difference	95% interval
AIME24	14	0.00268	[0.00152, 0.00412]
AIME25	15	0.00123	[0.00065, 0.00176]
AMC23	6	0.01102	[0.00455, 0.01709]

Mean whole-continuation confidence correlates positively with avg@32 on AIME24 ($\rho = 0.632$, CI [0.383, 0.818]) and AIME25 (0.705, CI [0.460, 0.854]). AMC23 is negative but uncertain (−0.200, CI [−0.565, 0.243]), despite its positive attempt-level coefficient. Thus attempt-level and problem-level associations answer different questions. AIME24 and AIME25 mean token count have strong negative avg@32 associations (−0.861 and −0.909).

Within-problem confidence differences are positive over 14, 15, and 6 mixed-outcome problems, respectively; Table 21 reports magnitudes and bootstrap intervals. Conditioning on a shared question reduces between-question variation but does not control attempt length or termination. Class views still lack repeated labelled attempts, so these repeated-attempt findings must not be generalized to class-specific effects.

5.9 Expert screen, storage, and implications

Table 22. Post-hoc largest absolute correctness coefficient per dataset in the expert-rate screen, Qwen3.5-35B-A3B. Point-biserial r and BH q within dataset/target; no expert bootstrap intervals. Numeric IDs pool six distinct layer-local modules. Repeated-attempt trace-level q -values are diagnostic, not cluster-adjusted inference.

Dataset	Pooled-ID rate	Traces	Errors	r	q
MATH-500	Top-1 61 / top-2 7	500	155	0.328	$3.36e - 11$
AIME24	Top-1 ID 123	960	143	-0.568	$1.87e - 80$
AIME25	Top-1 133 / top-2 85	960	187	0.556	$4.72e - 76$
Olympiad	Top-1 70 / top-2 181	675	353	0.312	$6.25e - 14$
AMC23	Top-8 ID 217	1280	42	-0.665	$2.83e - 161$
Minerva	Top-1 162 / top-2 165	272	199	0.409	$1.31e - 09$

The expert screen includes top-1 rates, any-top-eight rates, and up to 128 ordered top-1/top-2 pairs selected by routing support (minimum four traces). The largest absolute correctness coefficient per dataset is selected post hoc. Figure 11 shows all selected-pool IDs against correctness and total token count, without selecting a maximum. Numeric IDs pool six different layer-local modules; they do not identify a portable expert or establish an intervention target. Trace-level BH values on repeated-attempt datasets do not account for clustering.

Metric values associated with accuracy. Correlation signs alone do not specify the observed metric ranges associated with higher accuracy. Figure 12 therefore reports accuracy directly against router confidence, the selection-boundary margin, and the selected/rest mean probability gap. Bins are computed separately per benchmark; this is an exploratory whole-continuation analysis, not an online threshold-selection experiment.

Common router notation and unavailable extensions. Let $p_{(j)}$ denote the j th largest router probability, E the number of experts, and k the selected count ($E = 256$, $k = 8$ here). Router confidence is $C = 1 - H(p)/\log E$; leading-expert margin is $M = p_{(1)} - p_{(2)}$; selected probability mass is $S = \sum_{j \leq k} p_{(j)}$; selection-boundary margin is $B = p_{(k)} - p_{(k+1)}$; and selected/rest mean gap is $G = S/k - (1 - S)/(E - k)$. Thus M is not a selected-versus-rest statistic. These definitions apply across models with different native selection counts.

Rank-prefix curves (top 2, 4, 6, then all selected versus the rest) require additional probability summaries. Only the all-selected endpoint G is saved. Separate entropy of the selected and unselected distributions also cannot be recovered from global entropy, selected mass, or expert IDs. These extensions require new feature extraction and replay.

Replay covers 62,088,209 continuation tokens. Compact tensor artifacts occupy 5.970 GB versus a projected 1,907.350 GB of raw selected-layer payload; dataset files total 9.259 GB (decimal units). Storage and source-fingerprint checks are recorded in `report/correlation_run_audit.json`. No raw router/hidden tensors or expert weights are saved. The completed run supplies six-benchmark, full-reasoning, class, position, lexical, and repeated-attempt analyses. Next steps are independent label assessment, length-/termination-aware models, common-cohort class/window comparisons, and validation holding out whole problems. Separate held-out guiding results are reported in Section 7.

6 Three-model correlation results

Forward replay and saved correlation analyses are complete for Qwen3.5-35B-A3B, Gemma-4-26B-A4B, and GPT-OSS-20B. Each includes 4,647 attempts from 1,547 source problems on the same six benchmarks, with 100 complete avg@32 groups. All token-limit completions are retained. Completion of class analysis means analysis of available labels, not full annotation coverage. GPT-OSS class tokens occur only in MATH-500 and part of AIME24; the other four benchmarks have no class estimates. Qwen uses the local deterministic NVFP4 judge described above. Gemma and GPT-OSS use imported Qwen3.8-27B labels, with temperature 1, top- p 0.95, top- k 20, low reasoning effort and a 16,384-token judge budget. Consequently class comparisons also differ in annotation protocol and cohort. Generation uses Qwen GPTQ-Int4, Gemma NVFP4, and GPT-OSS checkpoints respectively; replay settings and selected decoder indices are below. Token counts and absolute position windows are model-specific. Correlations are marginal descriptive associations, not controlled model effects.

Table 23. Replay configuration and mean reasoning length used for position windows.

Model	Replay	Decoder indices	Mean tokens
Qwen3.5-35B-A3B	unsloth-4bit	27, 34, 35, 36, 38, 39	12,498.1
Gemma-4-26B-A4B	bnb-4bit	16, 19, 21, 22, 25, 27, 29	10,512.6
GPT-OSS-20B	mxfp4-bf16	15, 19, 21, 23	5,977.6

Table 24. All-attempt accuracy and termination by model. Invalid known-gold answers count as incorrect; repeated attempts are not independent problems.

Model	Dataset	Attempts	Correct	Accuracy	Limit hits
Qwen3.5-35B-A3B	MATH-500	500	345	0.690	17
Qwen3.5-35B-A3B	AIME24	960	817	0.851	105
Qwen3.5-35B-A3B	AIME25	960	773	0.805	165
Qwen3.5-35B-A3B	Olympiad	675	322	0.477	89
Qwen3.5-35B-A3B	AMC23	1280	1238	0.967	17
Qwen3.5-35B-A3B	Minerva	272	73	0.268	19
Gemma-4-26B-A4B	MATH-500	500	353	0.706	6
Gemma-4-26B-A4B	AIME24	960	852	0.887	54
Gemma-4-26B-A4B	AIME25	960	791	0.824	103
Gemma-4-26B-A4B	Olympiad	675	349	0.517	36
Gemma-4-26B-A4B	AMC23	1280	1249	0.976	3
Gemma-4-26B-A4B	Minerva	272	82	0.301	2
GPT-OSS-20B	MATH-500	500	332	0.664	4
GPT-OSS-20B	AIME24	960	681	0.709	99
GPT-OSS-20B	AIME25	960	668	0.696	118
GPT-OSS-20B	Olympiad	675	285	0.422	34
GPT-OSS-20B	AMC23	1280	1152	0.900	6
GPT-OSS-20B	Minerva	272	31	0.114	0

Table 25. Class token-bearing attempts; classes overlap. Zero coverage implies unavailable estimates.

Model	Class	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Qwen3.5-35B-A3B	Read	378	30	26	514	27	270
Qwen3.5-35B-A3B	Analyze	424	30	30	619	38	271
Qwen3.5-35B-A3B	Plan	407	28	28	507	28	271
Qwen3.5-35B-A3B	Implement	477	30	30	667	39	271
Qwen3.5-35B-A3B	Explore	260	28	27	474	22	251
Qwen3.5-35B-A3B	Verify	400	29	30	617	38	270
Qwen3.5-35B-A3B	Monitor	451	30	29	583	29	272
Gemma-4-26B-A4B	Read	393	28	29	528	34	271
Gemma-4-26B-A4B	Analyze	424	30	30	615	40	269
Gemma-4-26B-A4B	Plan	294	27	28	397	20	244
Gemma-4-26B-A4B	Implement	478	30	30	656	39	271

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Model	Class	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Gemma-4-26B-A4B	Explore	202	26	25	387	20	235
Gemma-4-26B-A4B	Verify	408	30	30	597	37	261
Gemma-4-26B-A4B	Monitor	280	30	29	461	28	251
GPT-OSS-20B	Read	474	68	0	0	0	0
GPT-OSS-20B	Analyze	433	68	0	0	0	0
GPT-OSS-20B	Plan	324	68	0	0	0	0
GPT-OSS-20B	Implement	500	68	0	0	0	0
GPT-OSS-20B	Explore	291	44	0	0	0	0
GPT-OSS-20B	Verify	435	68	0	0	0	0
GPT-OSS-20B	Monitor	496	68	0	0	0	0

Across models, confidence–correctness associations do not have a uniform sign. Qwen is positive on all six benchmarks; Gemma is negative on AMC23 and Minerva, and GPT-OSS is negative on MATH-500, AIME25, AMC23, and Minerva. The pointwise intervals in the comparison figures show substantial uncertainty for several of these estimates. Token-count associations are negative except for Gemma on AMC23. These differences motivate model-specific validation rather than assuming that a Qwen-derived routing rule transfers to other models.

6.1 Qwen3.5-35B-A3B

Table 26. Qwen3.5-35B-A3B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	−0.291 (500)	−0.726 (960)	−0.727 (960)	−0.387 (675)	−0.238 (1,280)	−0.293 (272)
Router confidence	0.199 (500)	0.533 (960)	0.493 (960)	0.243 (675)	0.175 (1,280)	0.112 (272)
Leading-expert margin	0.125 (500)	0.562 (960)	0.479 (960)	0.218 (675)	0.295 (1,280)	0.075 (272)
Selected probability mass	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Selection-boundary margin	0.054 (500)	−0.035 (960)	0.222 (960)	0.118 (675)	−0.010 (1,280)	0.039 (272)
Selected/rest mean gap	0.131 (500)	0.175 (960)	0.342 (960)	0.188 (675)	0.097 (1,280)	0.082 (272)
Hidden-state norm	0.203 (500)	−0.018 (960)	0.056 (960)	0.085 (675)	0.550 (1,280)	0.083 (272)
Hidden-state step distance	0.157 (500)	0.382 (960)	0.404 (960)	0.111 (675)	0.557 (1,280)	0.100 (272)
Hidden/router geometry	−0.152 (500)	−0.058 (960)	−0.077 (960)	−0.102 (675)	−0.229 (1,280)	−0.096 (272)
Top-1 switch rate	−0.071 (500)	−0.297 (960)	0.141 (960)	0.005 (675)	−0.160 (1,280)	−0.006 (272)
Adjacent selected-set overlap	0.047 (500)	0.022 (960)	−0.155 (960)	0.016 (675)	−0.062 (1,280)	−0.010 (272)

Confidence coefficients in benchmark order are 0.199, 0.533, 0.493, 0.243, 0.175, 0.112.

Table 27. Qwen3.5-35B-A3B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	−0.861 (30)	−0.909 (30)	−0.057 (40)
Router confidence	0.632 (30)	0.705 (30)	−0.200 (40)
Leading-expert margin	0.583 (30)	0.638 (30)	0.190 (40)
Selected probability mass	−0.112 (30)	0.388 (30)	−0.215 (40)
Selection-boundary margin	−0.367 (30)	0.135 (30)	−0.145 (40)
Selected/rest mean gap	−0.112 (30)	0.388 (30)	−0.215 (40)
Hidden-state norm	−0.384 (30)	−0.186 (30)	0.484 (40)
Hidden-state step distance	0.238 (30)	0.384 (30)	0.504 (40)
Hidden/router geometry	0.273 (30)	0.046 (30)	−0.331 (40)
Top-1 switch rate	−0.381 (30)	0.156 (30)	0.121 (40)
Adjacent selected-set overlap	−0.047 (30)	−0.196 (30)	−0.327 (40)

6.2 Gemma-4-26B-A4B

Table 28. Gemma-4-26B-A4B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	-0.155 (500)	-0.672 (960)	-0.709 (960)	-0.240 (675)	0.080 (1,280)	-0.205 (272)
Router confidence	0.080 (500)	0.392 (960)	0.313 (960)	0.027 (675)	-0.256 (1,280)	-0.037 (272)
Leading-expert margin	0.108 (500)	0.297 (960)	0.456 (960)	0.036 (675)	-0.243 (1,280)	0.010 (272)
Selected probability mass	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	-0.217 (1,280)	-0.061 (272)
Selection-boundary margin	-0.027 (500)	0.261 (960)	0.068 (960)	-0.043 (675)	-0.128 (1,280)	-0.028 (272)
Selected/rest mean gap	0.072 (500)	0.375 (960)	0.259 (960)	0.009 (675)	-0.217 (1,280)	-0.061 (272)
Hidden-state norm	0.036 (500)	0.161 (960)	0.007 (960)	-0.005 (675)	0.104 (1,280)	0.003 (272)
Hidden-state step distance	0.010 (500)	0.272 (960)	0.440 (960)	0.001 (675)	-0.141 (1,280)	0.059 (272)
Hidden/router geometry	-0.068 (500)	0.020 (960)	0.000 (960)	-0.019 (675)	-0.043 (1,280)	-0.076 (272)
Top-1 switch rate	0.052 (500)	-0.133 (960)	0.222 (960)	0.048 (675)	-0.116 (1,280)	0.100 (272)
Adjacent selected-set overlap	-0.021 (500)	0.129 (960)	-0.244 (960)	-0.049 (675)	0.038 (1,280)	-0.117 (272)

Confidence coefficients in benchmark order are 0.080, 0.392, 0.313, 0.027, -0.256, -0.037.

Table 29. Gemma-4-26B-A4B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	-0.744 (30)	-0.917 (30)	-0.099 (40)
Router confidence	0.473 (30)	0.468 (30)	0.071 (40)
Leading-expert margin	0.586 (30)	0.777 (30)	0.191 (40)
Selected probability mass	0.416 (30)	0.403 (30)	0.027 (40)
Selection-boundary margin	0.345 (30)	0.182 (30)	-0.063 (40)
Selected/rest mean gap	0.416 (30)	0.403 (30)	0.027 (40)
Hidden-state norm	0.026 (30)	-0.042 (30)	-0.060 (40)
Hidden-state step distance	0.410 (30)	0.547 (30)	-0.063 (40)
Hidden/router geometry	0.300 (30)	0.042 (30)	-0.004 (40)
Top-1 switch rate	-0.367 (30)	0.285 (30)	-0.215 (40)
Adjacent selected-set overlap	0.213 (30)	-0.321 (30)	0.059 (40)

6.3 GPT-OSS-20B

Table 30. GPT-OSS-20B: whole-continuation correctness r (finite attempts).

Metric	MATH-500	AIME24	AIME25	Olympiad	AMC23	Minerva
Token count	-0.221 (500)	-0.606 (960)	-0.646 (960)	-0.362 (675)	-0.169 (1,280)	-0.094 (272)
Router confidence	-0.061 (500)	0.210 (960)	-0.075 (960)	0.051 (675)	-0.077 (1,280)	-0.160 (272)
Leading-expert margin	0.041 (500)	0.137 (960)	-0.176 (960)	0.154 (675)	-0.018 (1,280)	-0.082 (272)
Selected probability mass	-0.052 (500)	0.128 (960)	-0.102 (960)	0.089 (675)	-0.033 (1,280)	-0.166 (272)
Selection-boundary margin	-0.036 (500)	-0.218 (960)	-0.091 (960)	-0.064 (675)	-0.005 (1,280)	-0.032 (272)
Selected/rest mean gap	-0.052 (500)	0.128 (960)	-0.102 (960)	0.089 (675)	-0.033 (1,280)	-0.166 (272)
Hidden-state norm	0.192 (500)	0.494 (960)	0.366 (960)	0.329 (675)	0.049 (1,280)	-0.006 (272)
Hidden-state step distance	-0.150 (500)	-0.186 (960)	-0.404 (960)	-0.210 (675)	-0.094 (1,280)	0.045 (272)
Hidden/router geometry	0.002 (500)	-0.239 (960)	-0.311 (960)	-0.076 (675)	-0.027 (1,280)	0.053 (272)
Top-1 switch rate	0.181 (500)	0.149 (960)	0.376 (960)	0.110 (675)	-0.006 (1,280)	0.331 (272)
Adjacent selected-set overlap	-0.198 (500)	-0.068 (960)	-0.278 (960)	-0.143 (675)	-0.052 (1,280)	-0.317 (272)

Confidence coefficients in benchmark order are -0.061, 0.210, -0.075, 0.051, -0.077, -0.160.

Table 31. GPT-OSS-20B: problem mean versus avg@32, Spearman ρ (problems).

Metric	AIME24	AIME25	AMC23
Token count	-0.889 (30)	-0.908 (30)	-0.338 (40)
Router confidence	0.337 (30)	0.104 (30)	0.186 (40)
Leading-expert margin	0.248 (30)	-0.028 (30)	0.262 (40)
Selected probability mass	0.186 (30)	0.046 (30)	0.272 (40)

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Metric	AIME24	AIME25	AMC23
Selection-boundary margin	−0.356 (30)	−0.273 (30)	0.060 (40)
Selected/rest mean gap	0.186 (30)	0.046 (30)	0.272 (40)
Hidden-state norm	0.733 (30)	0.655 (30)	0.338 (40)
Hidden-state step distance	−0.341 (30)	−0.655 (30)	−0.289 (40)
Hidden/router geometry	−0.658 (30)	−0.474 (30)	−0.177 (40)
Top-1 switch rate	0.162 (30)	0.546 (30)	−0.151 (40)
Adjacent selected-set overlap	−0.029 (30)	−0.340 (30)	−0.089 (40)

Aggregate correctness and metric-pair coefficients were independently recomputed from saved CSVs in all 60 model/scope combinations; repeated-problem mean correlations and within-problem contrasts were also checked. This validates numerical consistency, not judge accuracy or replay fidelity. Audit counts and numerical errors are saved in `report/multimodel_correlation_audit.json`; input SHA-256 hashes are in `report/multimodel_correlation_sources.json`. The appended atlases include every Gemma and GPT-OSS scope, including empty class panels. Full correctness tables are available in `report/multimodel_correlation_tables.tex`.

7 Held-out Routing Interventions

Calibration and evaluation population. Both policies split source problems within each benchmark, allocating 1,082 problems to calibration and 465 to evaluation (70/30, seed 42). Calibration uses 3,014 scored traces from 1,001 problems; 238 traces with missing saved correctness are excluded. This differs from the correlation analyzer, which counts known-gold missing answers as incorrect. The problem-weighted calibration accuracy is 0.5817 on this scored subset. Consequently, calibration targets may be sensitive to answer-extraction failures and must not be interpreted as estimates over all allocated traces. Each problem contributes total weight one, divided across its usable attempts. Evaluation retains 150 MATH-500, 9 AIME24, 9 AIME25, 203 OlympiadBench, 12 AMC23, and 82 Minerva problems. With 32 attempts on each AIME/AMC problem, this gives 1,395 paired attempts. No evaluation problem is used in policy calibration.

Identity policy. At each of the six selected Qwen layers, expert inclusion is weighted by its routing fraction and problem weight. Experts require support from at least four distinct problems and positive accuracy lift over the calibration baseline. At most eight identities per layer are selected; positive lifts are normalized to the largest retained lift. A signed strength scales the additive router-logit bias. Strength +1 promotes these identities; strength −1 suppresses the same identities. The negative control does not learn a separate set of incorrectly associated experts. Unlike the pooled-ID observational screen, policy targets are layer-qualified.

Margin policy. The policy calibrates the trace-mean selection-boundary margin $p_{(k)} - p_{(k+1)}$, with $k = 8$, using five problem-weighted quantile bins. Among bins with at least four supporting problems, it selects the highest-accuracy positive-lift bin, breaking ties by support and then lower margin. At inference, probability mixing moves the token-level gap toward the selected interval: uniform mixing reduces the gap, whereas mixing with a distribution uniform over the selected top- k increases it. Trace-level calibration and token-level application have different units. Both policies transfer routing targets from Unsloth 4-bit replay to GPTQ generation, which is an additional unisolated deployment difference.

Matched evaluation. Each policy is compared with its own baseline using identical problem IDs, prompts, prompt token IDs, per-attempt seeds, sampling settings, engine configuration, and recorded software versions. Generation uses temperature 0.6, top- p 0.95, and a 32,768-token limit. The identity +1 and margin comparisons use maximum concurrency 16; the identity −1 comparison uses 25. Baseline outputs therefore cannot be treated as interchangeable across comparisons. Only complete matched stochastic pairs are reported; the strength+2 condition has no complete matched evaluation artifact. Separate greedy and restricted-benchmark outputs are not

pooled here. The engine writes completed records incrementally and can resume without continuing a partially generated sequence. Baseline reuse requires matching contracts and a verified output hash.

Table 32. Complete matched Qwen3.5 GPTQ stochastic evaluations, 1,395 attempts on 465 held-out problems per comparison. B/G denotes baseline/guided; changes are percentage points. Flips are wrong-to-right/right-to-wrong. These are descriptive point estimates without cluster confidence intervals.

Policy	B accuracy	G accuracy	Change	Flips	Limit hits B/G	Mean tokens B/G
Identity +1	0.7391	0.7355	-0.36	39/44	140/146	13863/13706
Identity -1	0.7319	0.7398	+0.79	47/36	154/138	13830/13919
Margin 1	0.7384	0.7369	-0.14	39/41	144/143	13848/13840

Table 33. Qwen3.5 GPTQ guiding results by benchmark. Correct counts are baseline/guided; accuracy changes are percentage points. Repeated attempts from one source problem are dependent.

Policy	Dataset	Attempts	Correct B/G	Change
Identity +1	aime24	288	185/177	-2.78
Identity +1	aime25	288	246/246	+0.00
Identity +1	amc23	384	381/381	+0.00
Identity +1	math500	150	106/106	+0.00
Identity +1	minerva	82	16/18	+2.44
Identity +1	olympiad	203	97/98	+0.49
Identity -1	aime24	288	178/182	+1.39
Identity -1	aime25	288	242/248	+2.08
Identity -1	amc23	384	382/382	+0.00
Identity -1	math500	150	103/102	-0.67
Identity -1	minerva	82	20/19	-1.22
Identity -1	olympiad	203	96/99	+1.48
Margin 1	aime24	288	180/184	+1.39
Margin 1	aime25	288	243/246	+1.04
Margin 1	amc23	384	380/380	+0.00
Margin 1	math500	150	106/104	-1.33
Margin 1	minerva	82	20/18	-2.44
Margin 1	olympiad	203	101/96	-2.46

Observed effects and limits. Identity +1 changes 83 correctness outcomes and loses five net correct answers; margin guiding changes 80 and loses two. Identity -1 changes 83 and gains eleven. The corresponding pooled changes are -0.36, -0.14, and +0.79 percentage points. This does not establish a reliable benefit from promoting positive-lift experts or targeting the calibrated margin. The negative-strength point estimate is positive, but differences across strengths are not a controlled ranking because their baselines and concurrency differ. These are exploratory comparisons without paired source-problem confidence intervals or a separate strength-selection set. Pooled attempt accuracy also weights each AIME/AMC problem 32 times as heavily as each single-attempt problem; benchmark counts are reported to make this weighting explicit.

What the margin hook actually changes. Full diagnostics for the completed margin comparison record 20,657,039 hook token evaluations per selected layer. The top-eight identity set changes on 32.7–41.8% of these evaluations. Mean distance to the target interval falls from $2.19\text{--}4.71 \times 10^{-4}$ to $0.285\text{--}0.902 \times 10^{-4}$ across layers. Thus the numerical operation moves the intended gap, but it does not isolate a membership-preserving intervention after finite-precision log-probability conversion and casting. These diagnostics describe the hooked logits, not an independent recording of the kernel’s dispatched expert IDs. Their denominator can include prefill, padding, and recomputation and is not a generated-token count. The identity -1 comparison uses minimal diagnostics, so detailed hook statistics are not available on a common basis across all conditions.

8 Synthesis and Limitations

The frozen Qwen3.8 judge retains the 40-example seed because the optimized candidate fails the per-class recall gate. Its NVFP4 deployment reaches 0.710 accuracy on 407 reused validation units under both low and medium effort, while changing 10.1% of labels between settings. Neither prompt selection nor this deployment check establishes annotation accuracy on the generated benchmark traces. Imported Gemma and GPT-OSS labels additionally use a different sampling protocol and inference configuration.

All three models have linearly decodable information about upcoming Schoenfeld labels in the released response prefixes. Selected-layer AUROC ranges are 0.835–0.934 for Qwen, 0.767–0.906 for GPT-OSS, and 0.816–0.922 for Gemma. However, every response appears in both sentence partitions, and the reported layers are selected on the same test split. The results support within-corpus decodability rather than generalization to unseen responses. Qwen’s external 581-unit inspection has 50.3% primary and 60.8% compatible agreement under a provisional hybrid review; it is neither independent human validation nor evidence for the other encoders.

The three correlation corpora each cover 4,647 attempts, 1,547 problems, and 100 complete avg@32 groups. Associations depend on model, benchmark, and aggregation level. For Qwen, whole-continuation confidence correlates with avg@32 at $\rho = 0.632$ on AIME24 and 0.705 on AIME25, whereas AMC23 gives -0.200 with an interval containing zero. These estimates do not control length, termination, or difficulty. Problem-respecting bootstrap intervals address repeated attempts for the reported primary features; naive attempt-level significance screens remain exploratory. Sparse and unmatched class coverage further limits comparisons. No common causal interpretation of router confidence follows from these data.

The held-out interventions provide direct perturbation evidence for Qwen, but no consistent accuracy improvement: identity promotion and margin guiding lose five and two net correct answers, while identity suppression gains eleven out of 1,395 attempts. Each effect is relative to its own matched baseline. The policy calibration excludes unscored traces, transfers between quantizations, and selects targets from observational associations. Margin manipulation also changes expert membership in finite precision. These limitations prevent attributing a small aggregate effect exclusively to the intended routing statistic or claiming a portable improvement. Problem-cluster uncertainty and independent selection of intervention strength remain necessary.

9 Immediate Work

The remaining priorities are independent assessment of generated-trace labels; response-grouped probe validation with layer selection inside the training folds; matched annotation coverage across models; and length-/termination-aware correlation models evaluated on held-out problems. For routing interventions, uncertainty should respect source-problem clusters, and further controls should assess random expert targets, calibration sensitivity, and transfer across quantizations. Strength selection and final evaluation require separate problem sets. Class-conditioned routing remains unevaluated.

A Reproducibility Checklist

Current artifact inventory. The judge includes grouped prompt-selection studies, final-fit validation, and a frozen-prompt NVFP4 effort comparison. Boundary probes are complete for Qwen, GPT-OSS, and Gemma; external probe inspection covers Qwen only. Each correlation model has a complete six-benchmark corpus, with differing class-label coverage. Guiding tables use complete Qwen identity +1, identity −1, and margin-1 matched stochastic pairs. Incomplete conditions are excluded from numerical comparisons. Independent annotation assessment, response-grouped probe evaluation, and problem-cluster intervention uncertainty remain outstanding.

Regenerating probe, effort, and guiding tables. Run `python3 report/generate_current_results.py` from the repository root. It reads saved probe summaries and the judge-effort CSV, checks that all reported guiding manifests are complete and share comparison contracts, and verifies unique matching IDs, prompts, prompt tokens, per-attempt sampling settings, and Boolean outcomes. It then recomputes pooled and benchmark-specific correctness, flips, token counts, and truncation counts. `report/current_results_sources.json` records input SHA-256 hashes and the derived guiding statistics. The generator does not run inference, rejudge answers, or calculate uncertainty intervals.

A.1 Completed run and independent checks

The numerical source root is `results/correlation_pipeline/reasoning-vllm-v1`. Under it, `analysis/unsloth--Qwen3.5-35B-A3B/correlations.json` records completed analysis of 4,647 attempts. Source hashes are saved in `report/correlation_tables_sources.json`; the audit separately records the analysis SHA-256. Companion CSVs contain trace, problem, and expert features. The `views-v1` directory contains full reasoning, seven classes, ten position bins, overflow, and coverage summaries. Saved analysis uses 500 bootstrap replicates, seed 42, minimum expert support of four traces, and at most 128 ordered expert pairs.

The position-reference corpus has 4,647 traces and mean reasoning length 12,498.062 tokens. Its SHA-256 is 99b71a50bf12f39ecec05f9034b3b07ff57236d2e53a7343f01caa791e108c1e. View population version 2 pools all reasoning tokens for full and position views, and available tagged sentence tokens for class views. All attempts, including limit hits, are retained with pairwise omission of missing features.

Independent audit. `report/audit_tagged_correlation_run.py` matches all generation records to replay fingerprints, verifies available annotation fingerprints and sentence offsets, checks selected layers and compact values, and validates inclusion, correctness propagation, token-limit flags, and view coverage. It independently recalculates saved binary and trace-level metric-pair coefficients across 20 scopes from CSV observations (52,199 coefficients). It additionally checks 984 problem-mean Spearman and within-problem difference estimates for whole continuation and full reasoning. It also checks reported expert maxima and their continuous-metric associations. Counts and numerical discrepancies are recorded in `report/correlation_run_audit.json`.

These checks do not independently rejudge mathematical answers or recompute model activations, bootstrap intervals, or BH adjustments.

Regenerating the report. From the repository root, using Python with NumPy, SciPy, and Matplotlib:

```
python3 report/audit_tagged_correlation_run.py \
  > /tmp/correlation_run_audit.json
cp /tmp/correlation_run_audit.json report/correlation_run_audit.json
MPLCONFIGDIR=/tmp/moe-report-matplotlib \
  python3 report/generate_correlation_tables.py
cd report
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The renderer writes coverage, correctness, class, lexical-event, expert, and repeated-attempt tables, a compact plot atlas, and standalone PDF/PNG figures. The atlas replaces the large numeric appendix in the compiled report. The renderer reuses saved feature correlations and additionally computes descriptive sampled-label counts, adjacent-sentence transitions, class-share associations, and binned outcomes from fingerprinted inputs. These new summaries and bin counts/ranges are saved in `report/descriptive_plot_statistics.json`; router-value intervals use 500 source-problem bootstrap resamples with seed 42. No model is fitted. Original experiment outputs are unchanged.

Scope of reproducibility. Mutable local Docker tags and checkpoint names do not fix all underlying binaries. Exact reproduction also requires resolved model revisions, image digests, package versions, and the matching code revision. Saved contracts and source fingerprints identify this report’s numerical population.

Three-model report update. Run `python3 report/generate_multimodel_correlations.py` from the repository root, then rebuild with the `latexmk` command above. This reads the Qwen run above and the corresponding reasoning-vllm-v1 runs under `results/correlation_pipeline/gemma-nvfp4-nf4` and `results/correlation_pipeline/gpt-oss-20b`. The generator verifies coverage and recomputes reported aggregate coefficients from saved CSVs, then writes comparison tables, figures, and both additional model atlases. Source hashes and check results are recorded separately in `report/multimodel_correlation_sources.json` and `report/multimodel_correlation_audit.json`.

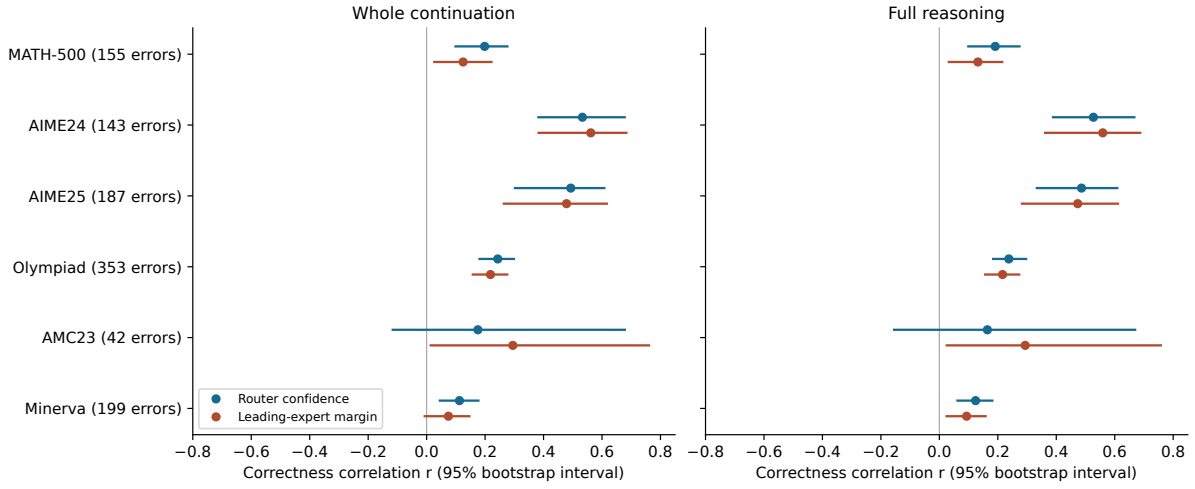


Fig. 1. Qwen3.5 router confidence and leading-expert margin versus correctness, with saved pointwise 95% problem-bootstrap intervals across all six benchmarks. Both views retain all attempts, including limit hits; full reasoning excludes non-reasoning continuation tokens. Error counts refer to attempts, not independent source problems.

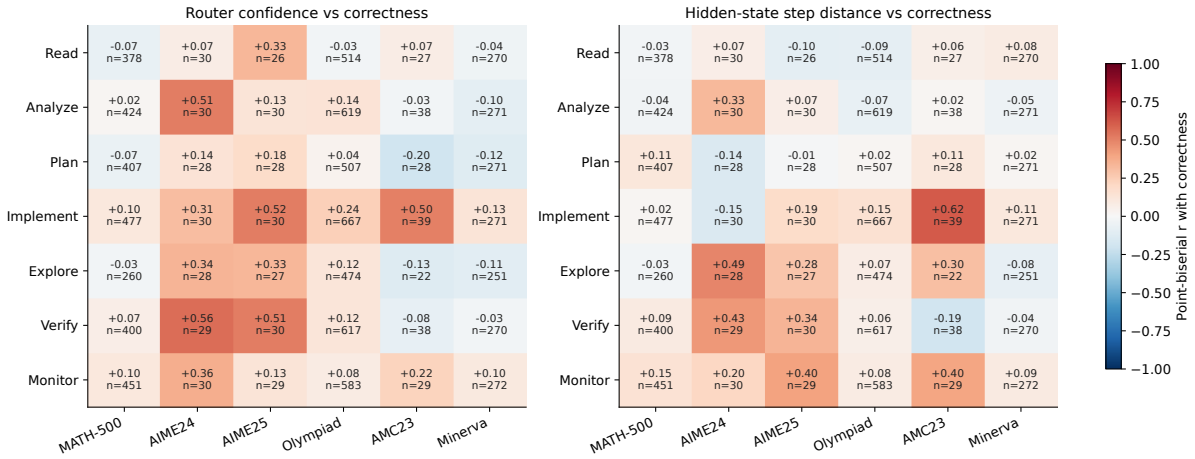


Fig. 2. Class-conditioned confidence and hidden-state step distance versus whole-attempt correctness. Cells show point-biserial r and finite trace count. Classes use sampled Qwen3.8 labels on Qwen3.5 generations, with different token-bearing cohorts. No significance markings are implied.

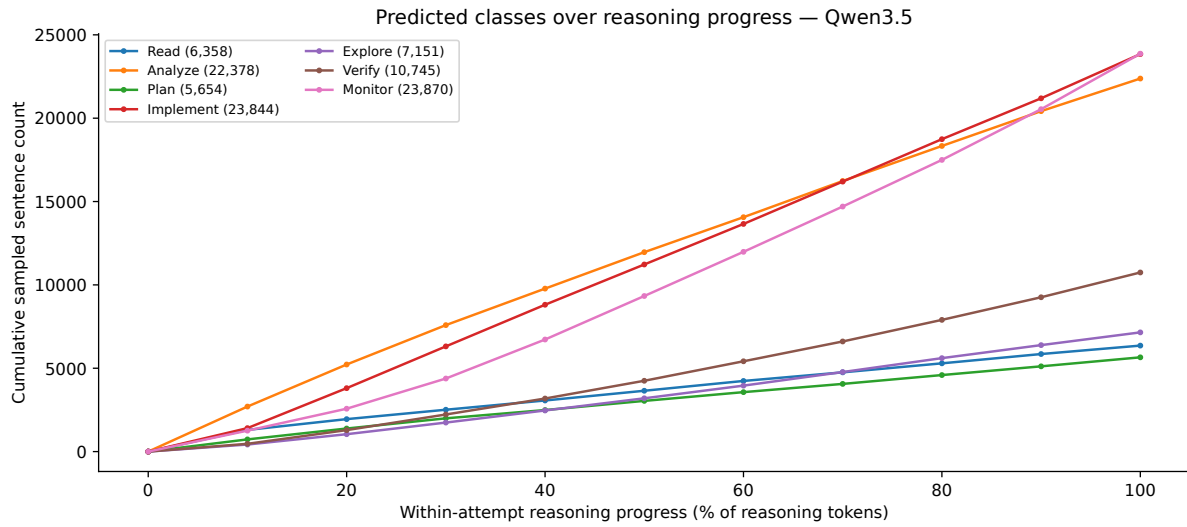


Fig. 3. Cumulative counts of sampled sentences by predicted class, pooled across benchmarks. Counts start at zero; a sentence is counted when its last reasoning token is reached. Progress is normalized by each attempt's full reasoning-token length, unlike the absolute windows in Figure 8. Legend totals equal the rightmost endpoints. These are raw sampled counts, not cumulative percentages or estimates of full-corpus counts; longer traces and benchmark sampling quotas affect them.

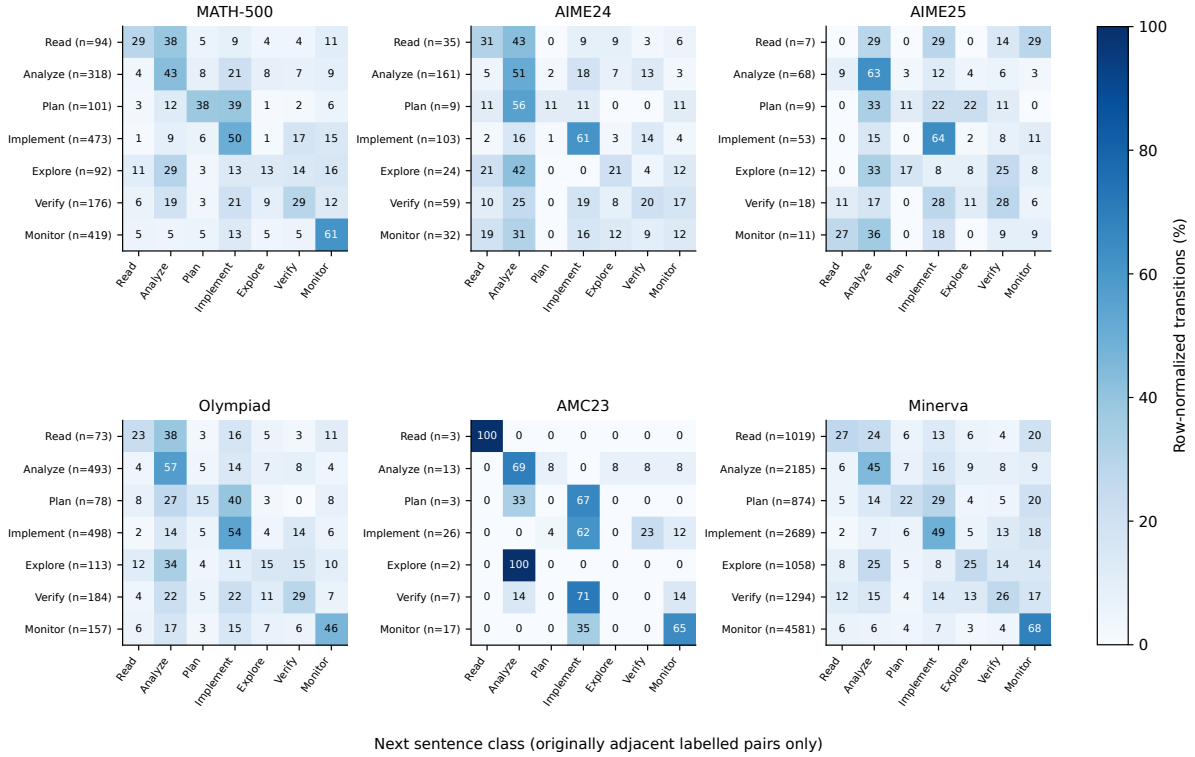


Fig. 4. Observed class-state transitions by benchmark. Rows are the current class, columns the next class; cells are row percentages and row labels give pair counts. Only originally consecutive sentence indices with both labels are counted. Gaps in sampling are never bridged. Self-transitions are retained; empty rows are undefined. This describes the observed adjacent-pair sample, not the complete transition process or an effect of changing state.

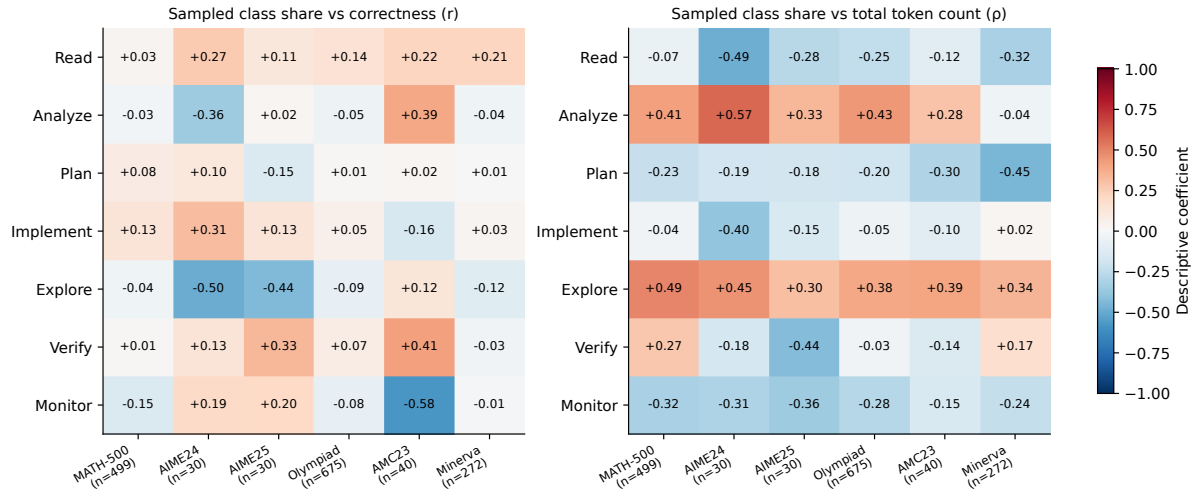


Fig. 5. Sampled class proportions versus whole-attempt correctness (point-biserial r) and total continuation-token count (Spearman ρ). All labelled attempts enter each class, including observed zero proportions. These differ from correlations of router features within class-selected tokens. No significance claim is made; class shares sum to one and cannot be interpreted as independent effects.

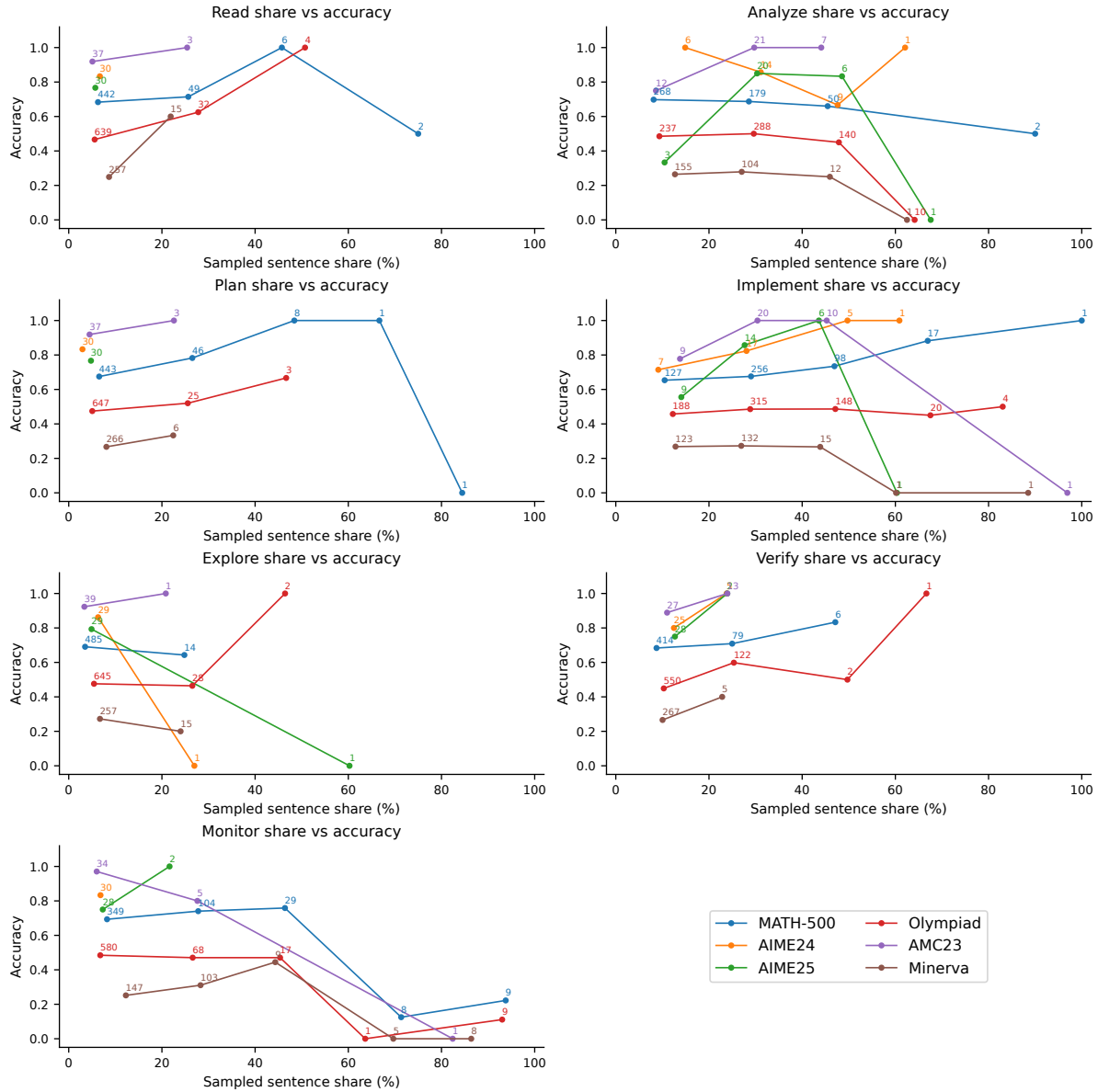


Fig. 6. Actual outcome levels versus sampled class proportions. Fixed share bins are $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 80)$, and $[80, 100]$ percent. Each point is at the mean observed share and shows accuracy; small labels give attempt counts. Empty bins are omitted. Colours distinguish benchmarks; lines only connect occupied bins. Sparse bins, sentence sampling, length, and difficulty can explain apparent trends; these descriptive plots do not identify optimal class proportions or a validated accuracy-length Pareto frontier.

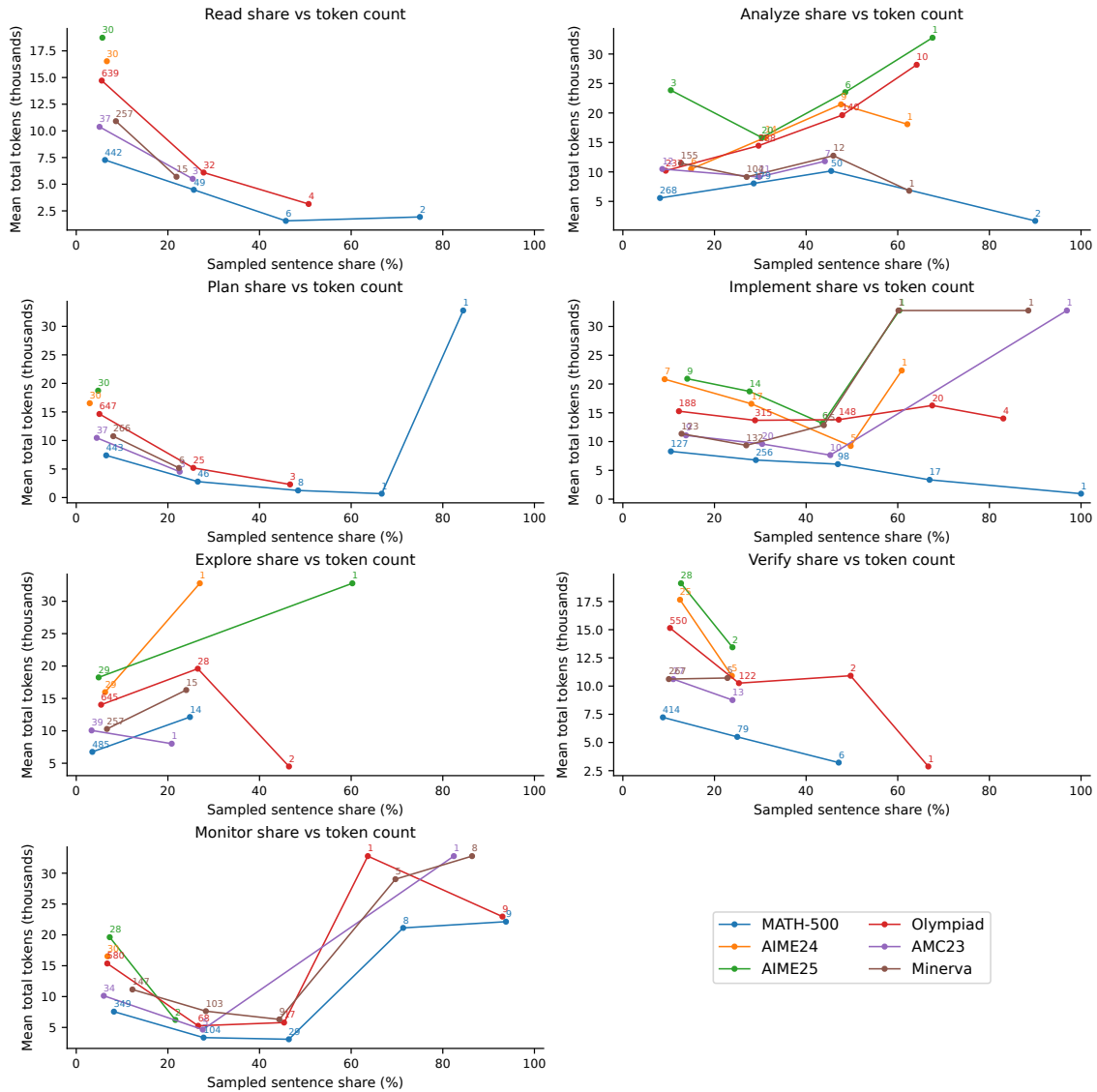


Fig. 7. Mean total continuation tokens versus sampled class share. Bin boundaries, colours, sample-count labels, and sampling limitations are as in Figure 6. Token count includes the final answer and token-limit completions; lines are descriptive, not intervention effects.

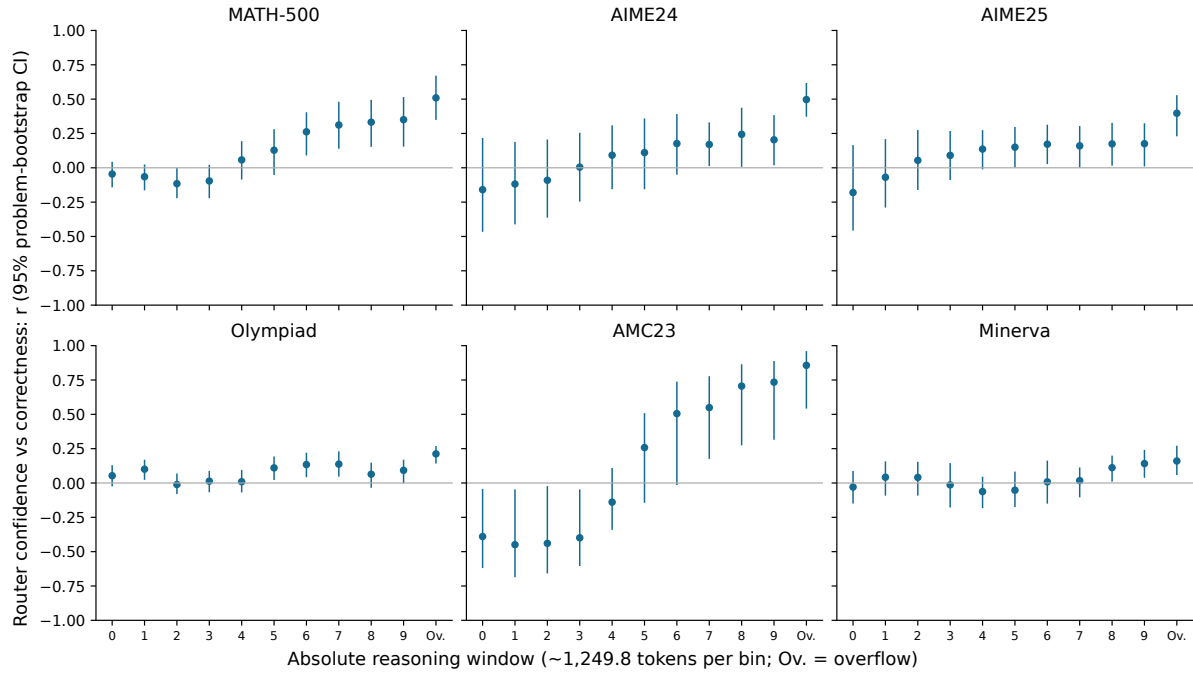


Fig. 8. Router confidence versus whole-attempt correctness in absolute reasoning windows, with saved pointwise 95% source-problem bootstrap intervals. Each benchmark has its own panel. Missing estimates remain blank. Later windows use changing token-bearing cohorts; these are not within-trace causal effects.

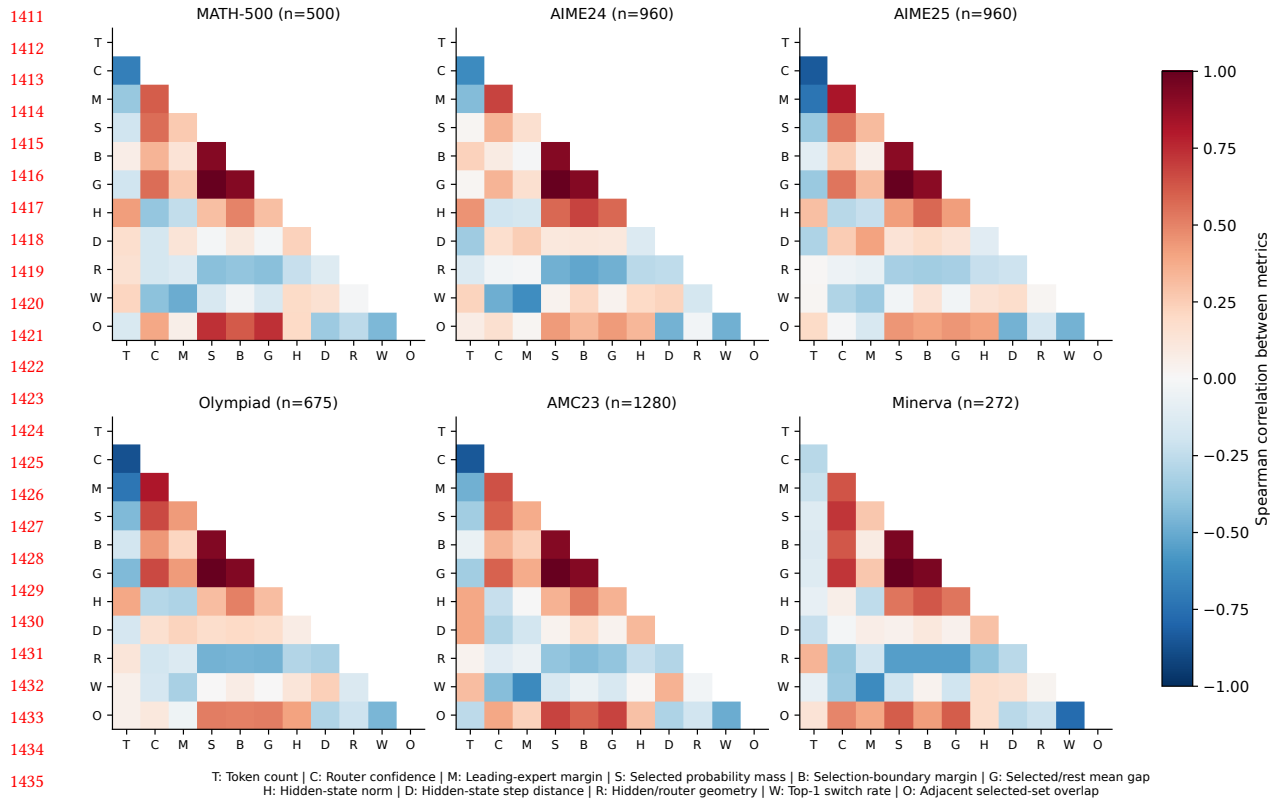


Fig. 9. Whole-continuation metric–metric Spearman correlations across all six benchmarks. Lower triangles display each association once; diagonals and upper triangles are omitted. Repeated attempts and limit hits remain included. These are descriptive metric relationships, not correctness effects.

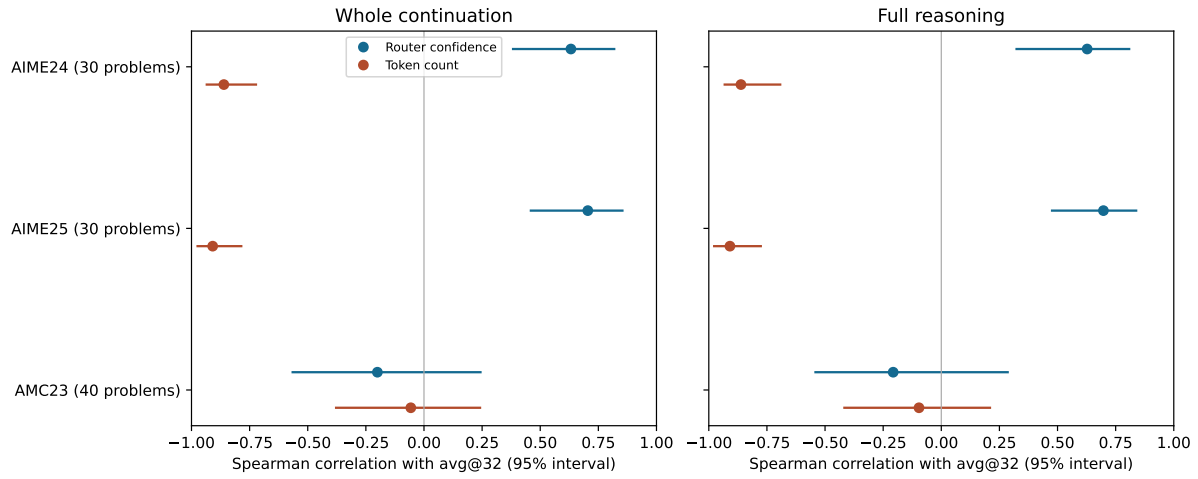


Fig. 10. Mean confidence and token count versus problem-level avg@32, with saved 95% whole-problem bootstrap intervals. Whole-continuation and full-reasoning views use the same complete 30/30/40 problem groups. These estimates remain unadjusted for difficulty, length, and termination.

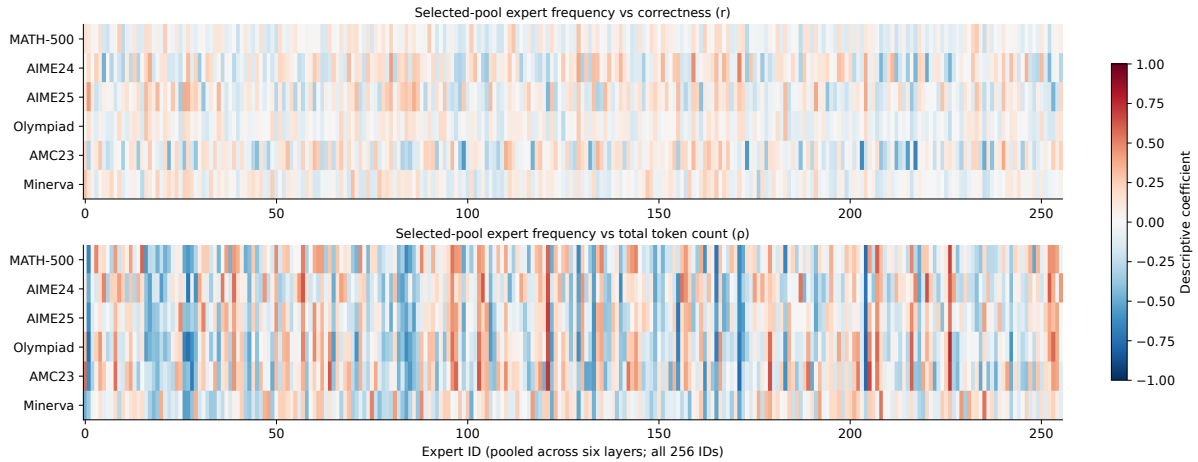


Fig. 11. Selected-pool membership frequency for every expert ID versus correctness (r , top) and total continuation tokens (ρ , bottom). All 256 IDs are shown in their original order for each benchmark. Rates pool tokens and six layers; the same ID refers to distinct layer-local modules. These associations do not establish that activating an expert will improve accuracy or reduce length.

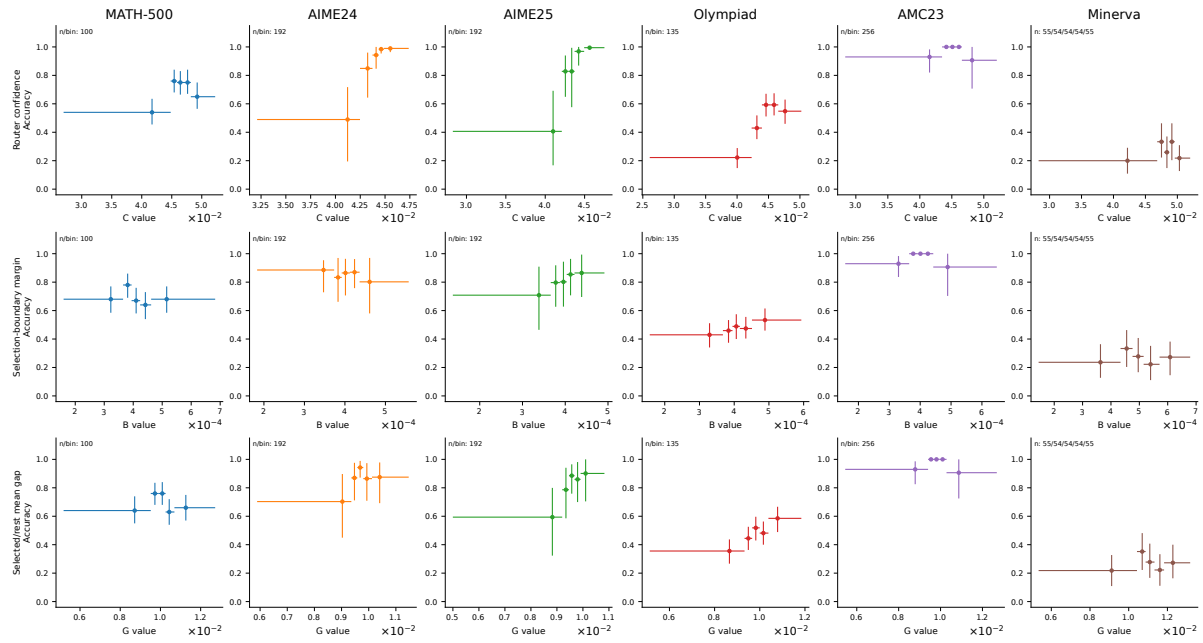


Fig. 12. Accuracy at observed router-metric values. Each benchmark uses up to five quantile bins; horizontal bars show actual value ranges, points show mean values and accuracy, and panel labels give attempt counts per bin (left to right, or a shared count when equal). Vertical bars are pointwise 95% percentile intervals from 500 whole-source-problem bootstrap resamples within each fixed bin (seed 42). Degenerate intervals can occur in all-correct/all-incorrect bins and do not imply certainty. All attempts and limit hits are retained. Bins can share source problems, so their estimates are not independent; neither bin selection nor length and termination confounding is addressed by these intervals.

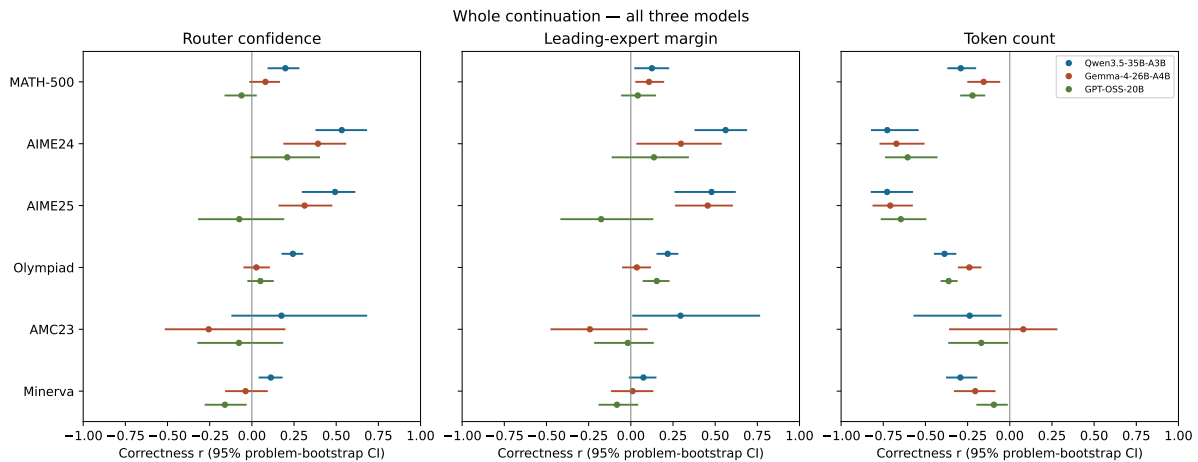


Fig. 13. Whole continuation: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

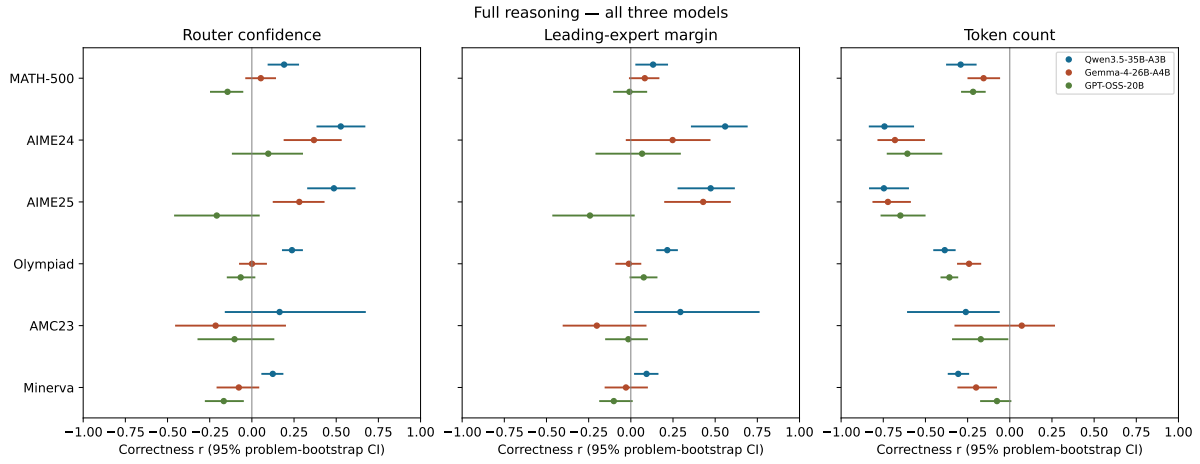


Fig. 14. Full reasoning: saved correctness correlations and pointwise source-problem bootstrap intervals. Overlap or separation of intervals is not a paired test of model differences.

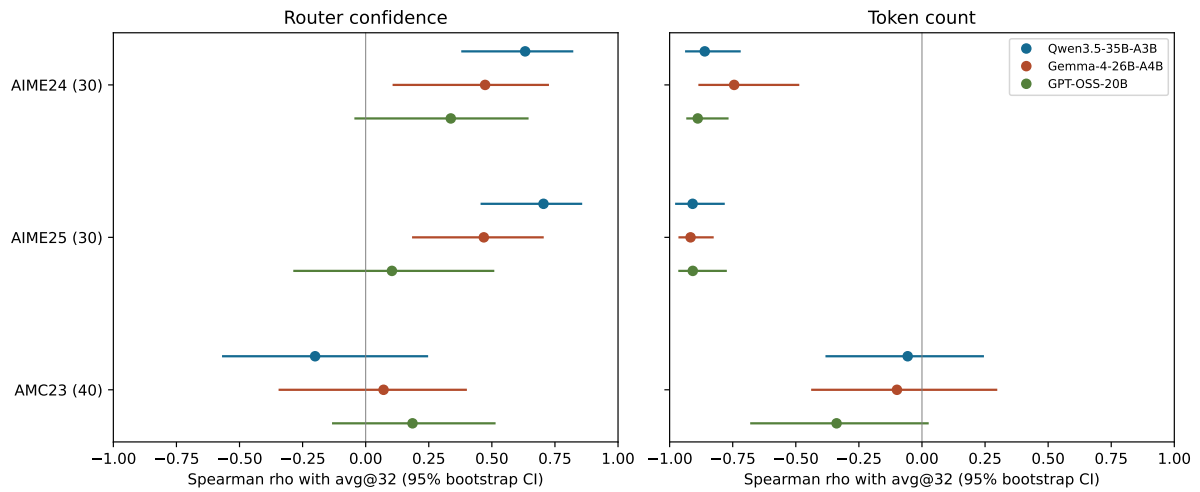


Fig. 15. All models: whole-continuation problem means versus avg@32, with saved pointwise problem-bootstrap intervals. All 32 attempts enter each problem mean.

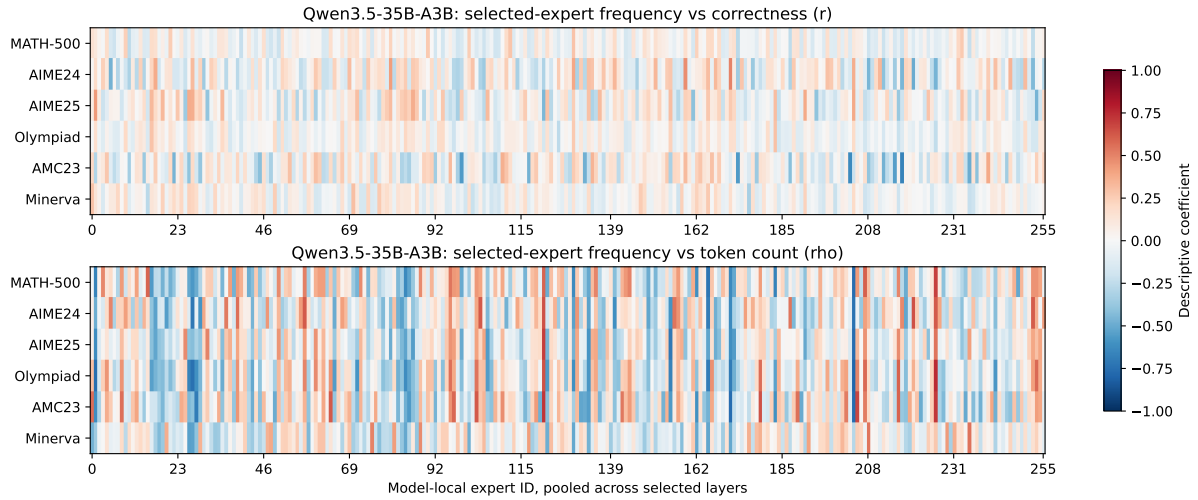


Fig. 16. Qwen3.5-35B-A3B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

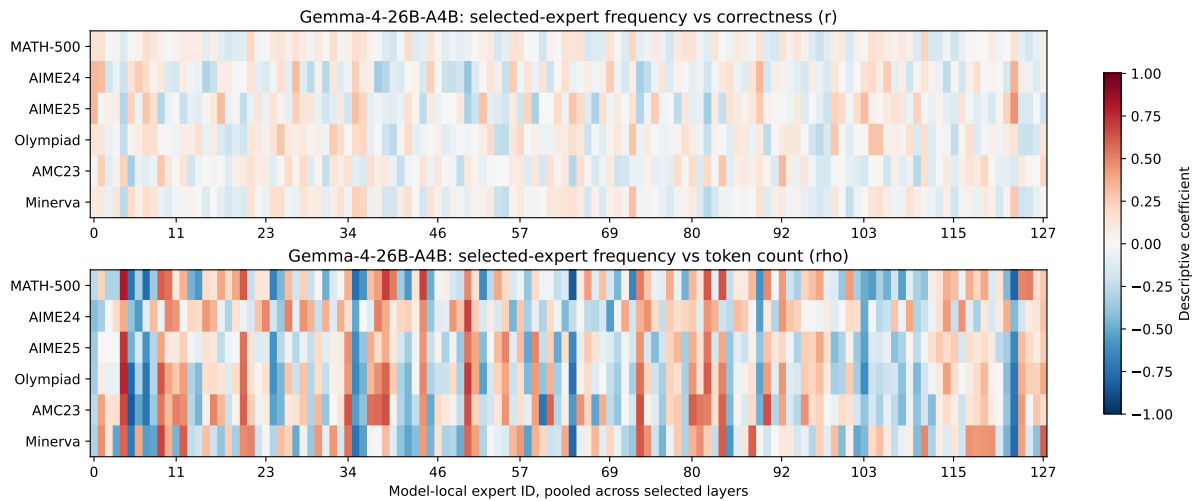


Fig. 17. Gemma-4-26B-A4B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

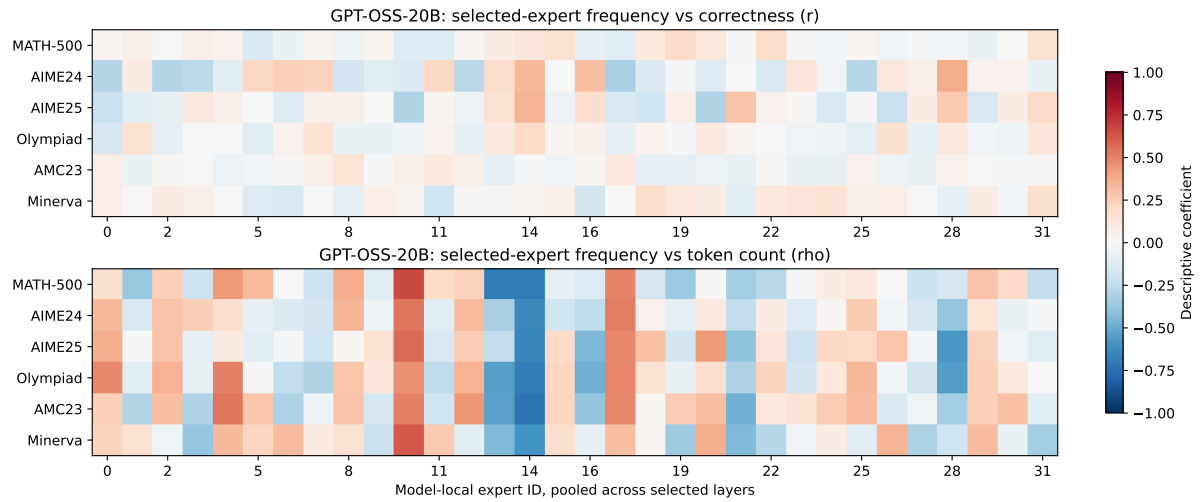


Fig. 18. GPT-OSS-20B: selected-pool expert frequency correlations with correctness and token count. IDs are model-local and pooled across selected layers; they do not identify matching experts across models. Missing coefficients are blank.

B Aggregate correlation plot atlas

Each page fixes a token scope. Lower triangles show metric–metric Spearman ρ ; the last column (Acc) shows point-biserial r with whole-attempt correctness. A dash means undefined or unavailable, not zero. Panels are descriptive; no significance threshold is implied. Sample sizes and saved intervals remain in the fingerprinted source JSON. Legacy lexical targets are omitted. Detailed numeric tables remain in `correlation_tables.tex` as a separate generated artifact.

Table 34. Common metric notation; $k = 8$ for this model.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap
Ch	Character count
St	Step count

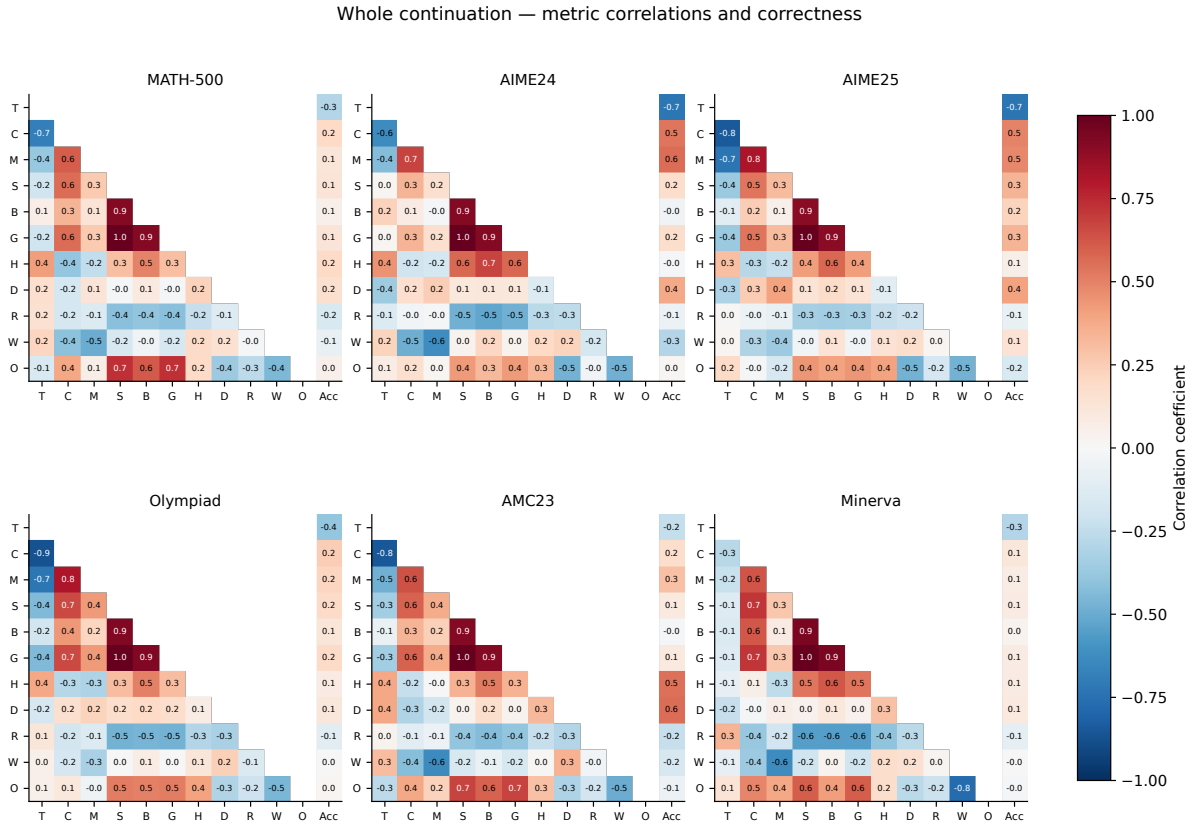


Fig. 19. Whole continuation: metric relationships and correctness. Codes are defined in Table 34.

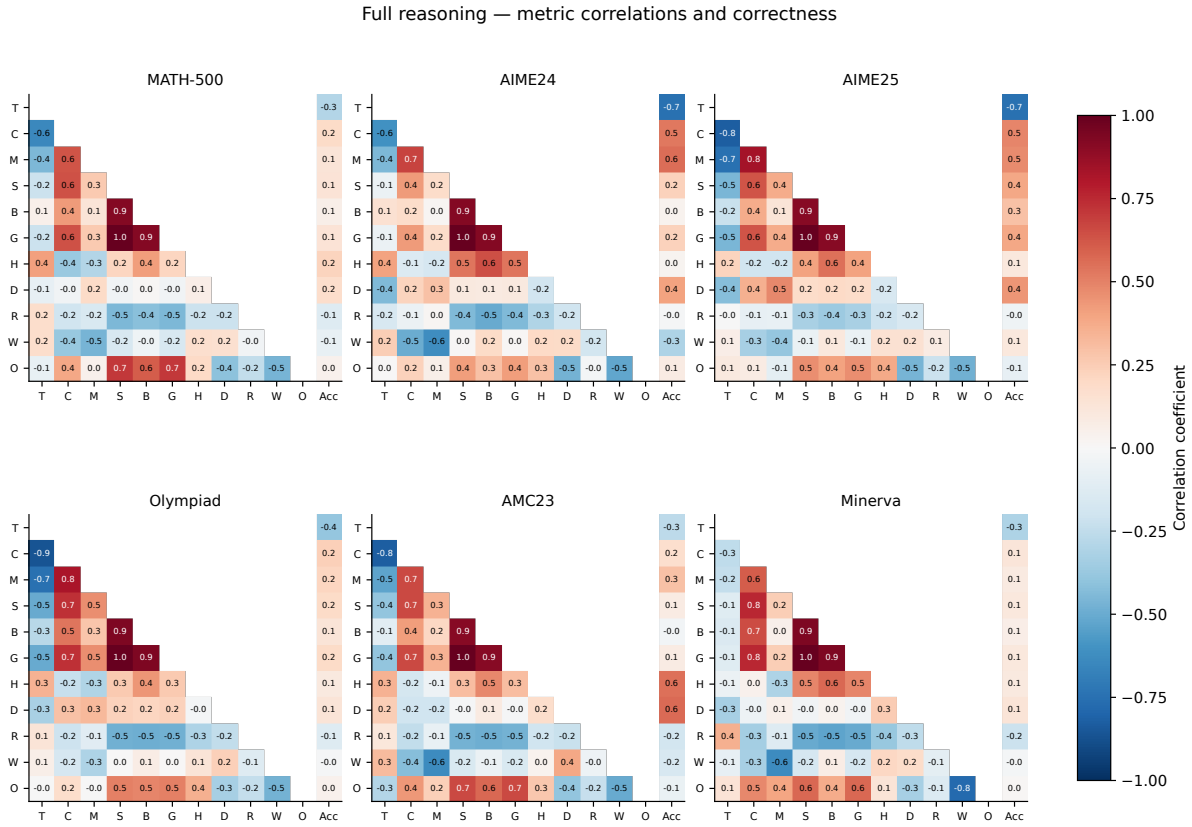


Fig. 20. Full reasoning: metric relationships and correctness. Codes are defined in Table 34.

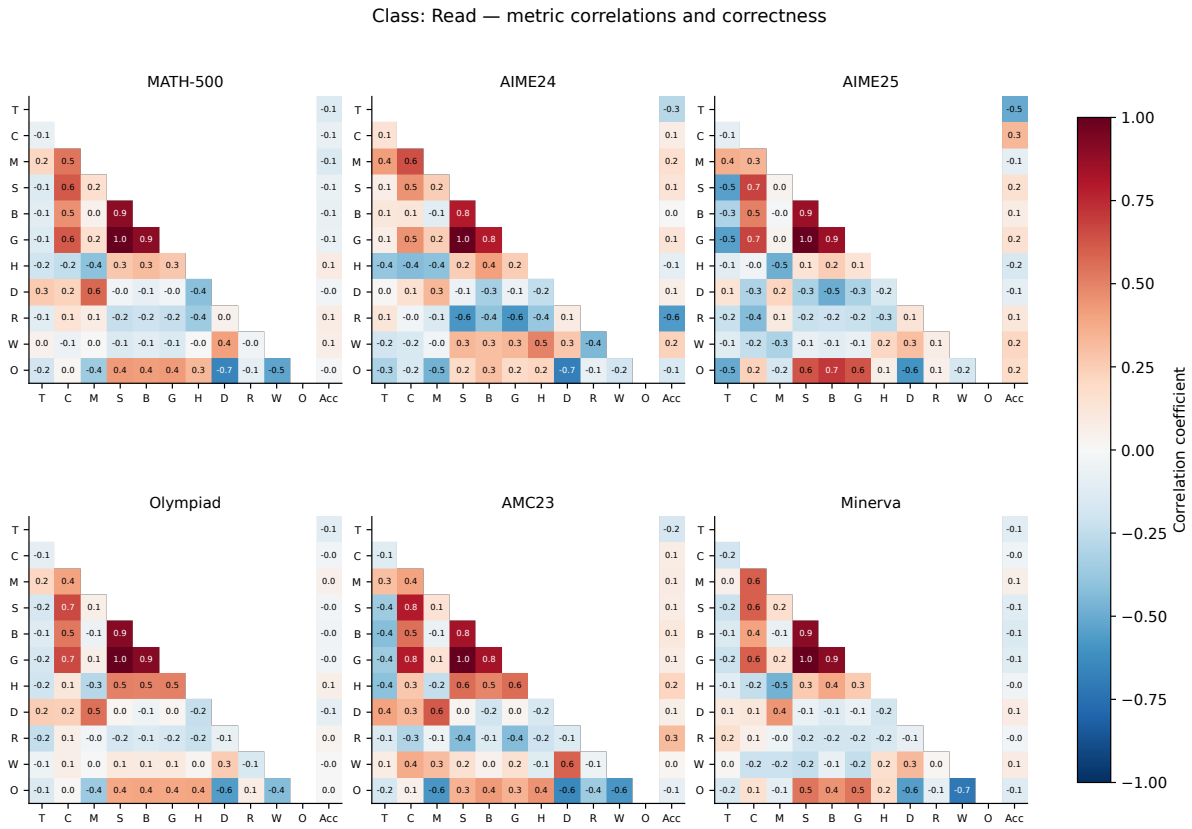


Fig. 21. Class: Read: metric relationships and correctness. Codes are defined in Table 34.

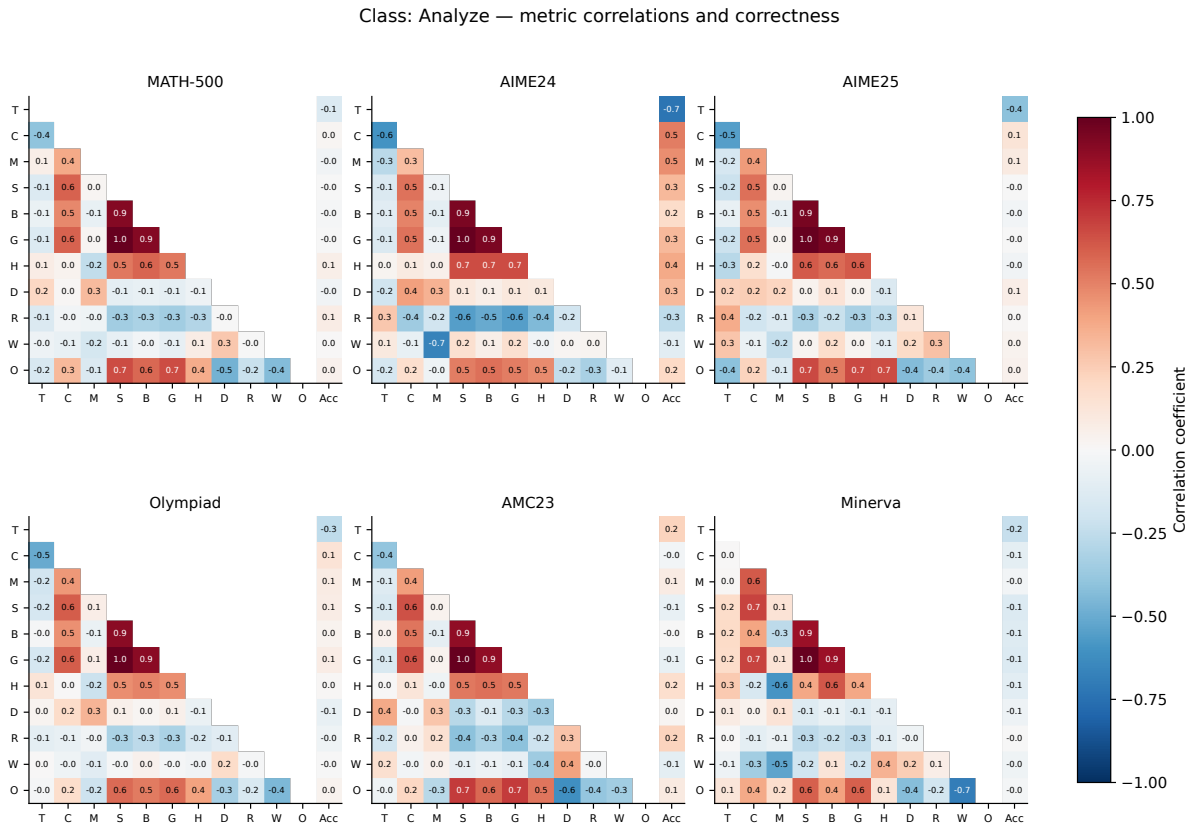


Fig. 22. Class: Analyze: metric relationships and correctness. Codes are defined in Table 34.

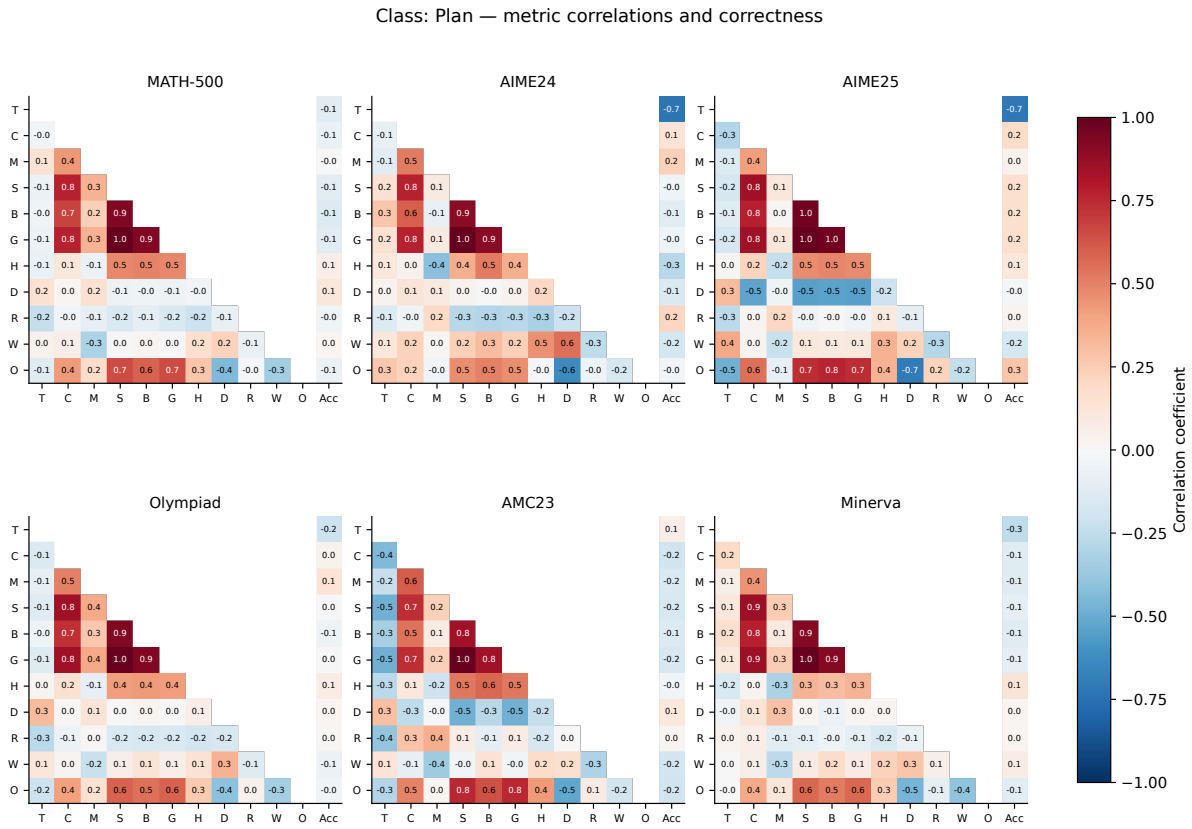


Fig. 23. Class: Plan: metric relationships and correctness. Codes are defined in Table 34.

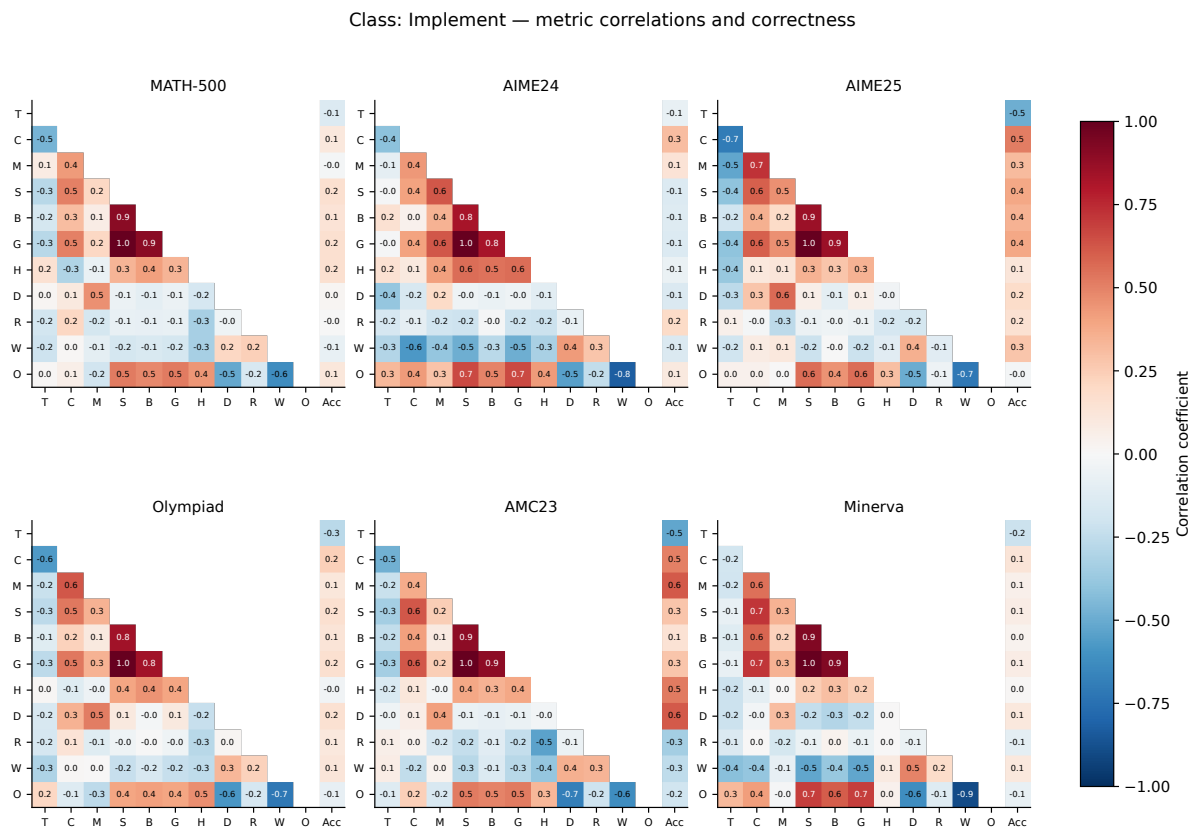


Fig. 24. Class: Implement: metric relationships and correctness. Codes are defined in Table 34.

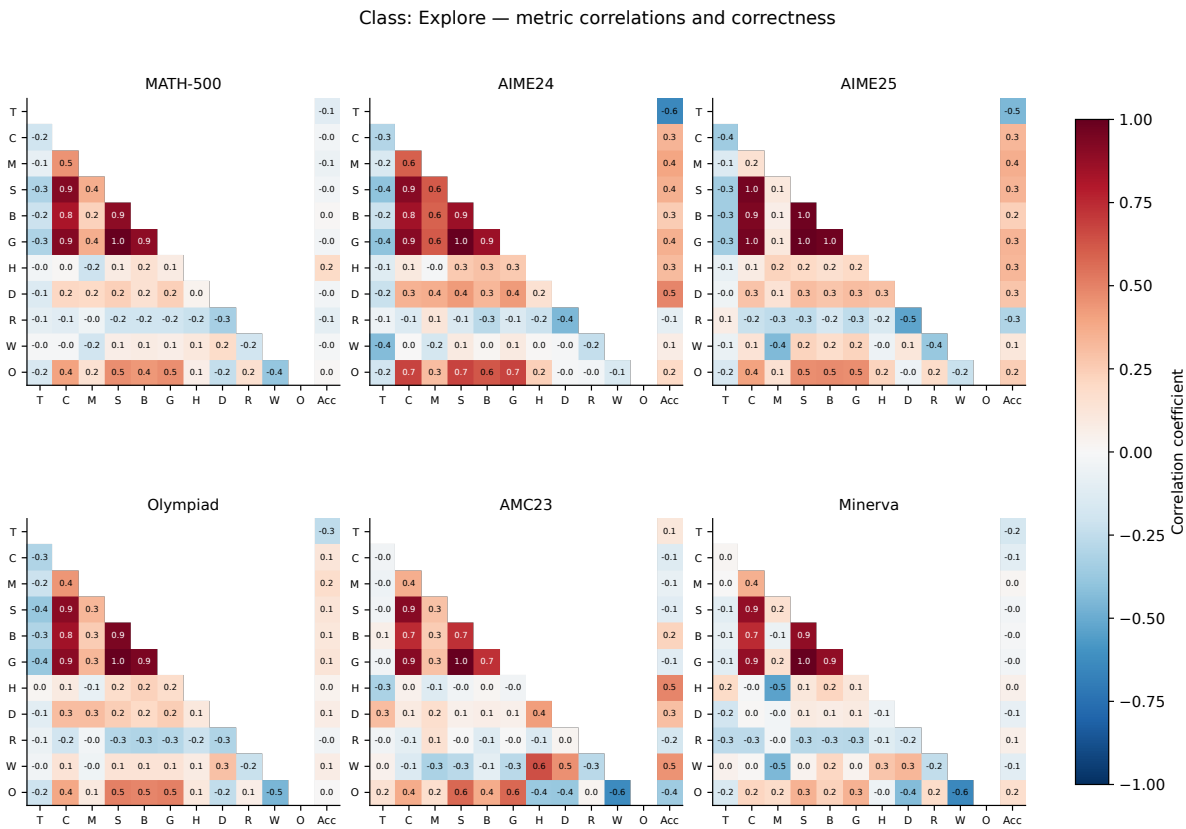


Fig. 25. Class: Explore: metric relationships and correctness. Codes are defined in Table 34.



Fig. 26. Class: Verify: metric relationships and correctness. Codes are defined in Table 34.

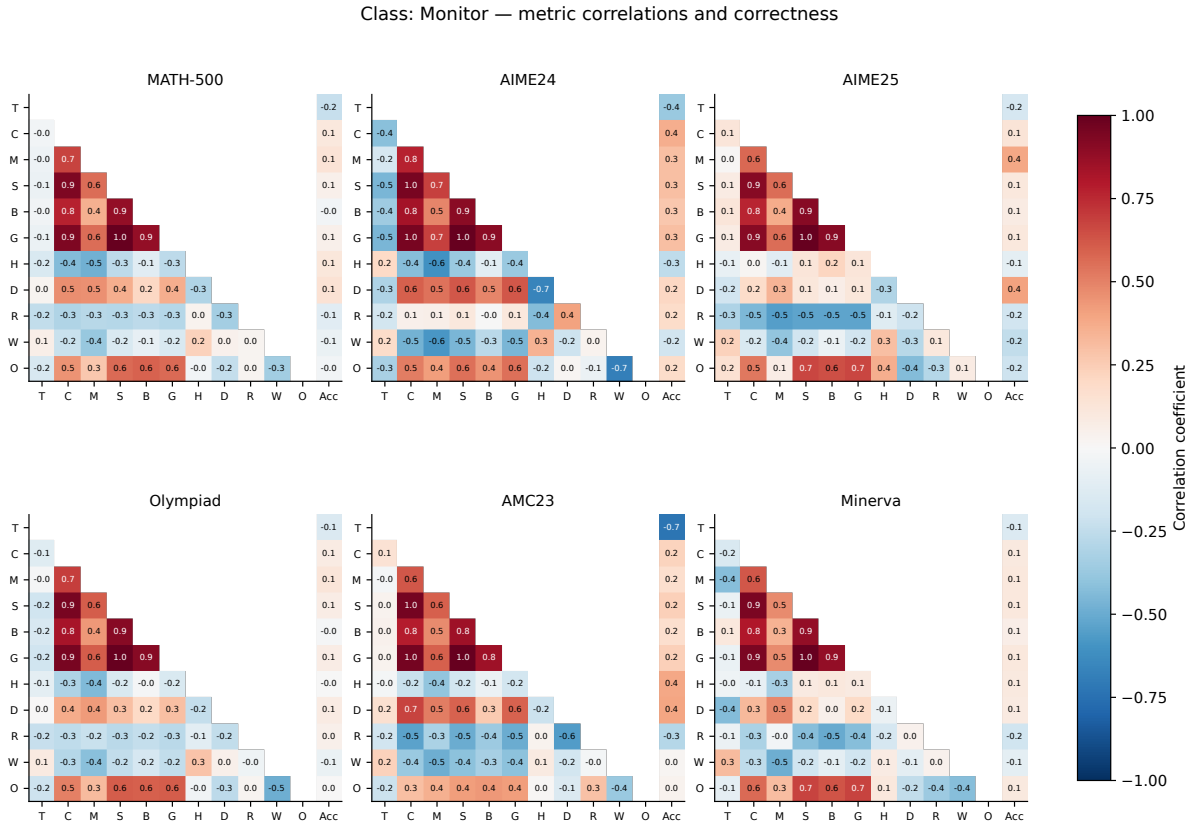


Fig. 27. Class: Monitor: metric relationships and correctness. Codes are defined in Table 34.

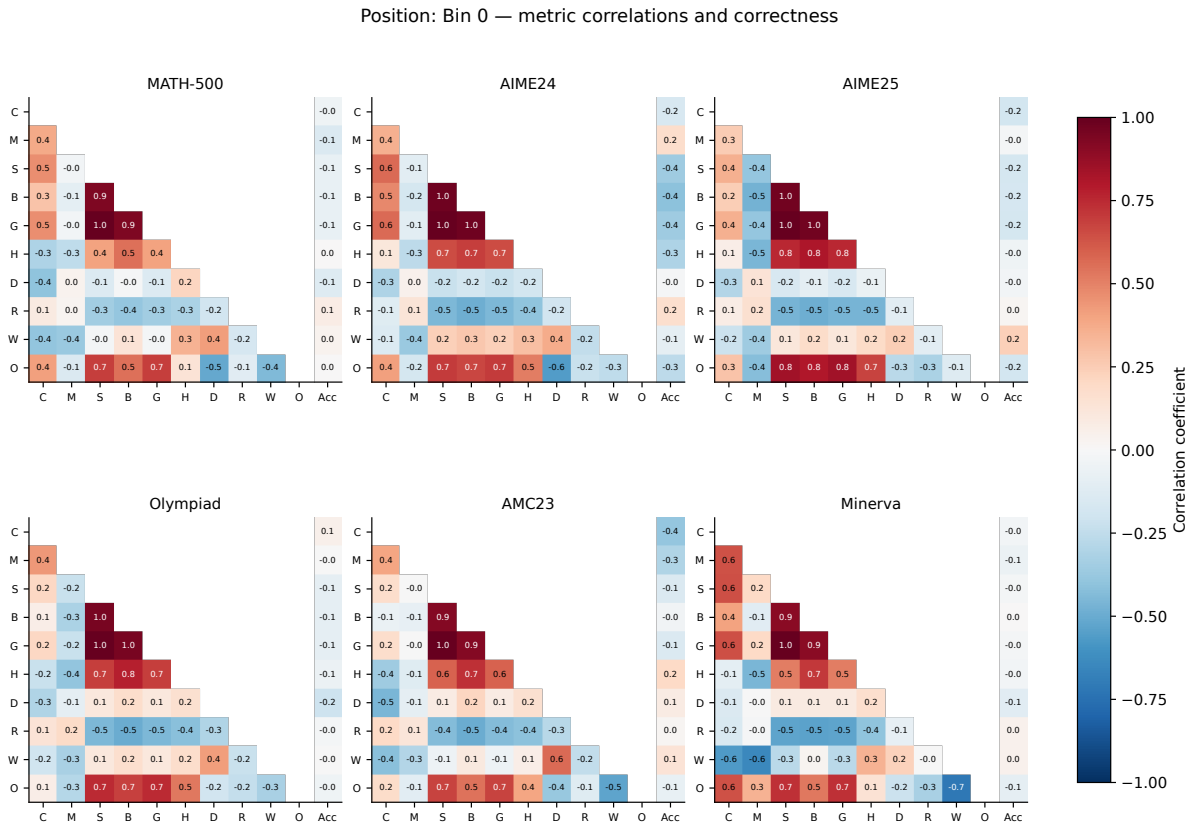


Fig. 28. Position: Bin 0: metric relationships and correctness. Codes are defined in Table 34.

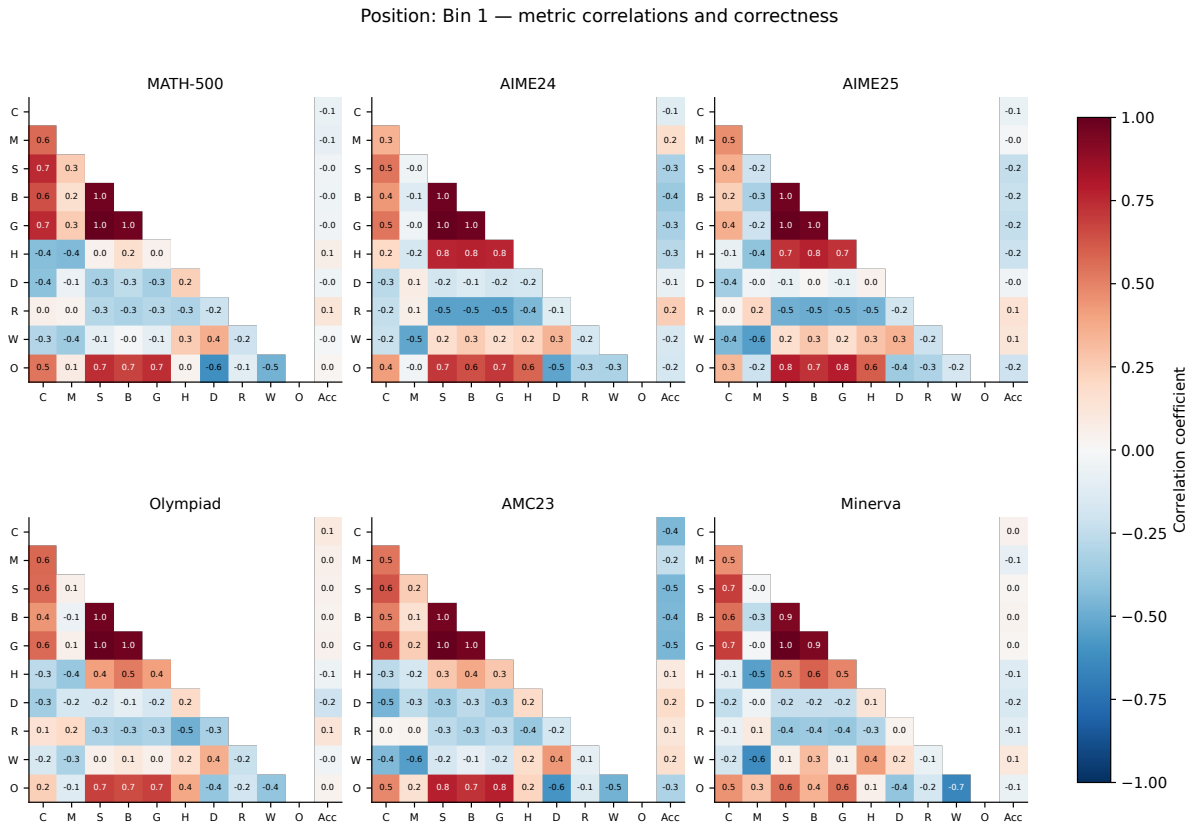


Fig. 29. Position: Bin 1: metric relationships and correctness. Codes are defined in Table 34.

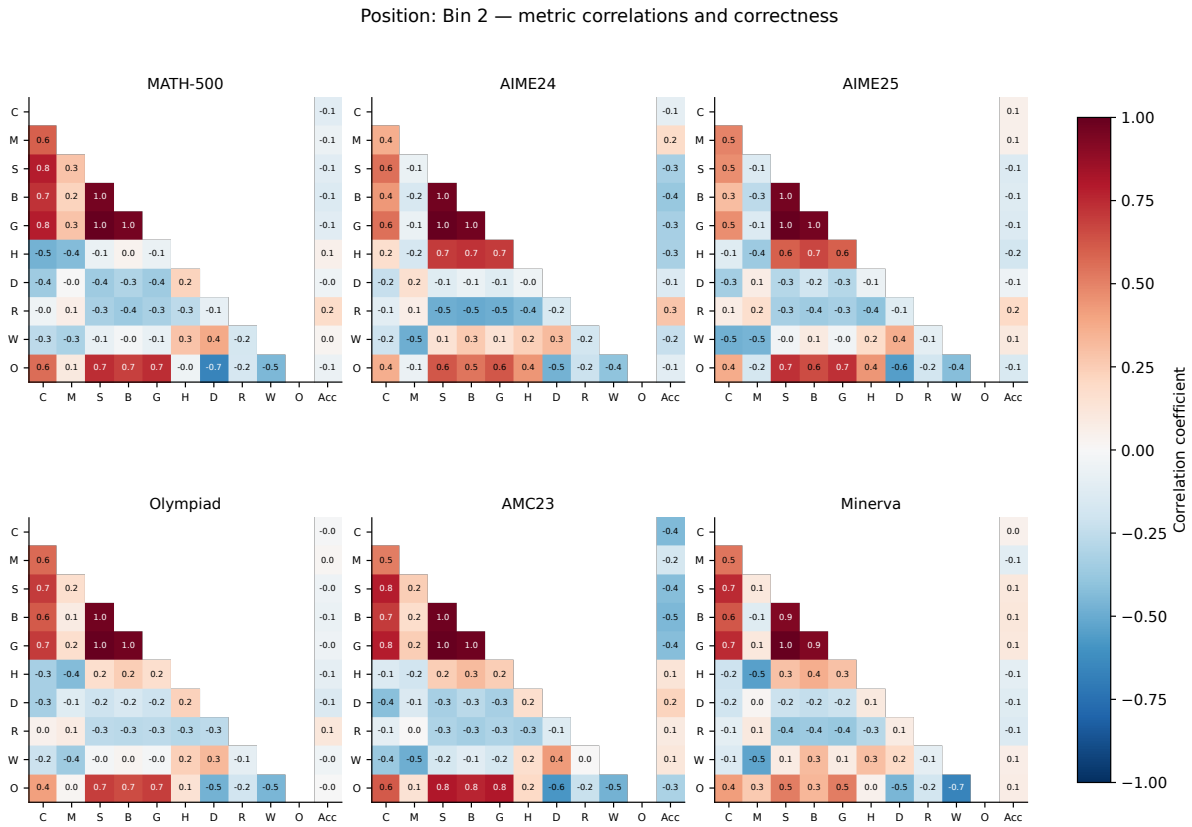


Fig. 30. Position: Bin 2: metric relationships and correctness. Codes are defined in Table 34.

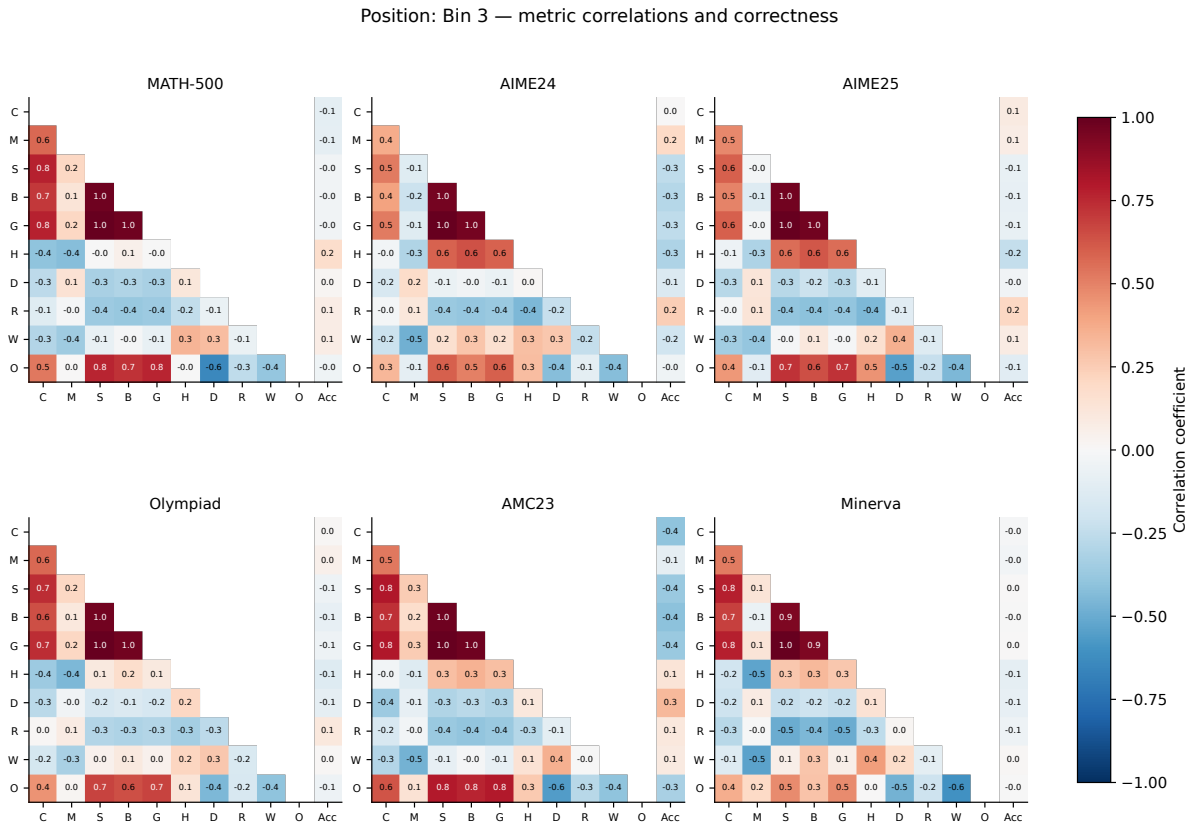


Fig. 31. Position: Bin 3: metric relationships and correctness. Codes are defined in Table 34.

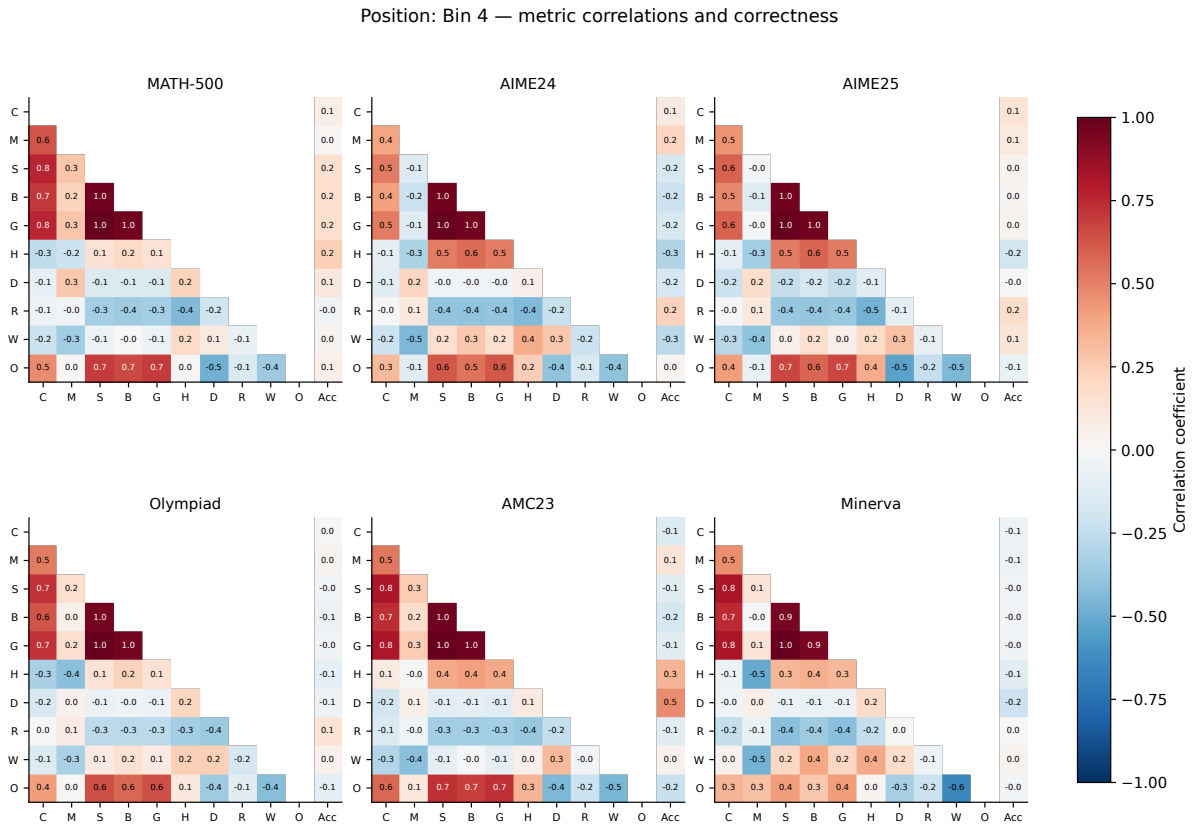


Fig. 32. Position: Bin 4: metric relationships and correctness. Codes are defined in Table 34.

Position: Bin 5 — metric correlations and correctness

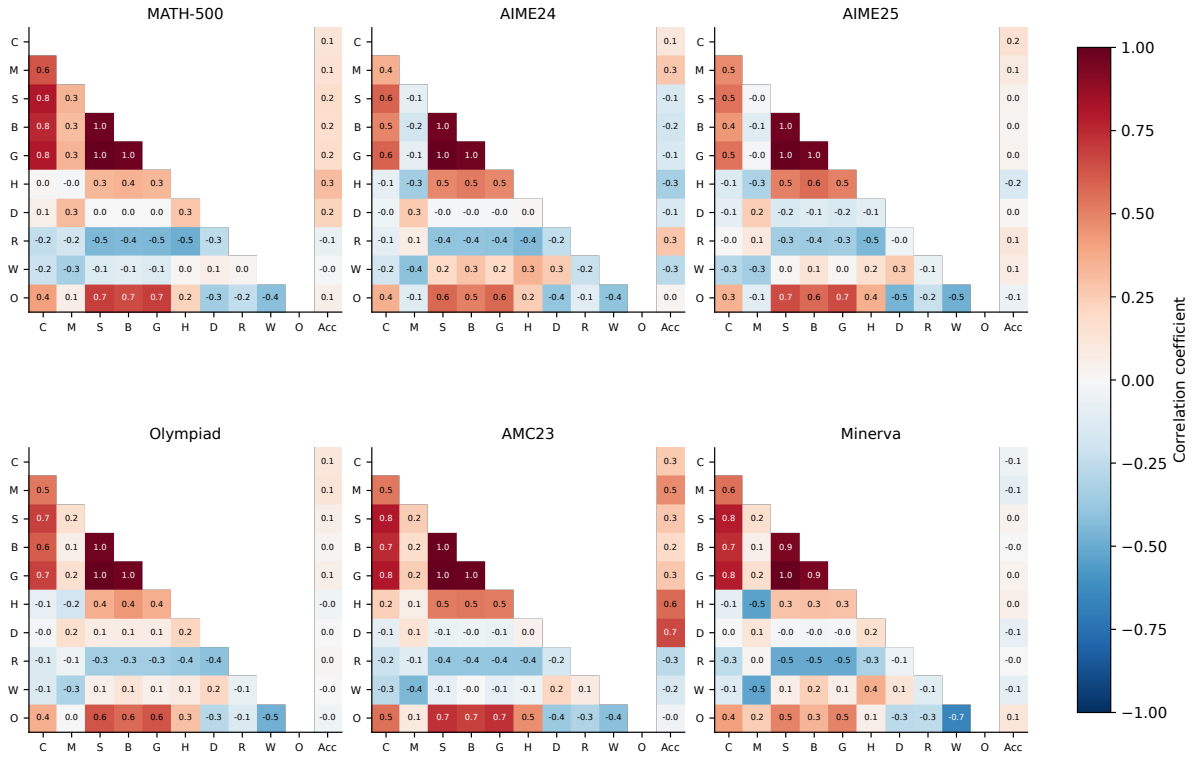


Fig. 33. Position: Bin 5: metric relationships and correctness. Codes are defined in Table 34.

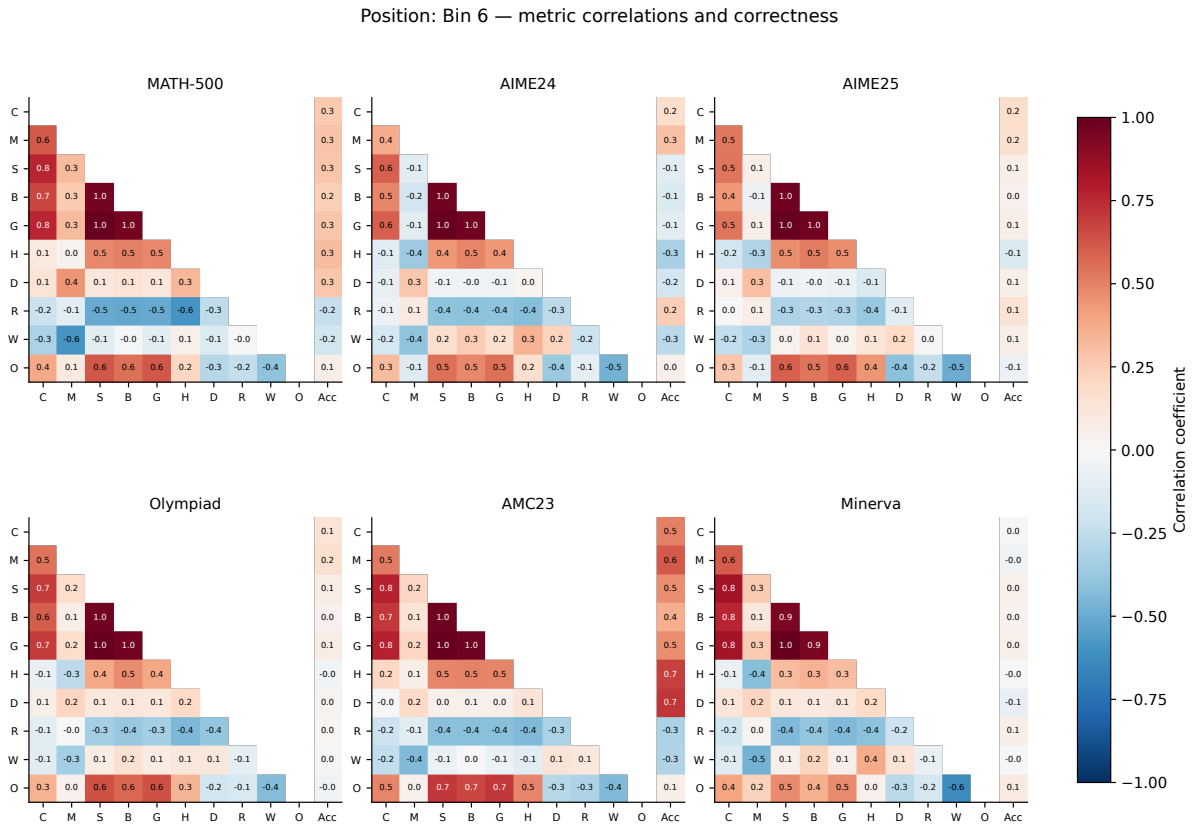


Fig. 34. Position: Bin 6: metric relationships and correctness. Codes are defined in Table 34.

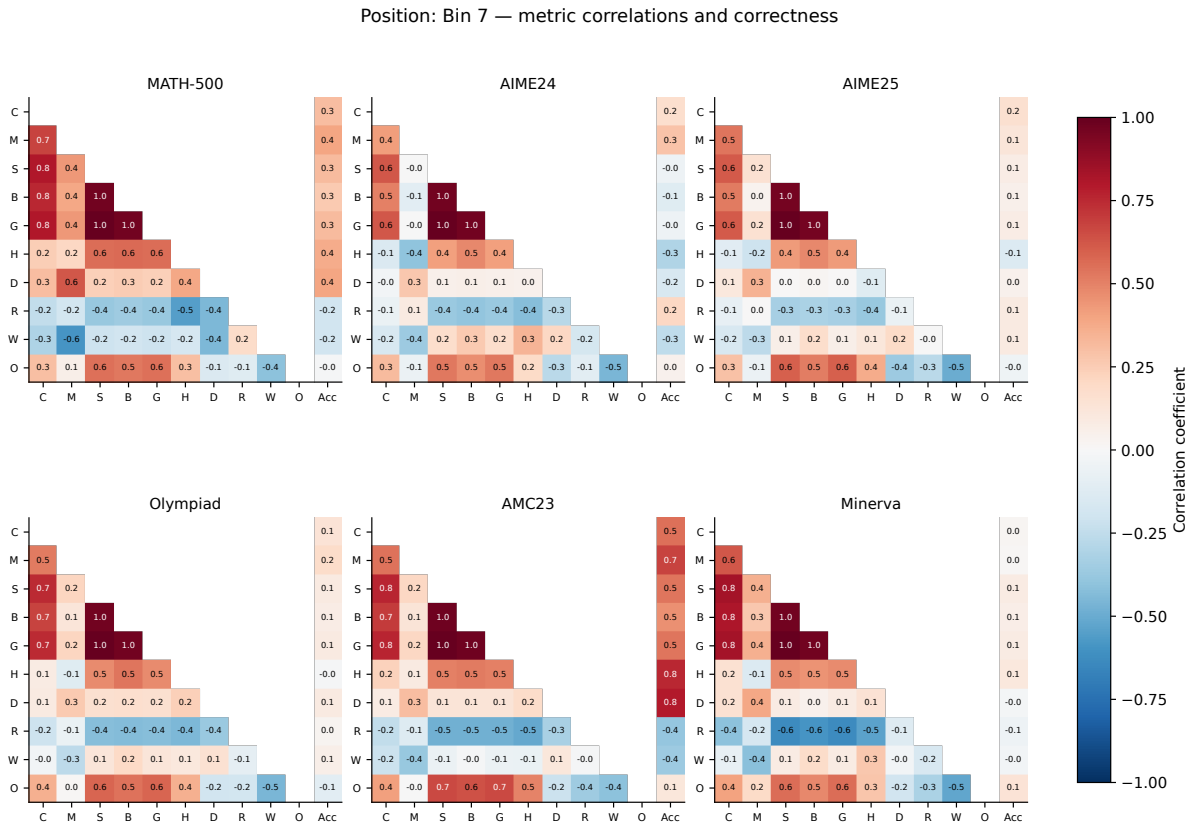


Fig. 35. Position: Bin 7: metric relationships and correctness. Codes are defined in Table 34.

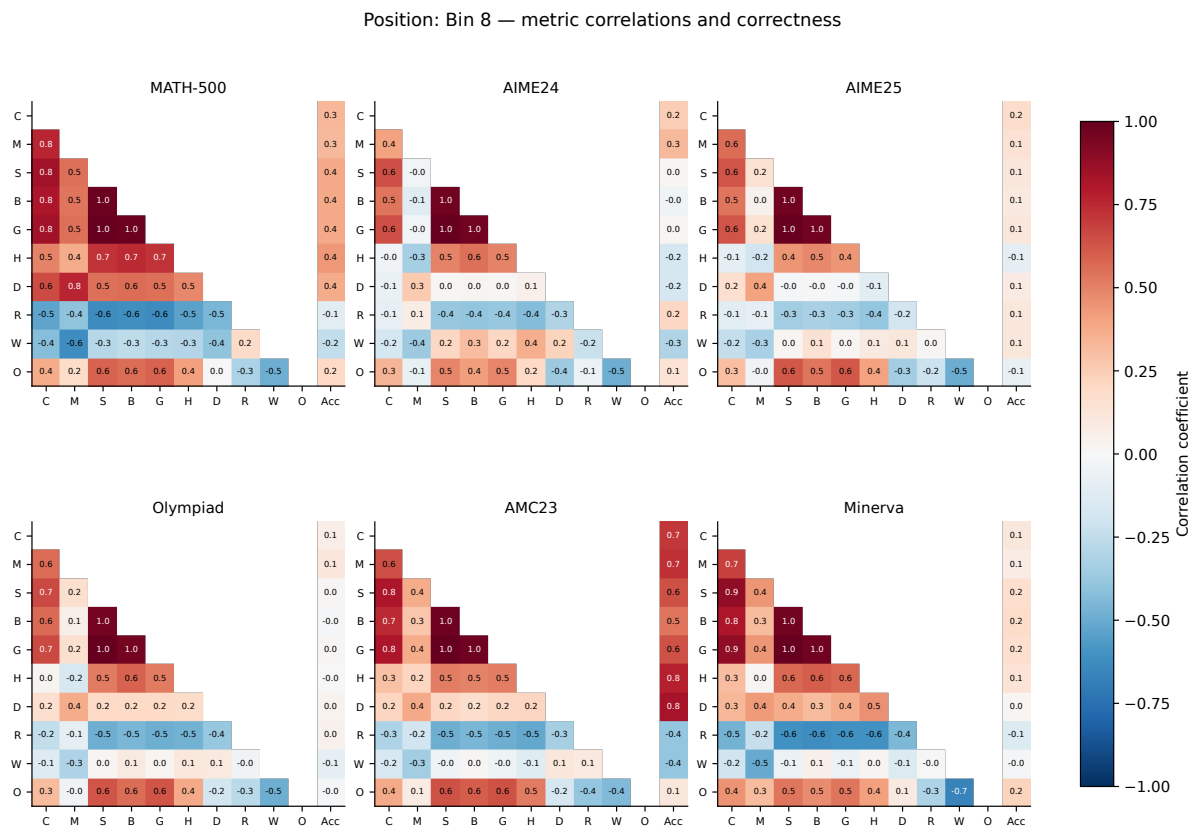


Fig. 36. Position: Bin 8: metric relationships and correctness. Codes are defined in Table 34.

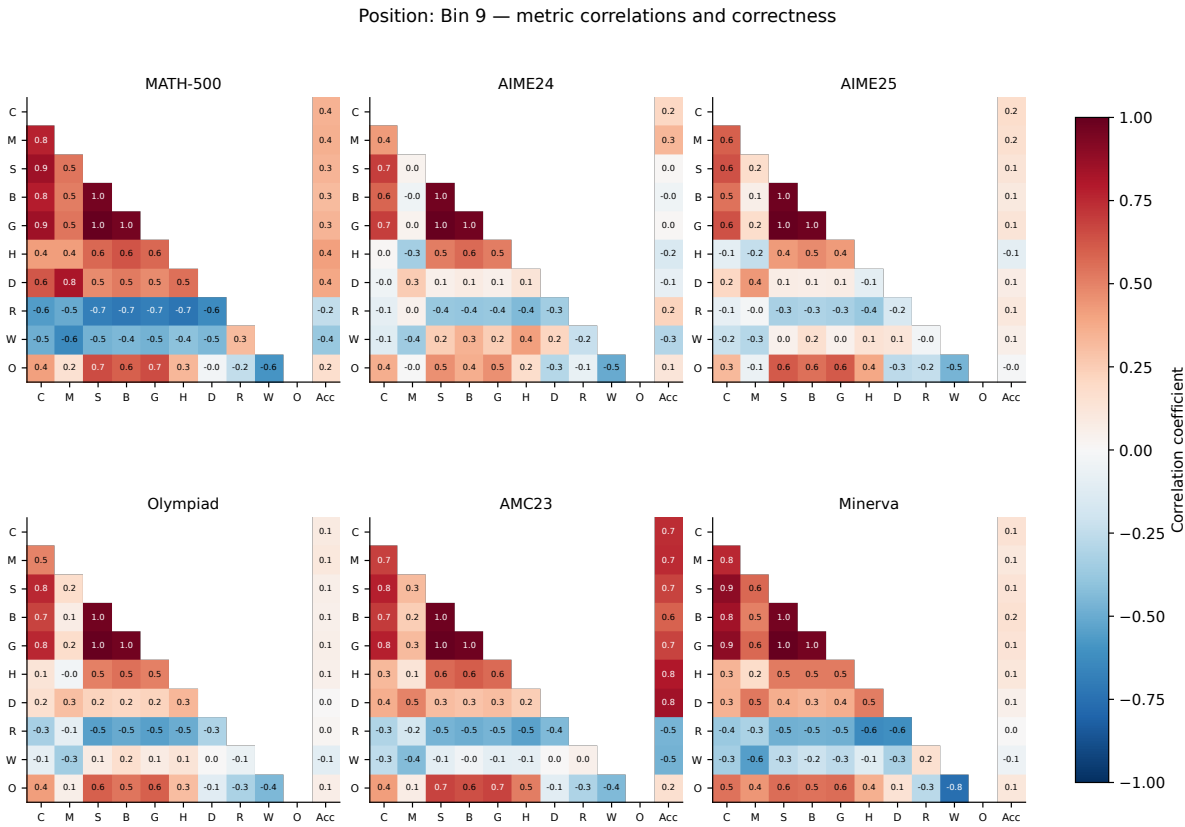


Fig. 37. Position: Bin 9: metric relationships and correctness. Codes are defined in Table 34.

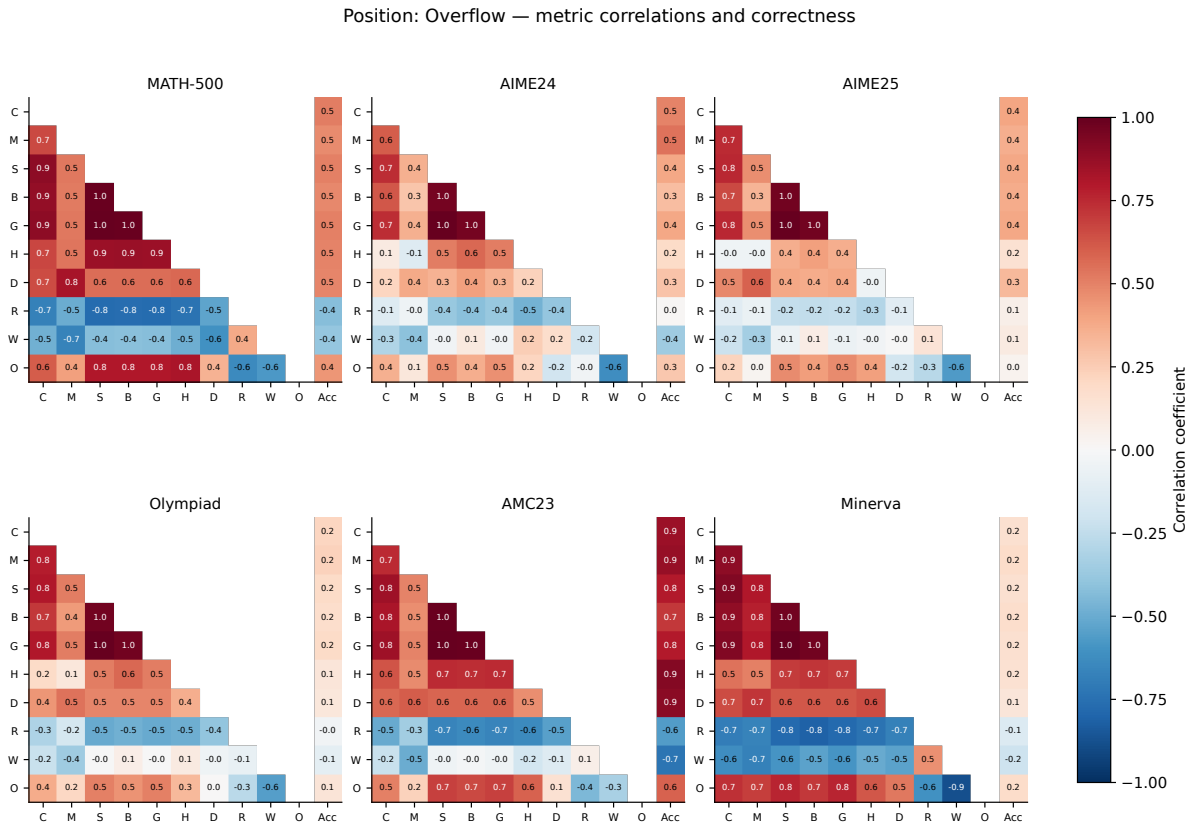


Fig. 38. Position: Overflow: metric relationships and correctness. Codes are defined in Table 34.

C Gemma and GPT-OSS correlation plot atlas

Lower triangles show metric–metric Spearman ρ ; Acc shows point-biserial r with correctness. Missing or undefined values are dashes, not zero. Token-bearing counts are upper bounds on pairwise finite counts. Each model has whole-continuation, full-reasoning, seven class, and eleven position scopes. Qwen plots appear in the preceding atlas. These are descriptive screens without significance selection.

Table 35. Metric key for all models.

Code	Metric
T	Token count
C	Router confidence
M	Leading-expert margin
S	Selected probability mass
B	Selection-boundary margin
G	Selected/rest mean gap
H	Hidden-state norm
D	Hidden-state step distance
R	Hidden/router geometry
W	Top-1 switch rate
O	Adjacent selected-set overlap

Gemma-4-26B-A4B: Whole continuation

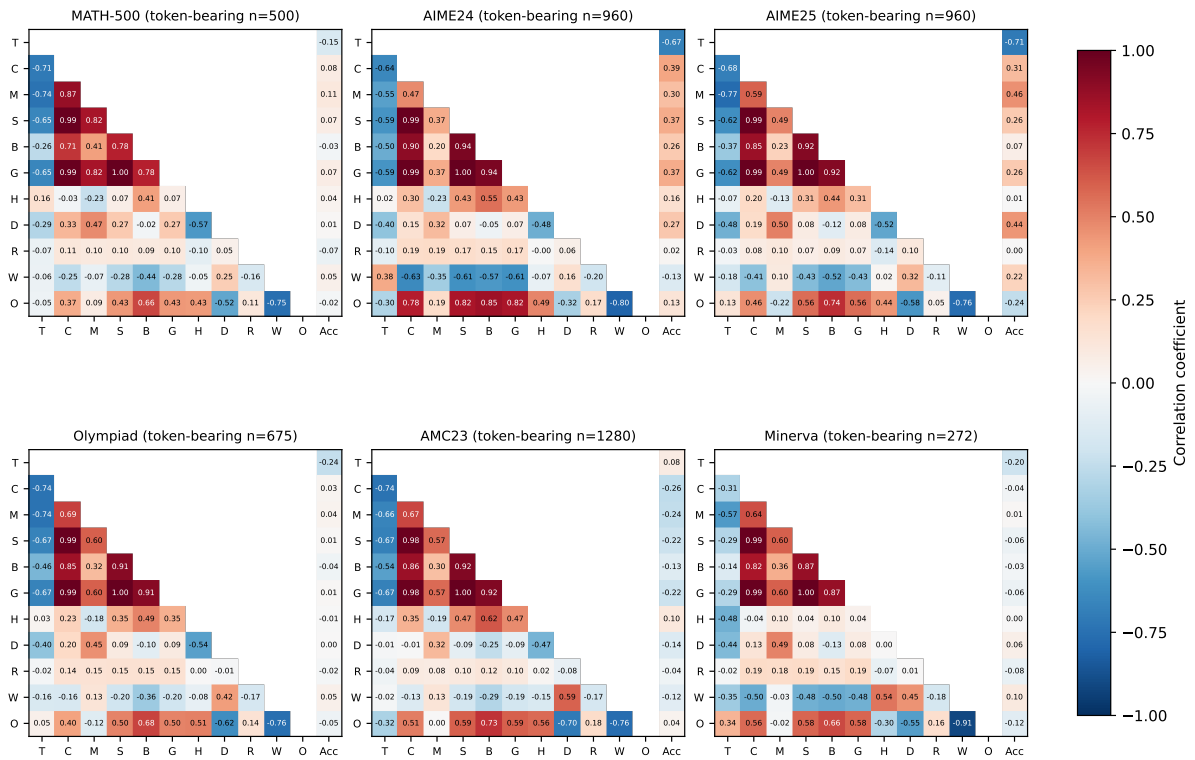


Fig. 39. Gemma-4-26B-A4B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Full reasoning

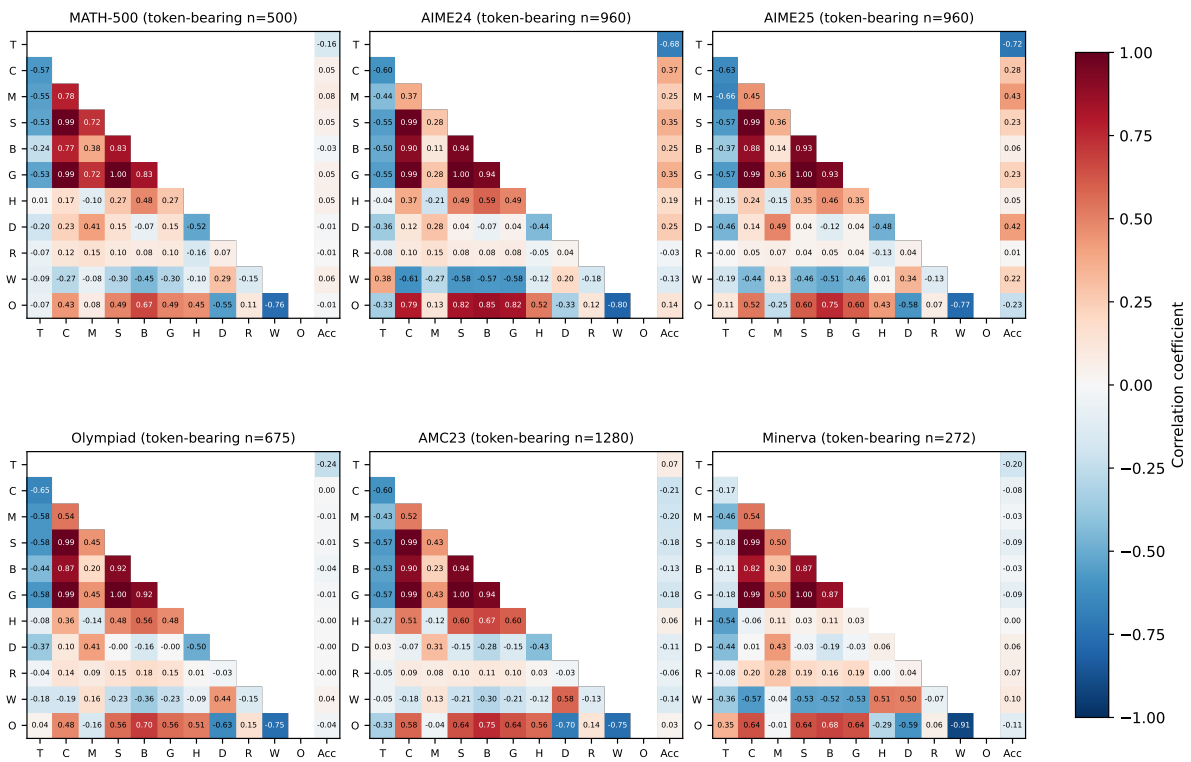


Fig. 40. Gemma-4-26B-A4B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Read

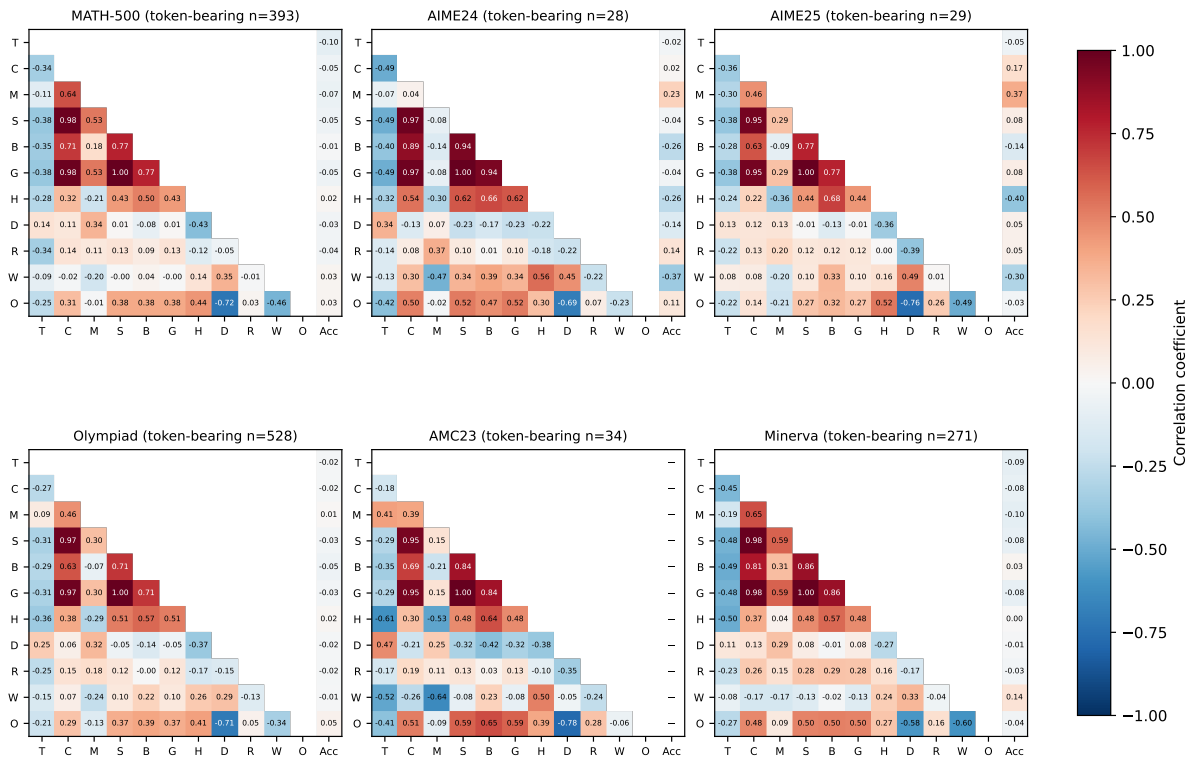


Fig. 41. Gemma-4-26B-A4B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Analyze

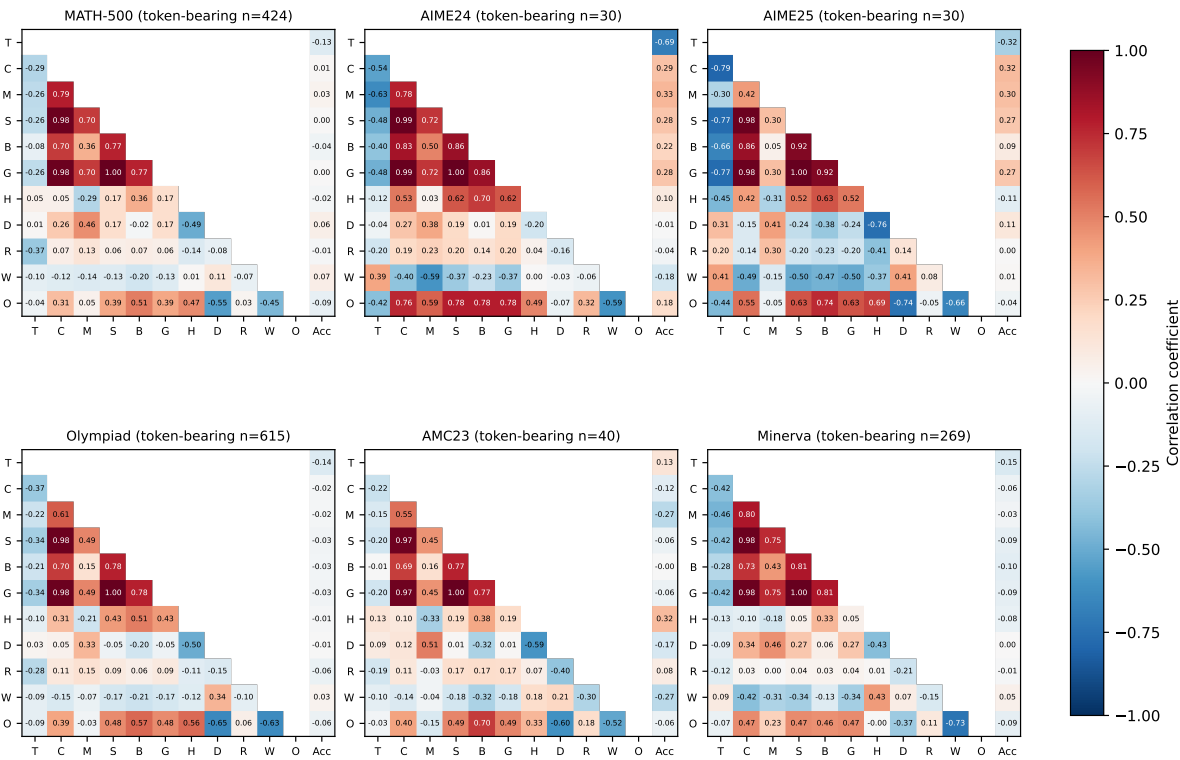


Fig. 42. Gemma-4-26B-A4B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Plan

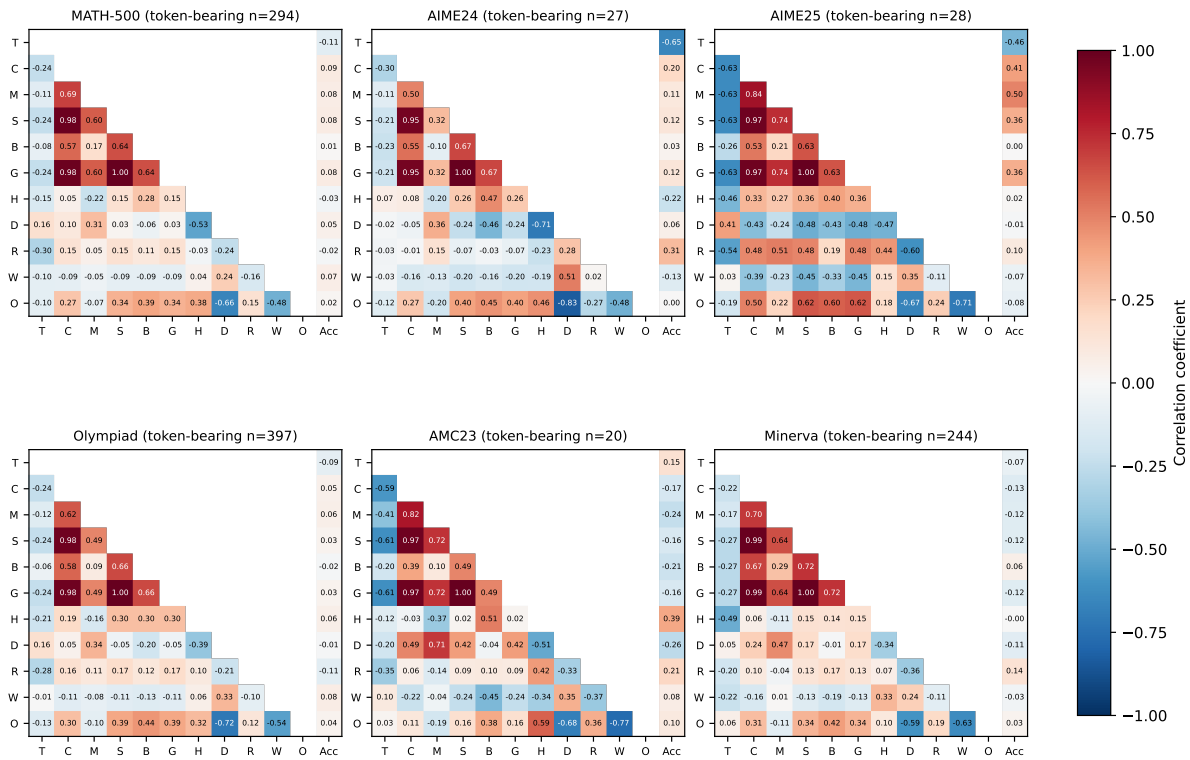


Fig. 43. Gemma-4-26B-A4B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Implement

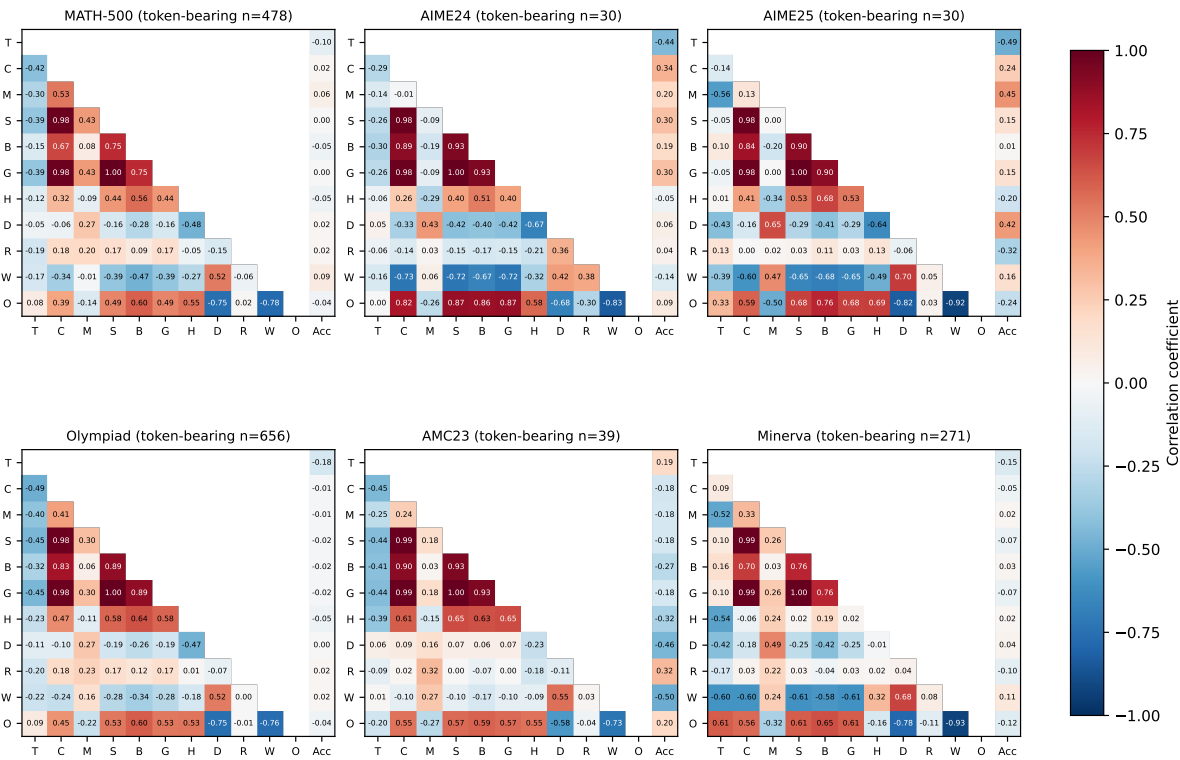


Fig. 44. Gemma-4-26B-A4B / Class: Implement. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Explore

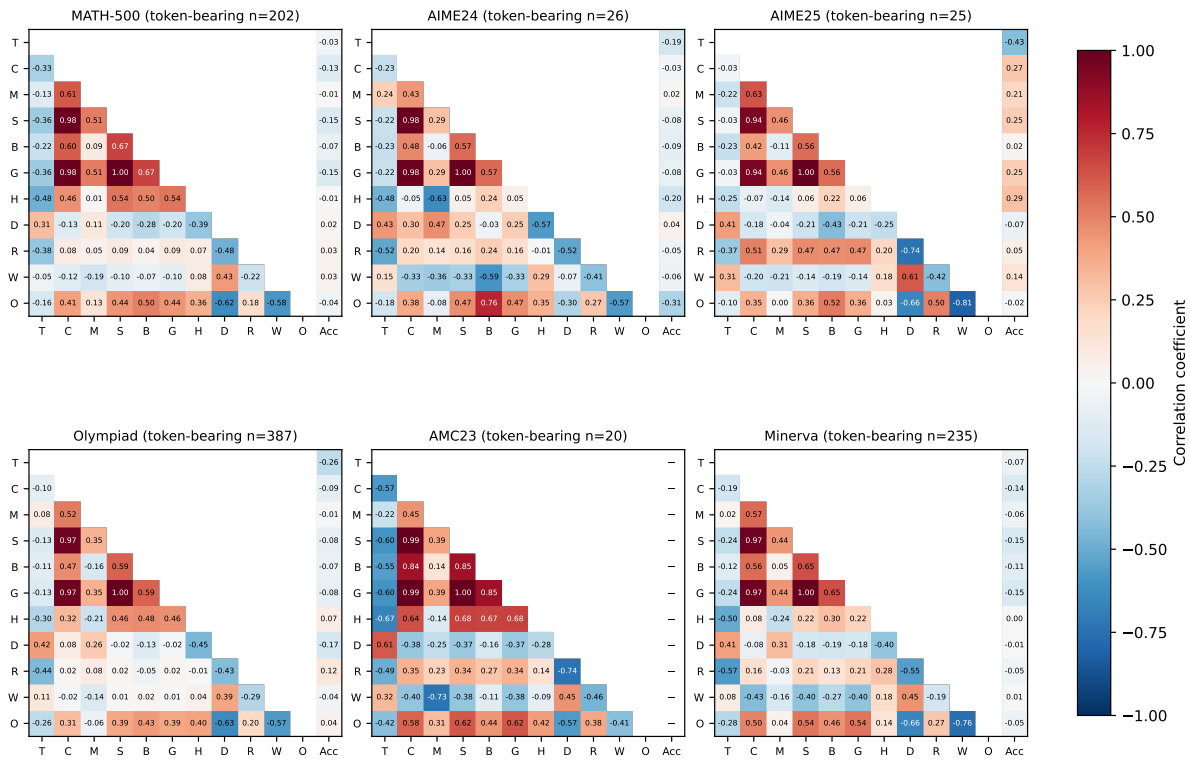


Fig. 45. Gemma-4-26B-A4B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Verify

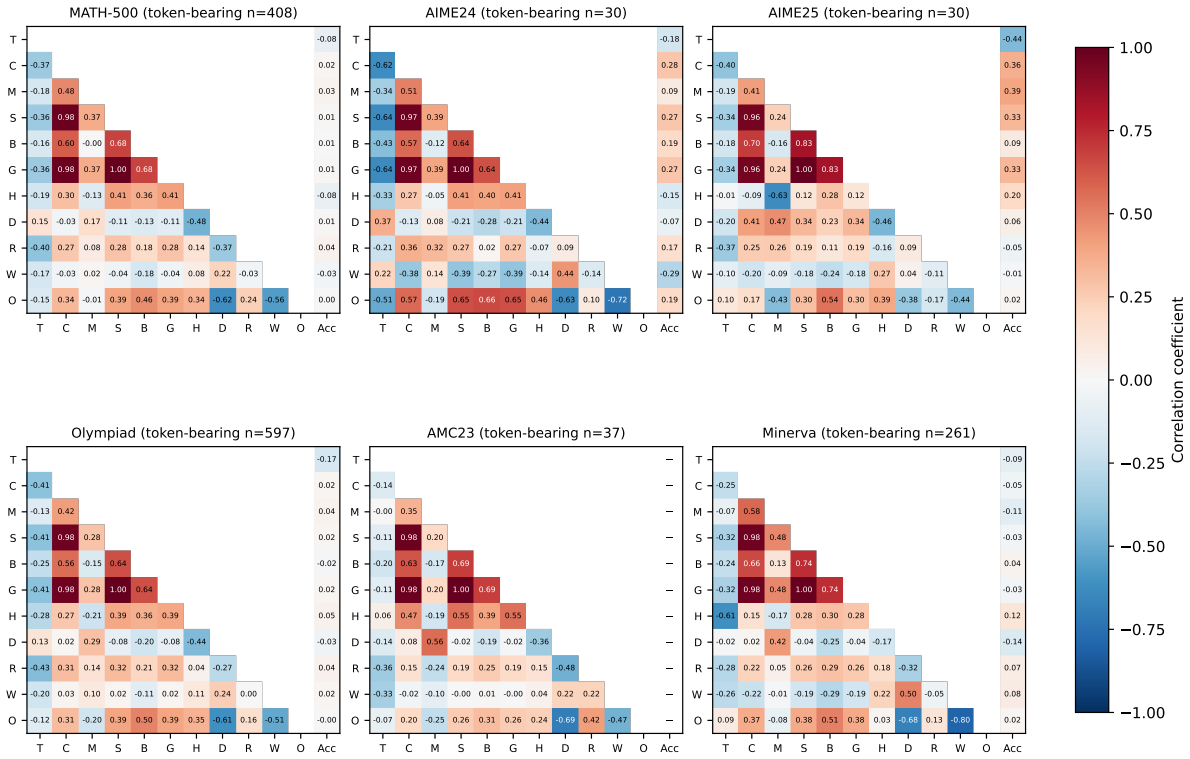


Fig. 46. Gemma-4-26B-A4B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Class: Monitor

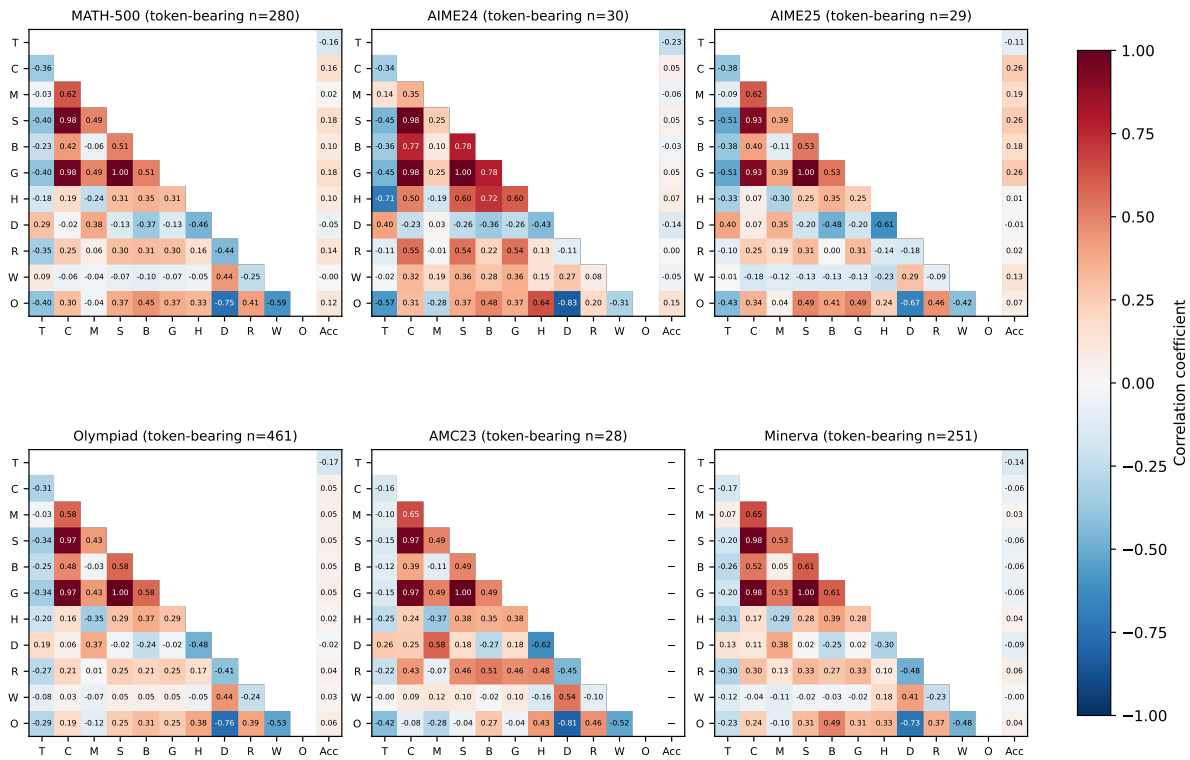


Fig. 47. Gemma-4-26B-A4B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 00

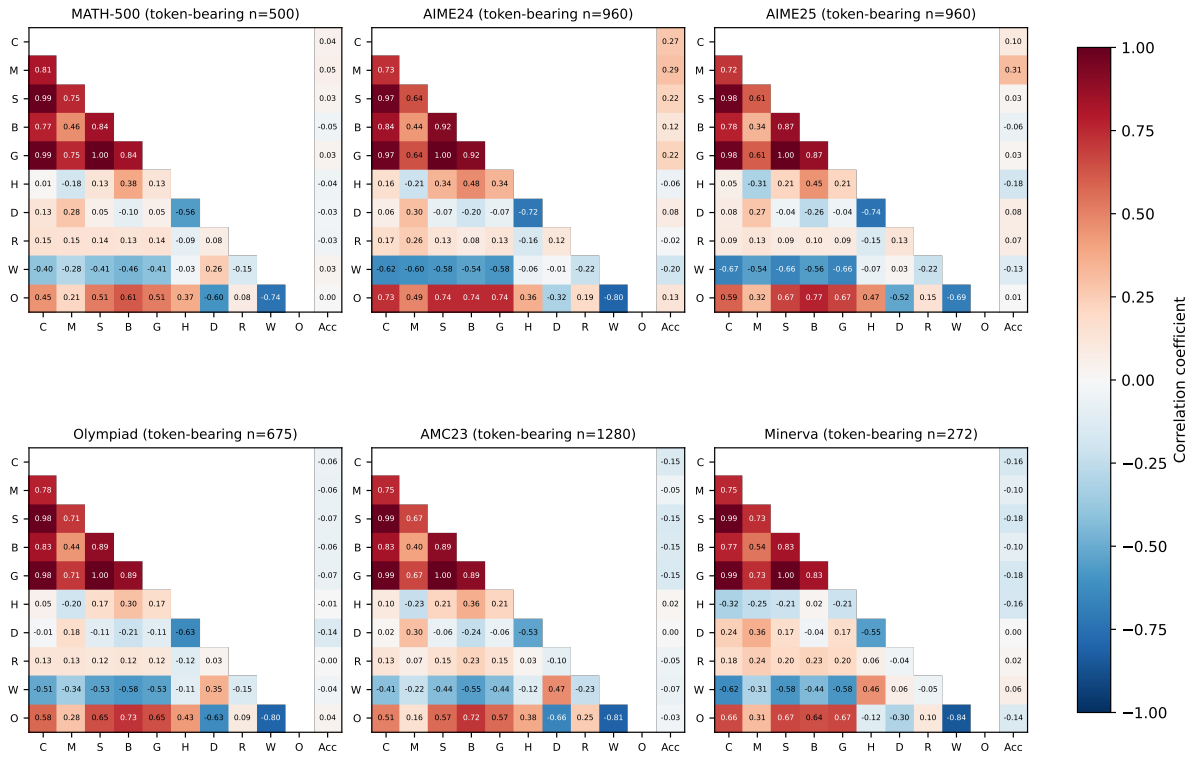


Fig. 48. Gemma-4-26B-A4B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 01

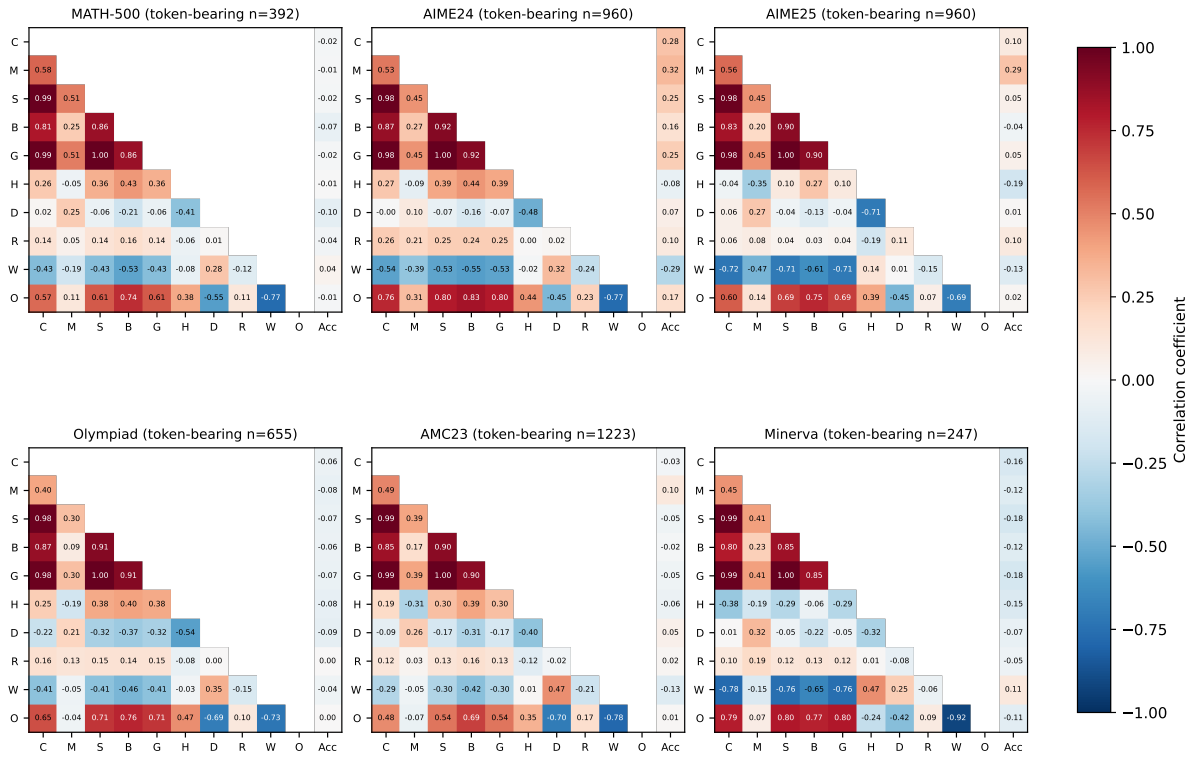


Fig. 49. Gemma-4-26B-A4B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 02

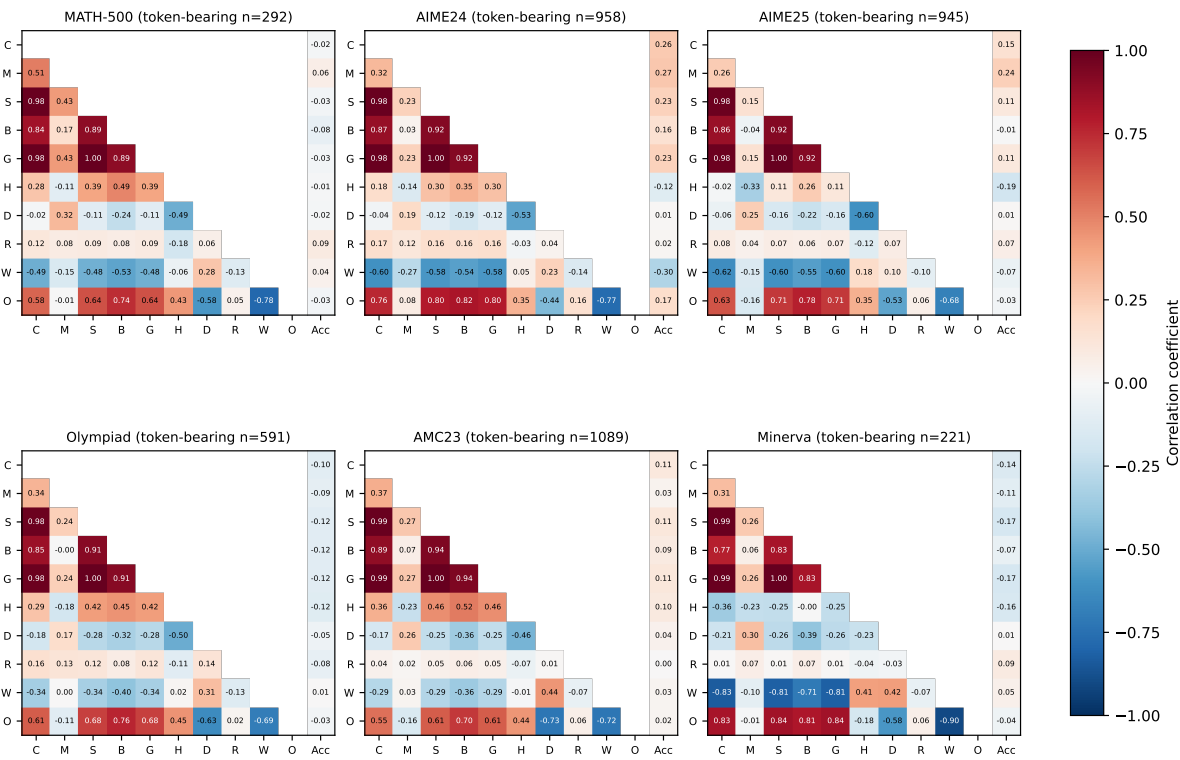


Fig. 50. Gemma-4-26B-A4B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 03

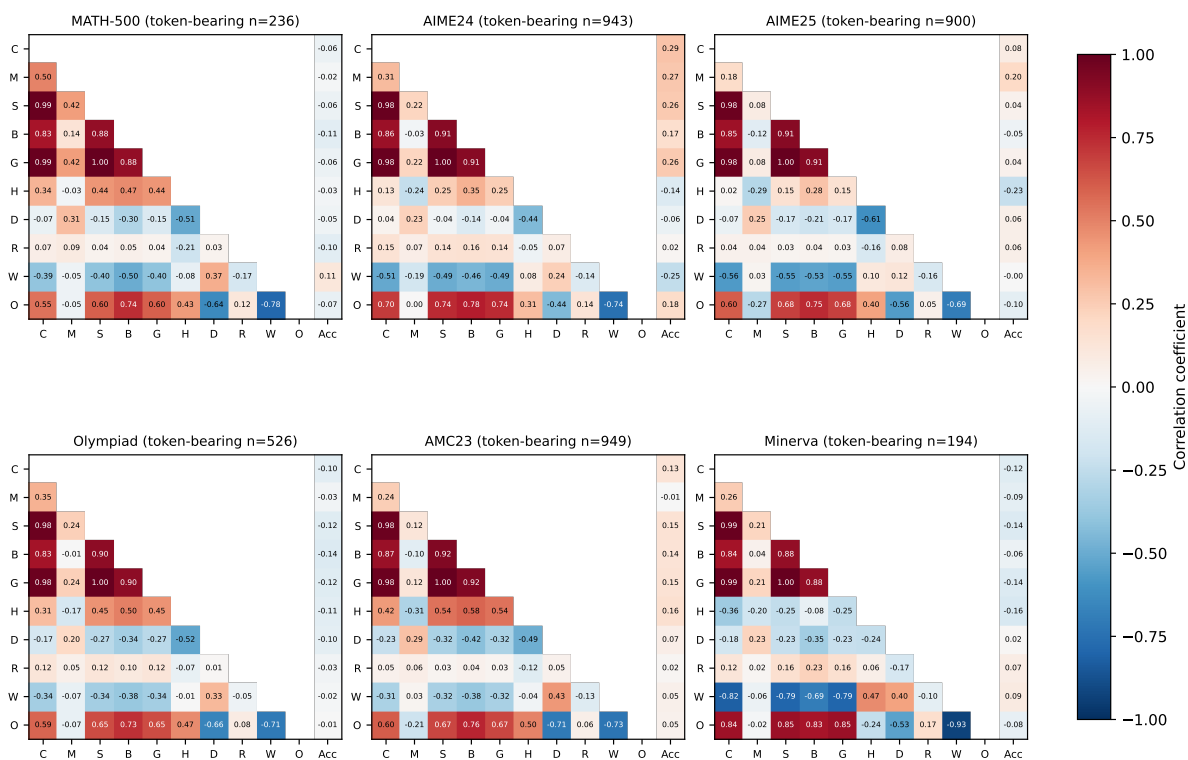


Fig. 51. Gemma-4-26B-A4B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 04

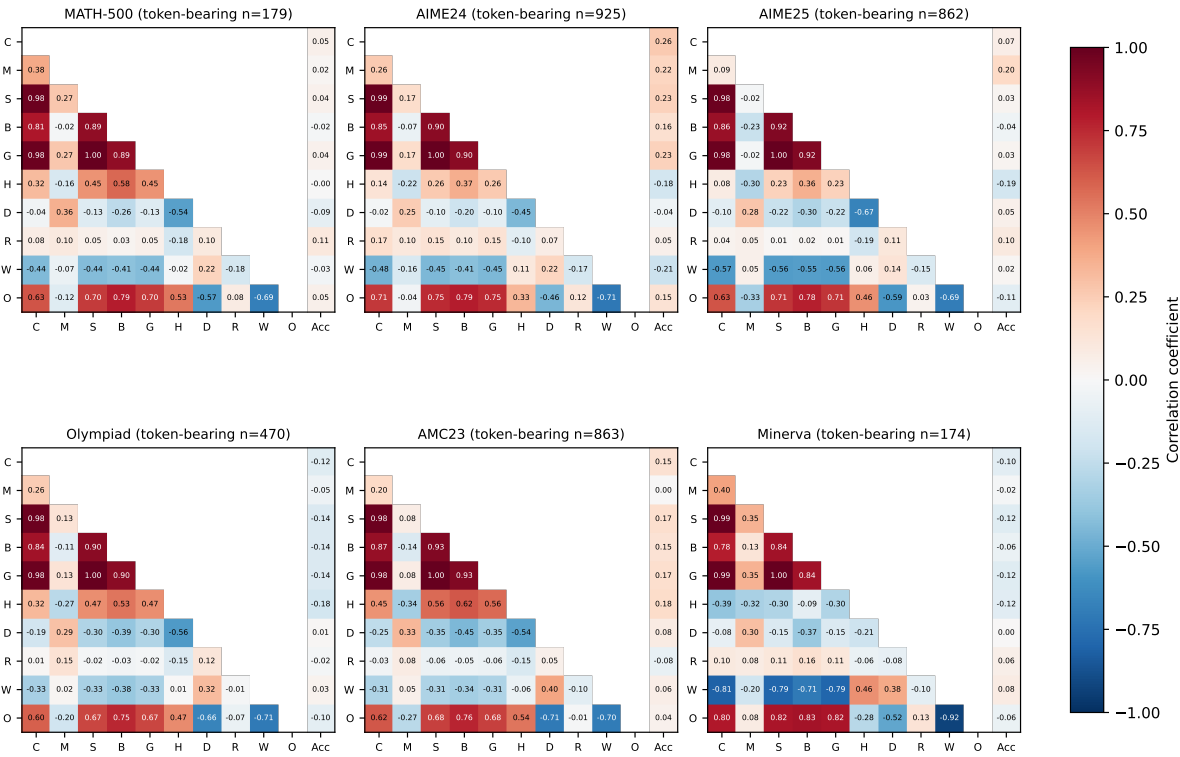


Fig. 52. Gemma-4-26B-A4B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 05

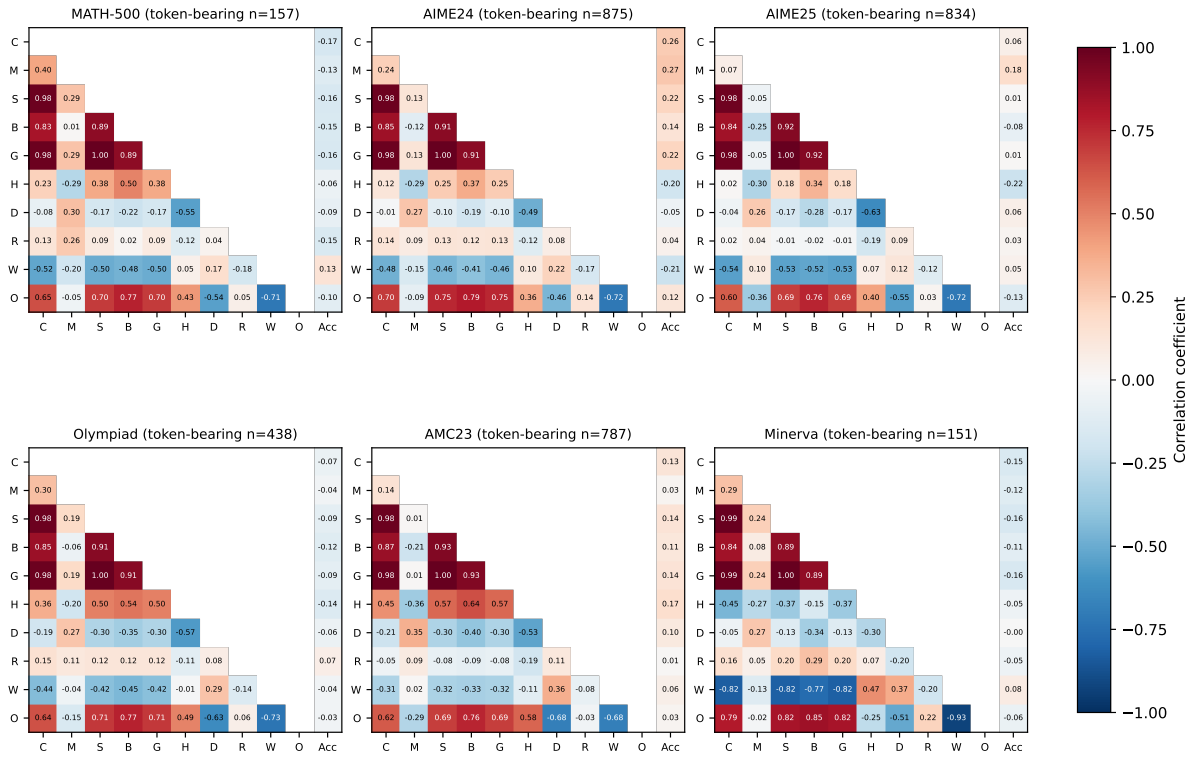


Fig. 53. Gemma-4-26B-A4B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 06

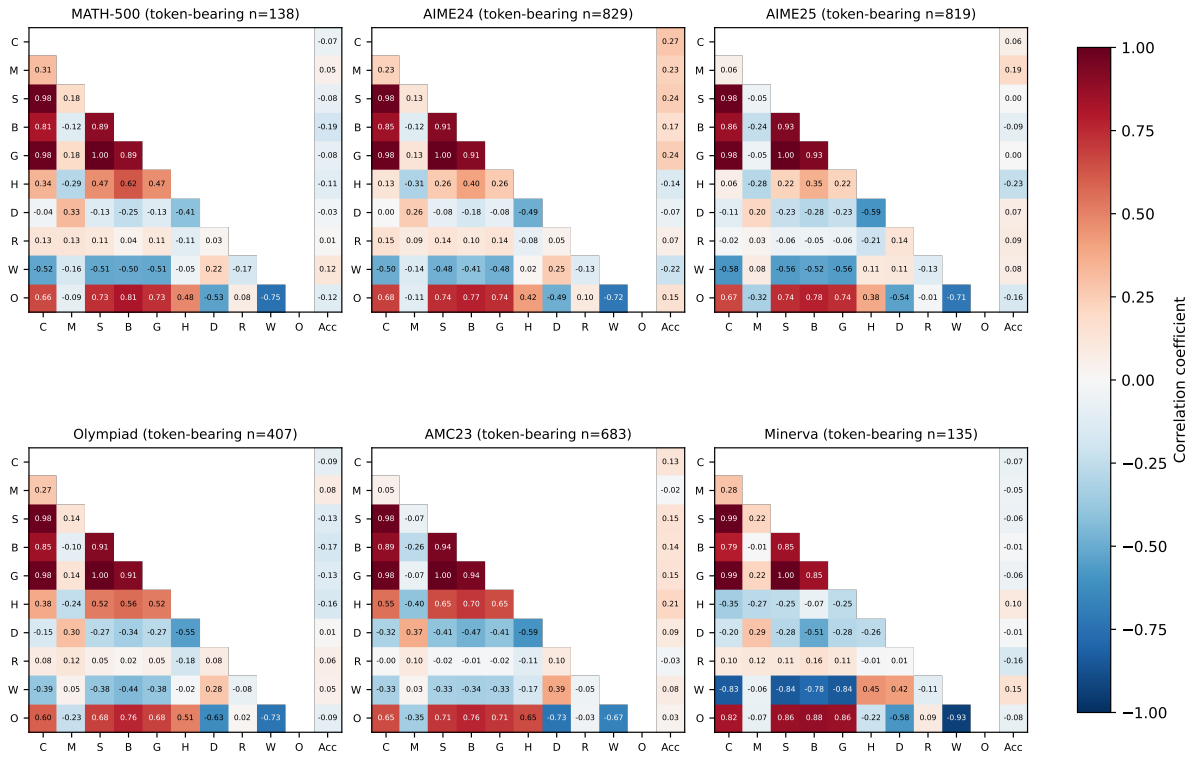


Fig. 54. Gemma-4-26B-A4B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 07

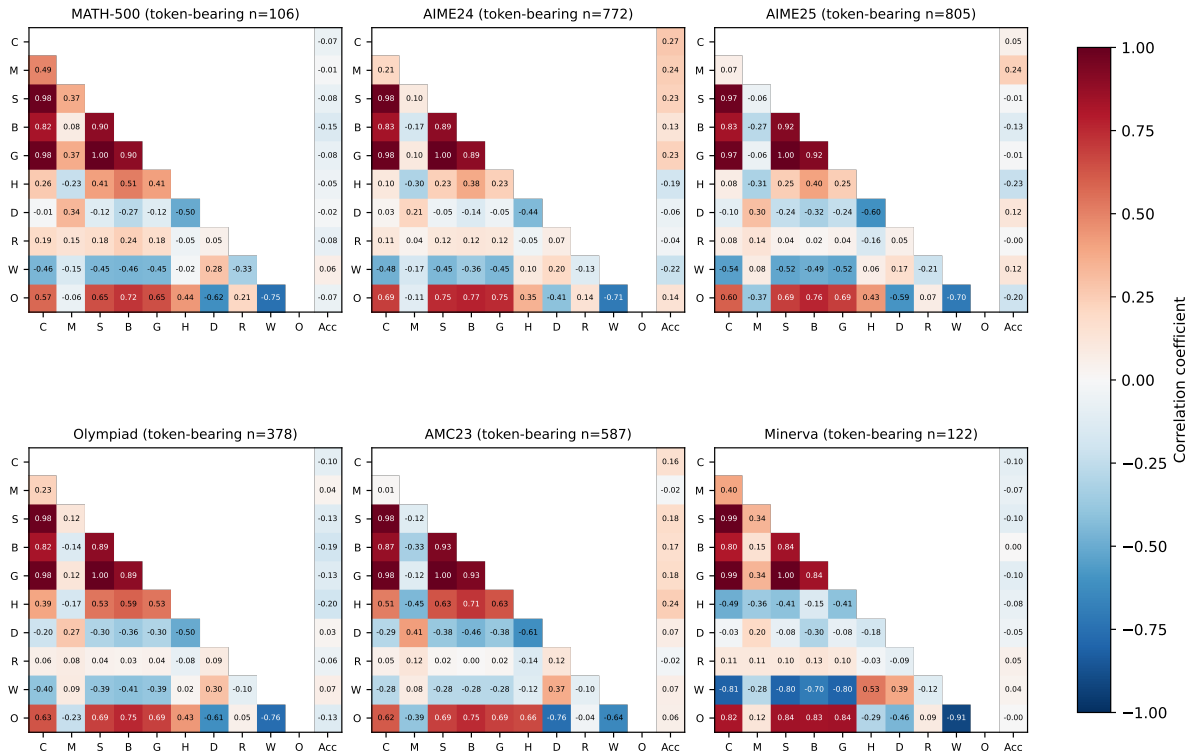


Fig. 55. Gemma-4-26B-A4B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 08

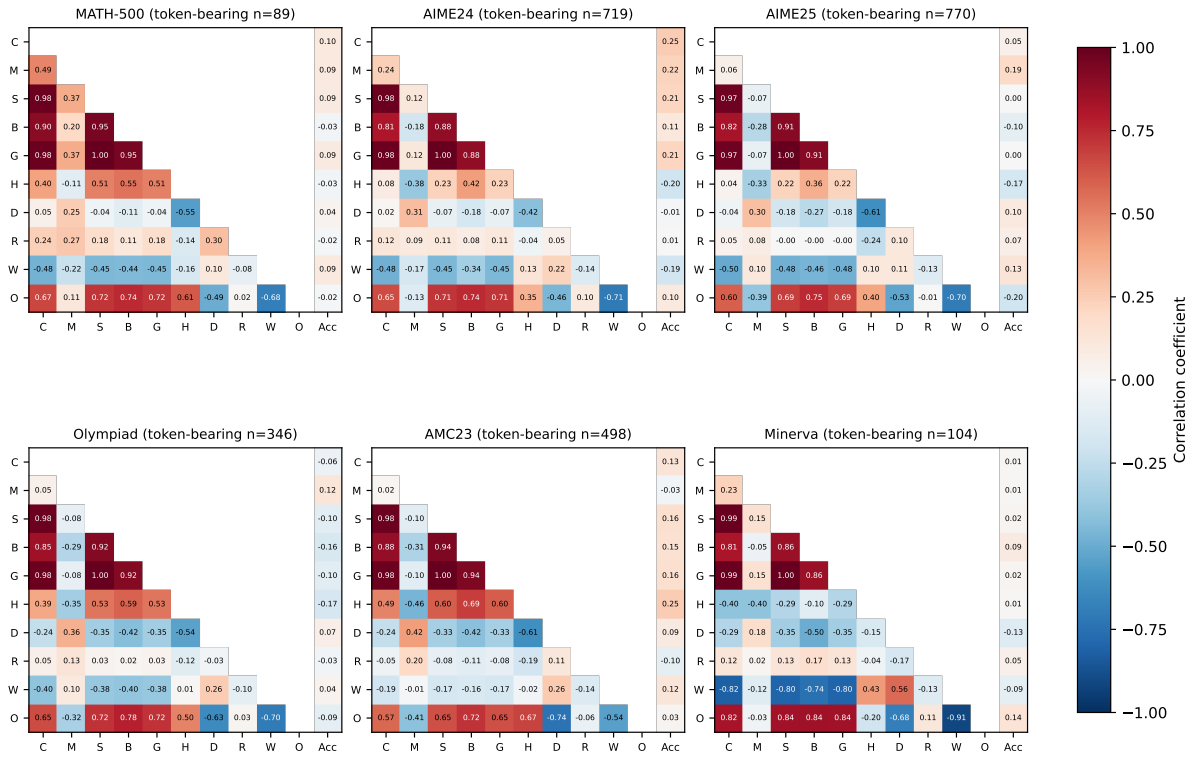


Fig. 56. Gemma-4-26B-A4B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: bin 09

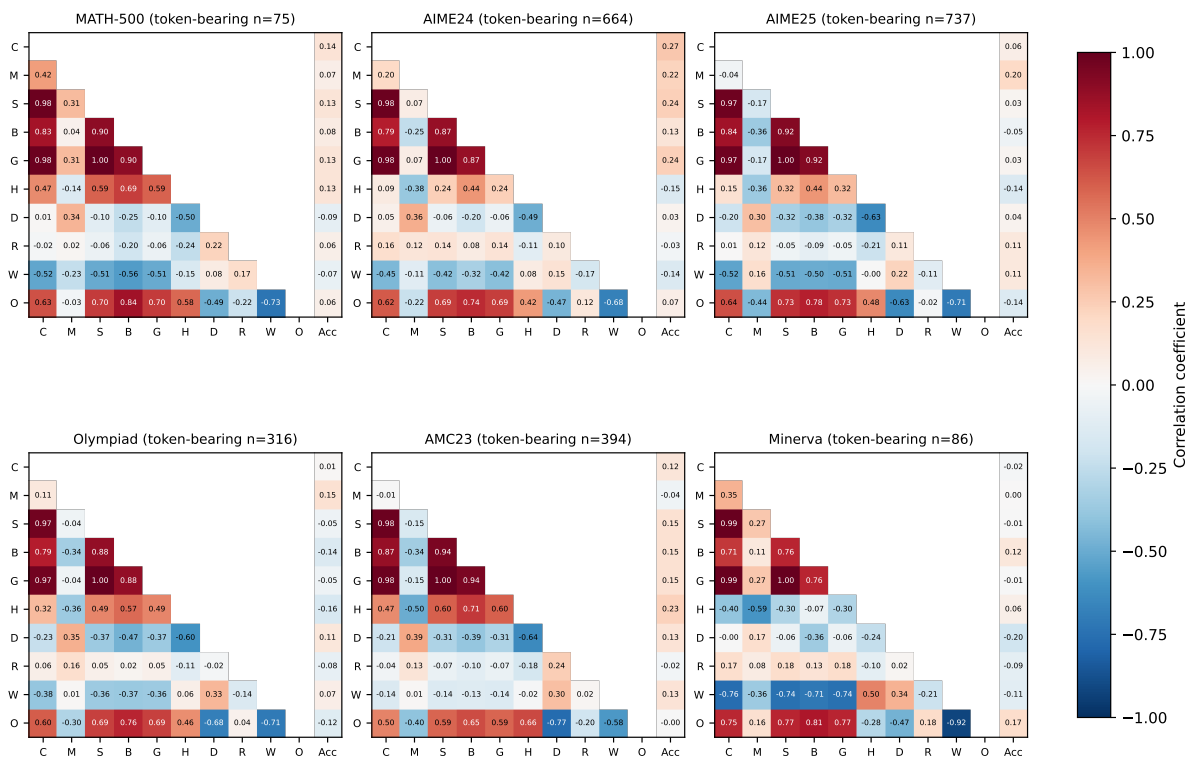


Fig. 57. Gemma-4-26B-A4B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

Gemma-4-26B-A4B: Position: overflow

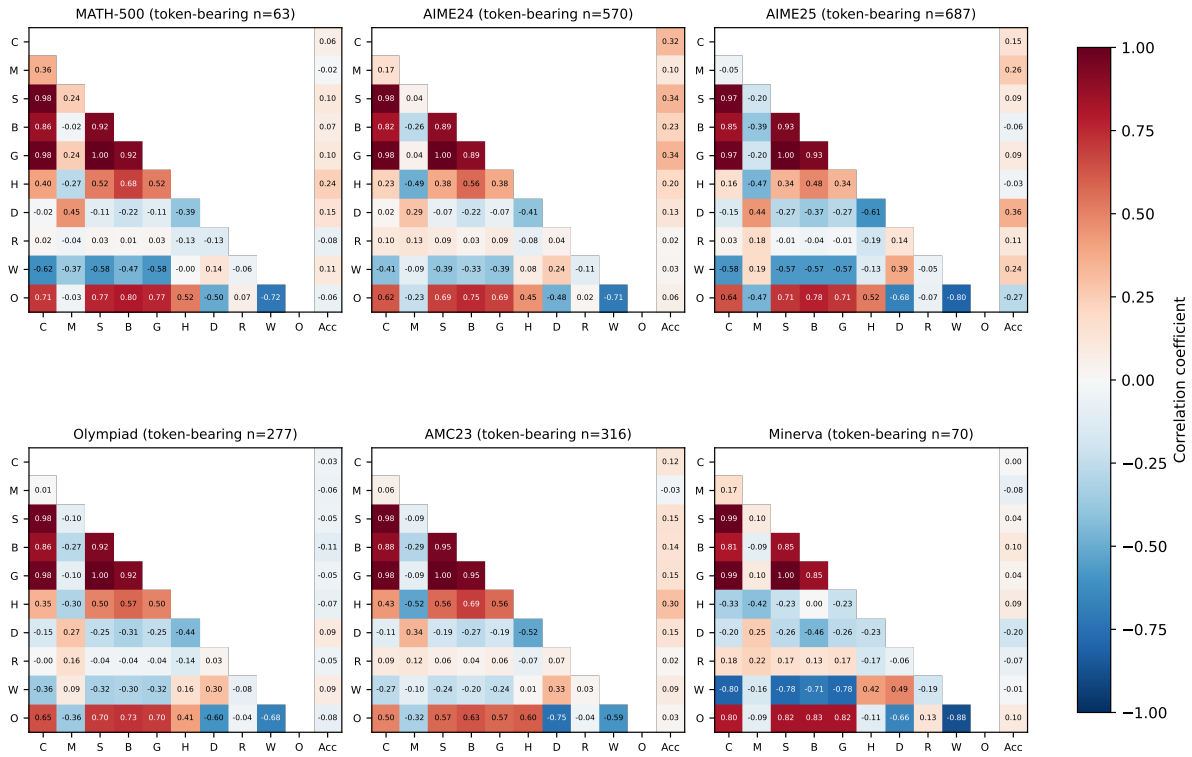


Fig. 58. Gemma-4-26B-A4B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Whole continuation

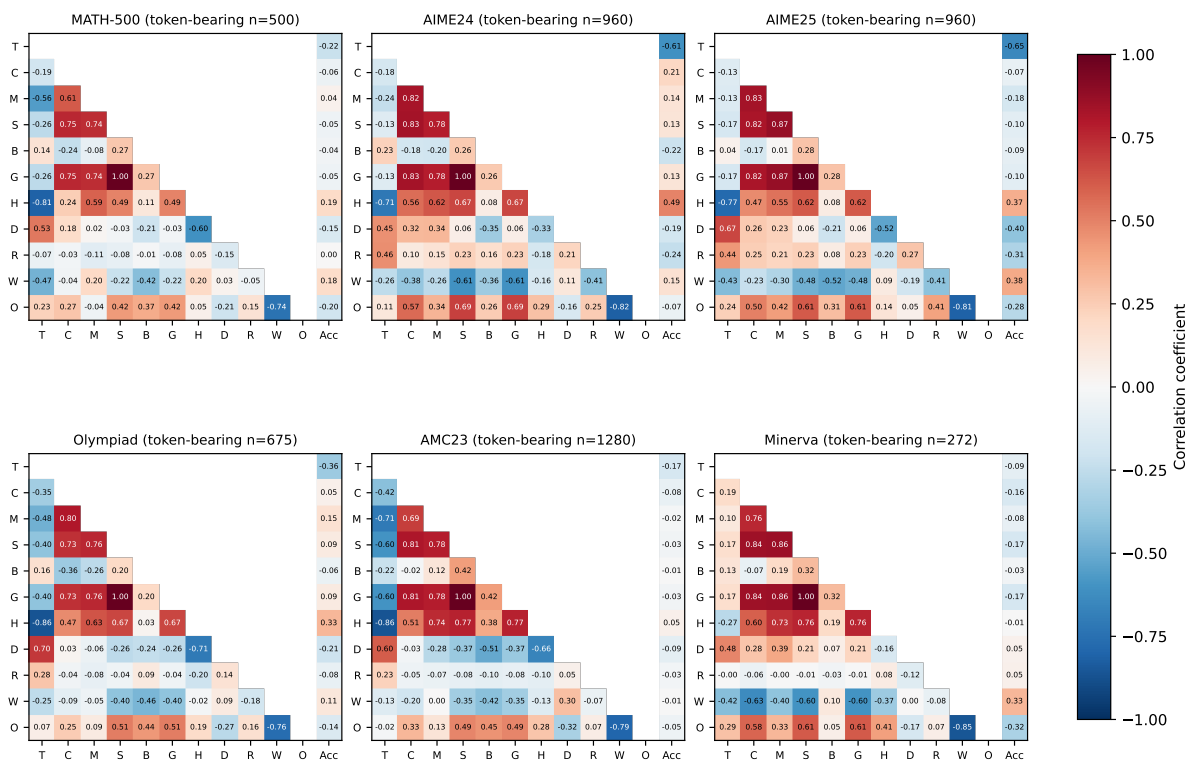


Fig. 59. GPT-OSS-20B / Whole continuation. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Full reasoning

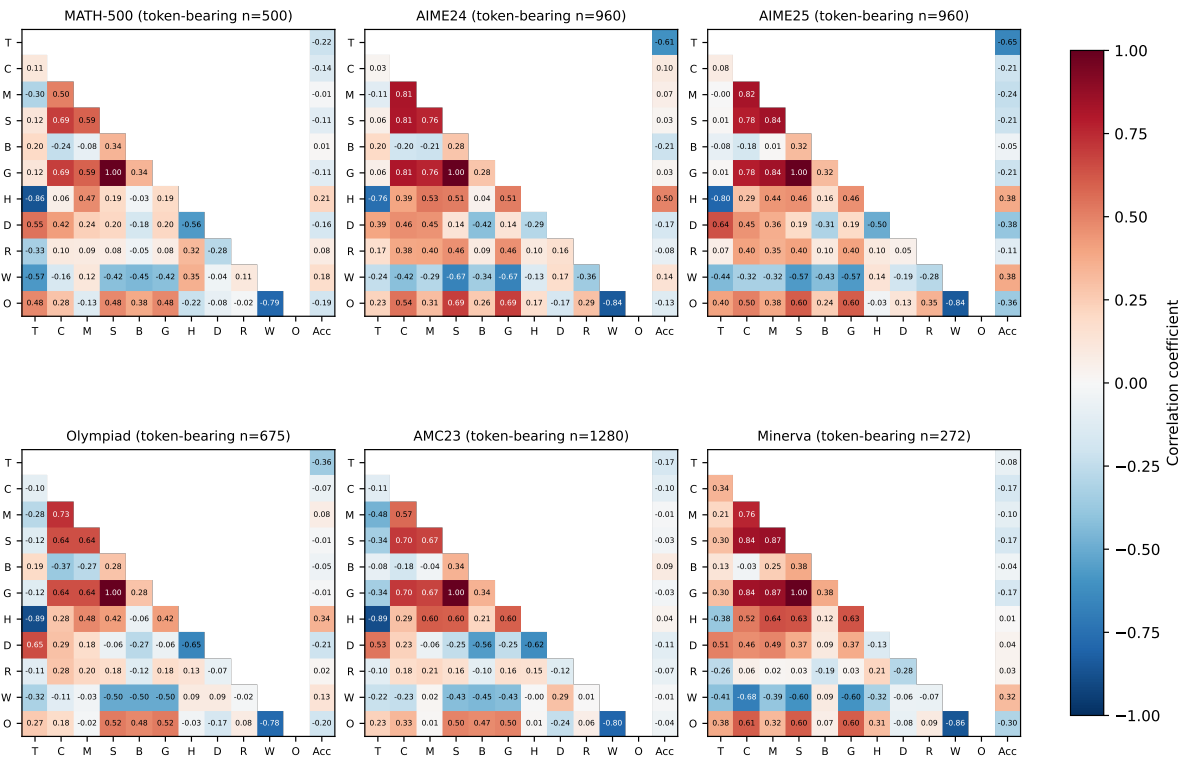


Fig. 60. GPT-OSS-20B / Full reasoning. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Read

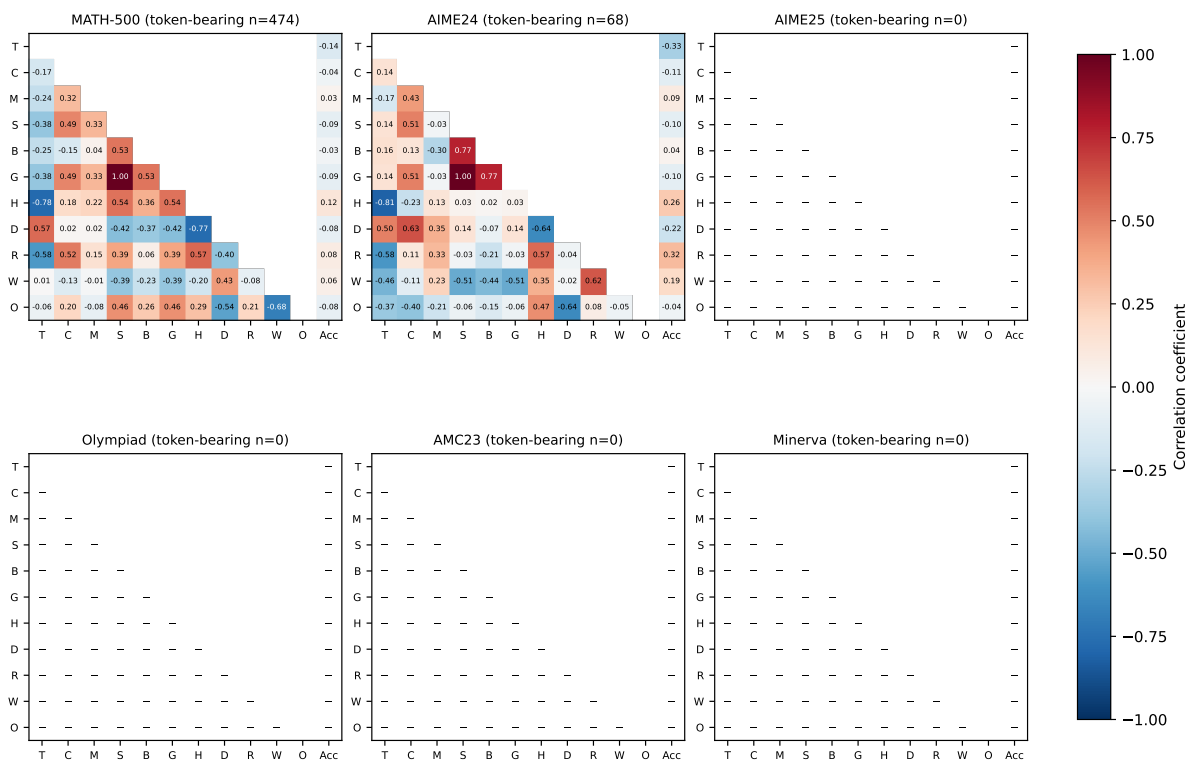


Fig. 61. GPT-OSS-20B / Class: Read. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Analyze

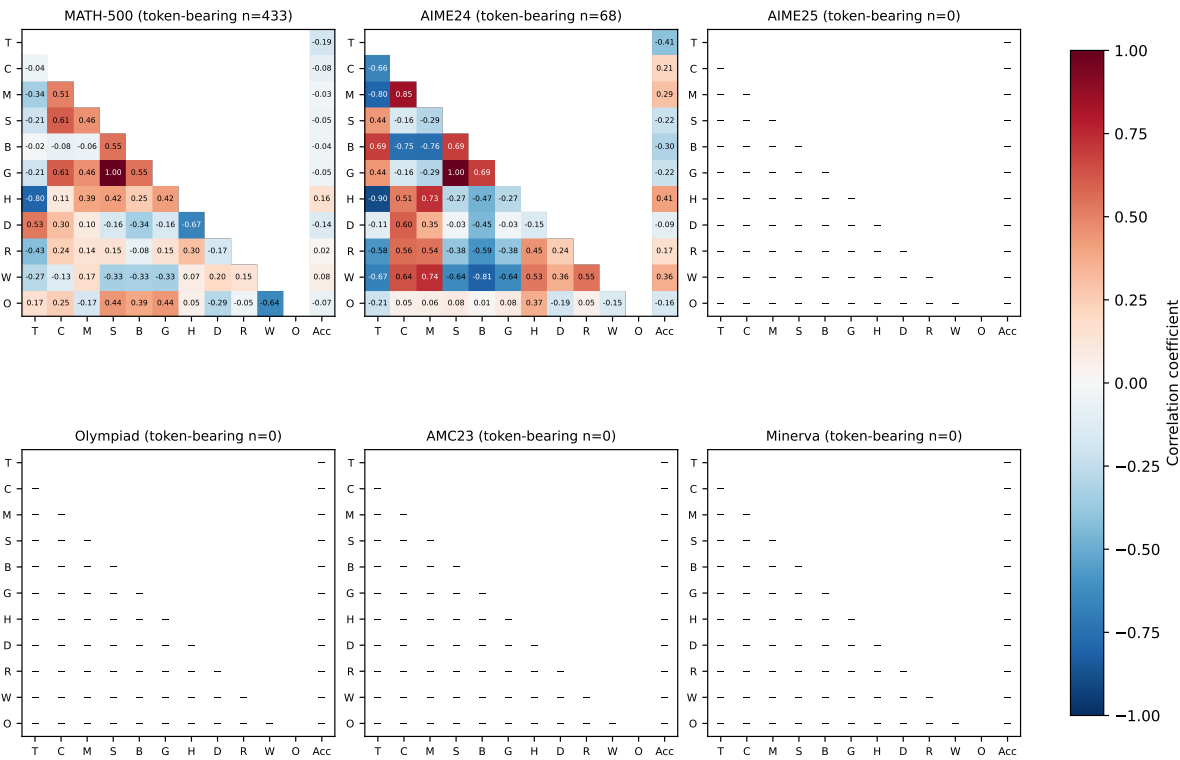


Fig. 62. GPT-OSS-20B / Class: Analyze. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Plan

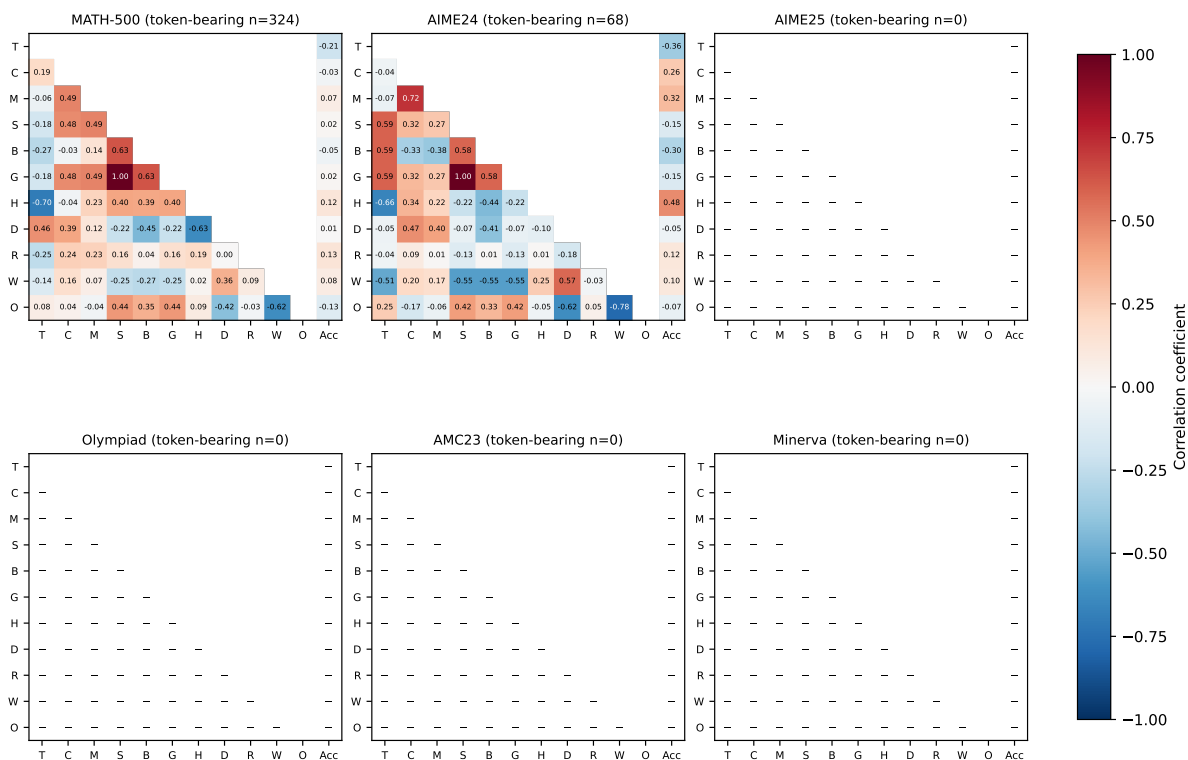


Fig. 63. GPT-OSS-20B / Class: Plan. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Explore

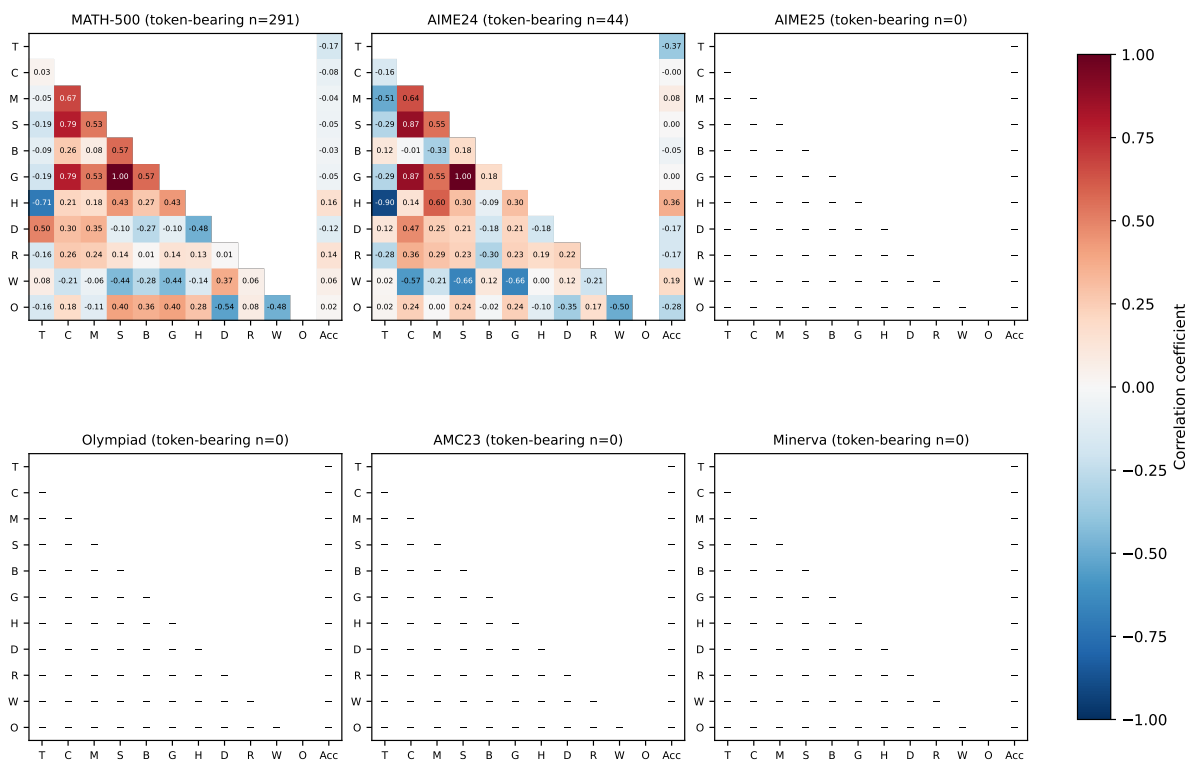


Fig. 65. GPT-OSS-20B / Class: Explore. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Verify

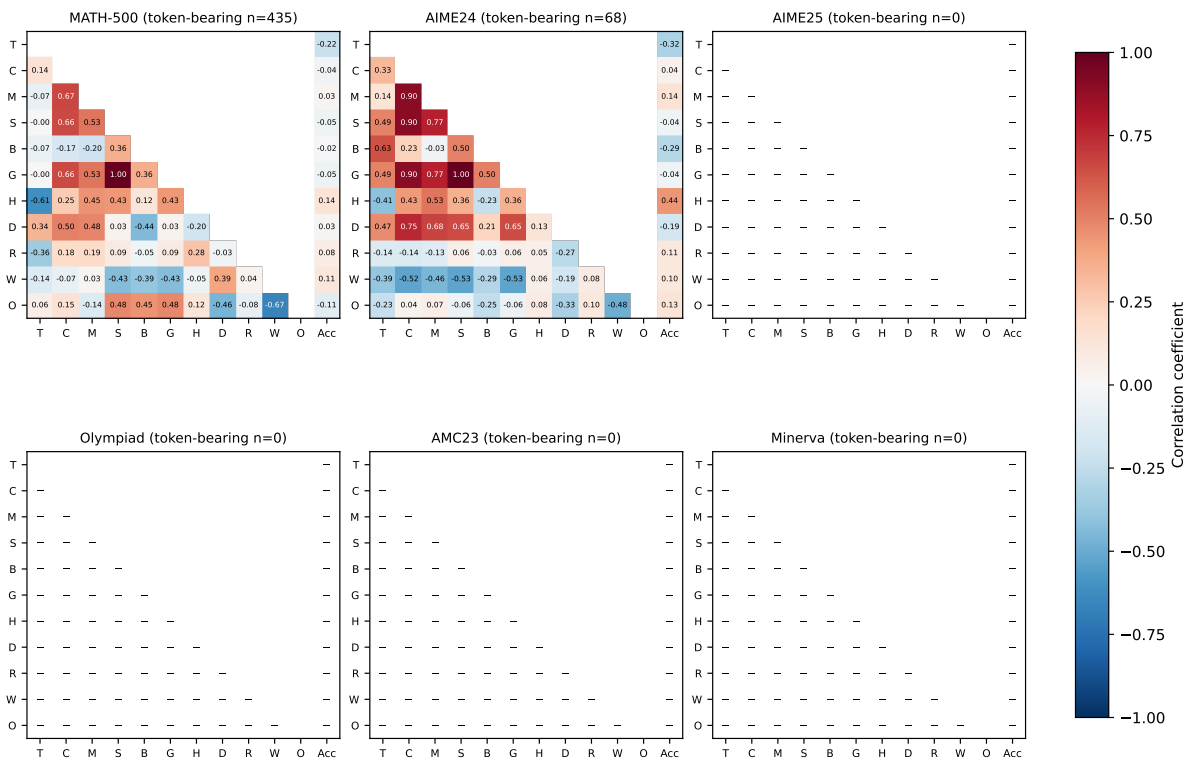


Fig. 66. GPT-OSS-20B / Class: Verify. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Class: Monitor

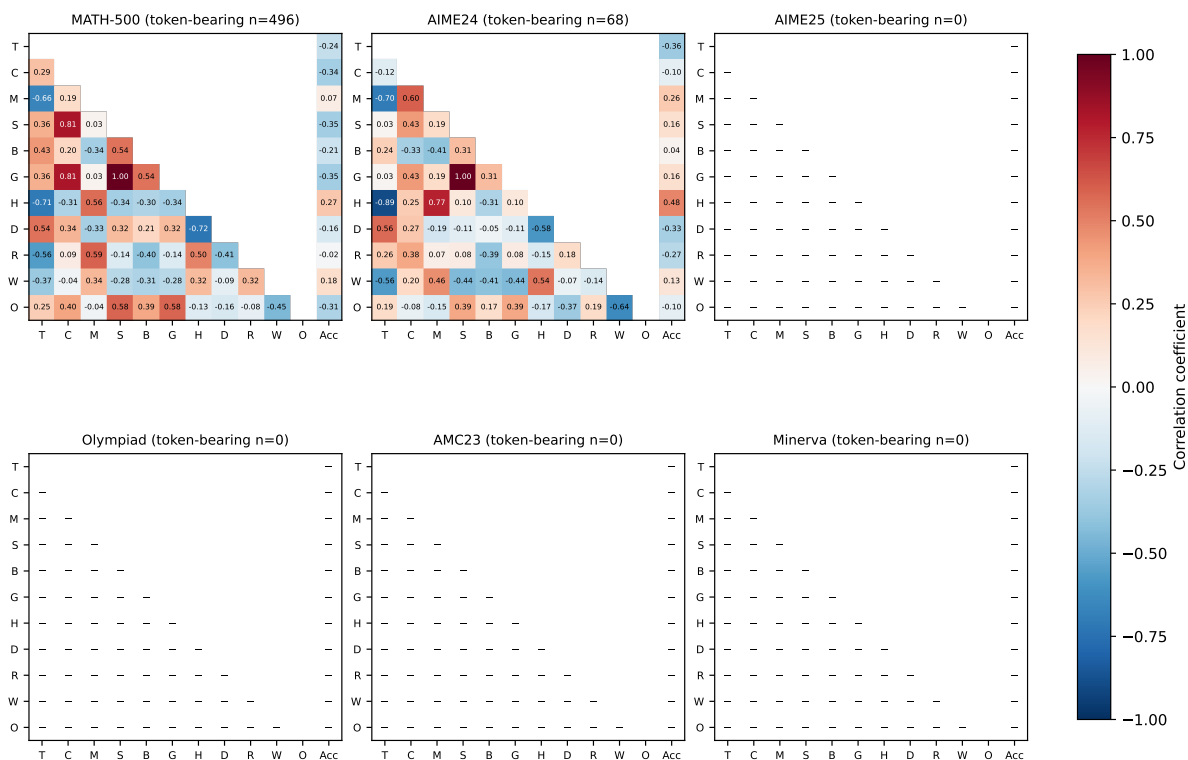


Fig. 67. GPT-OSS-20B / Class: Monitor. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 00

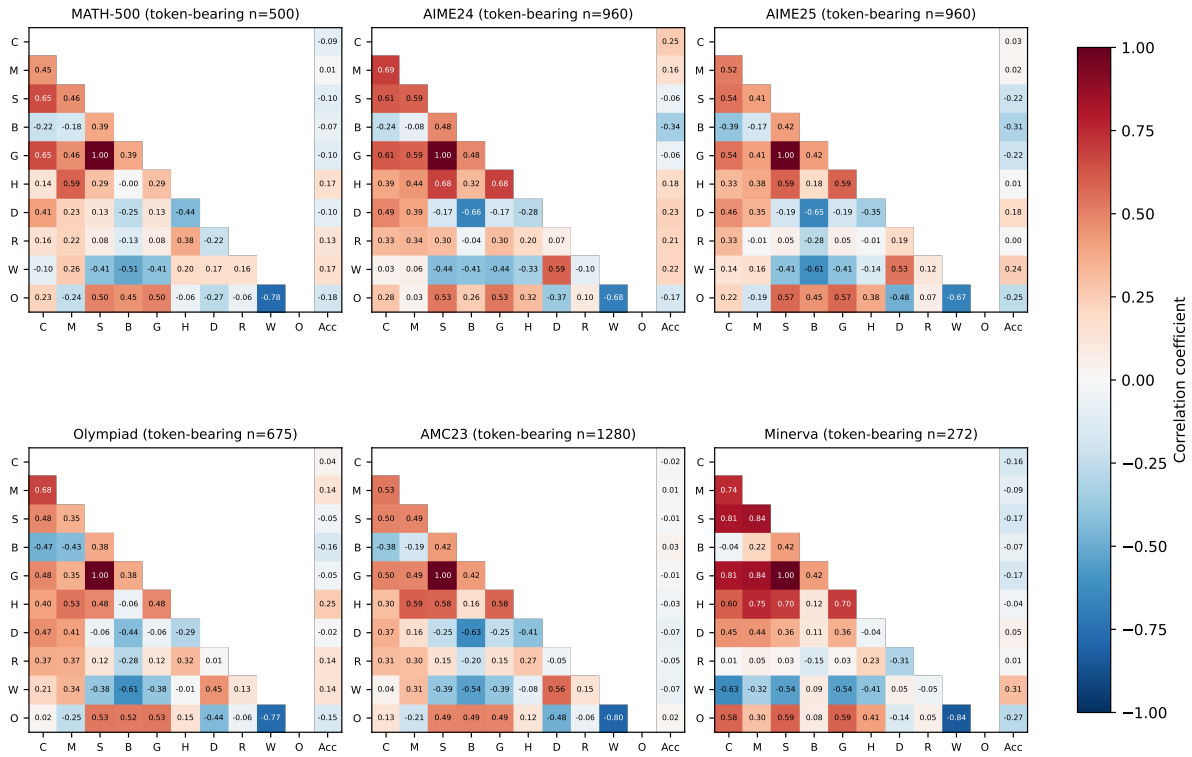
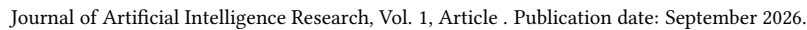


Fig. 68. GPT-OSS-20B / Position: bin 00. Metric correlations and correctness; model-specific cohorts and tokenization.

Fig. 69. GPT-OSS-20B / Position: bin 01. Metric correlations and correctness; model-specific cohorts and tokenization.



GPT-OSS-20B: Position: bin 02

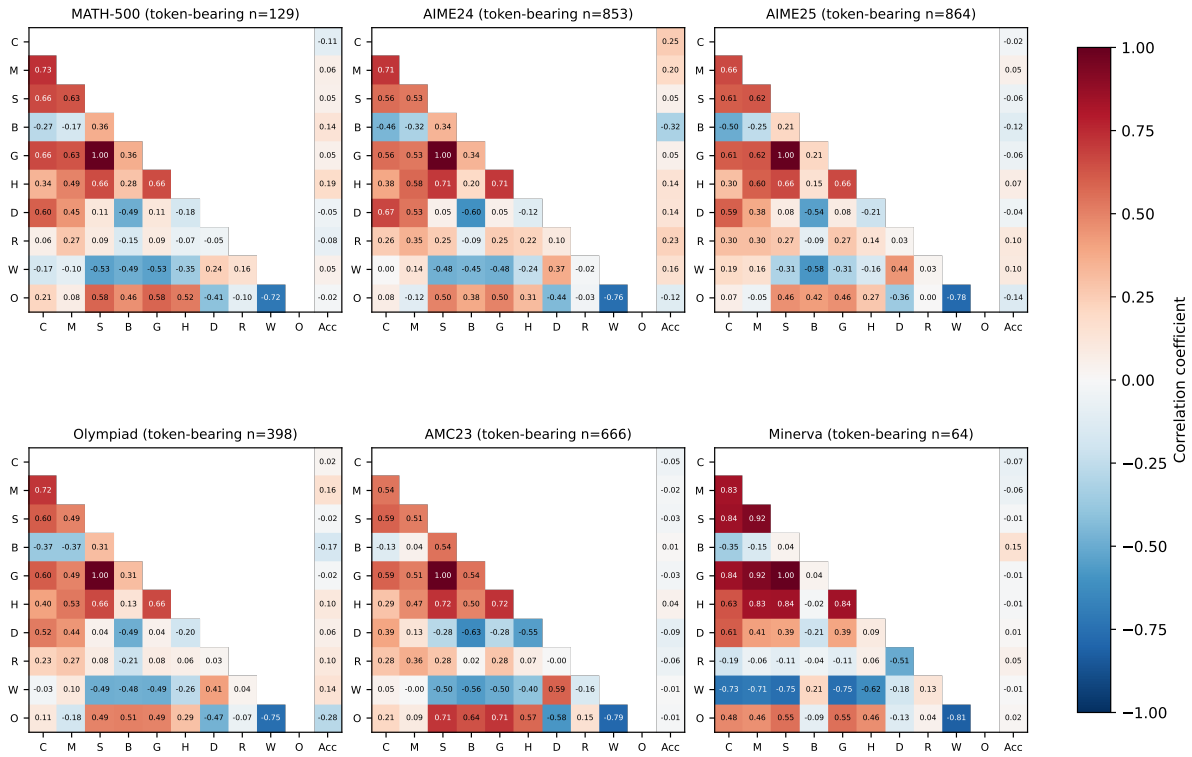


Fig. 70. GPT-OSS-20B / Position: bin 02. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 03

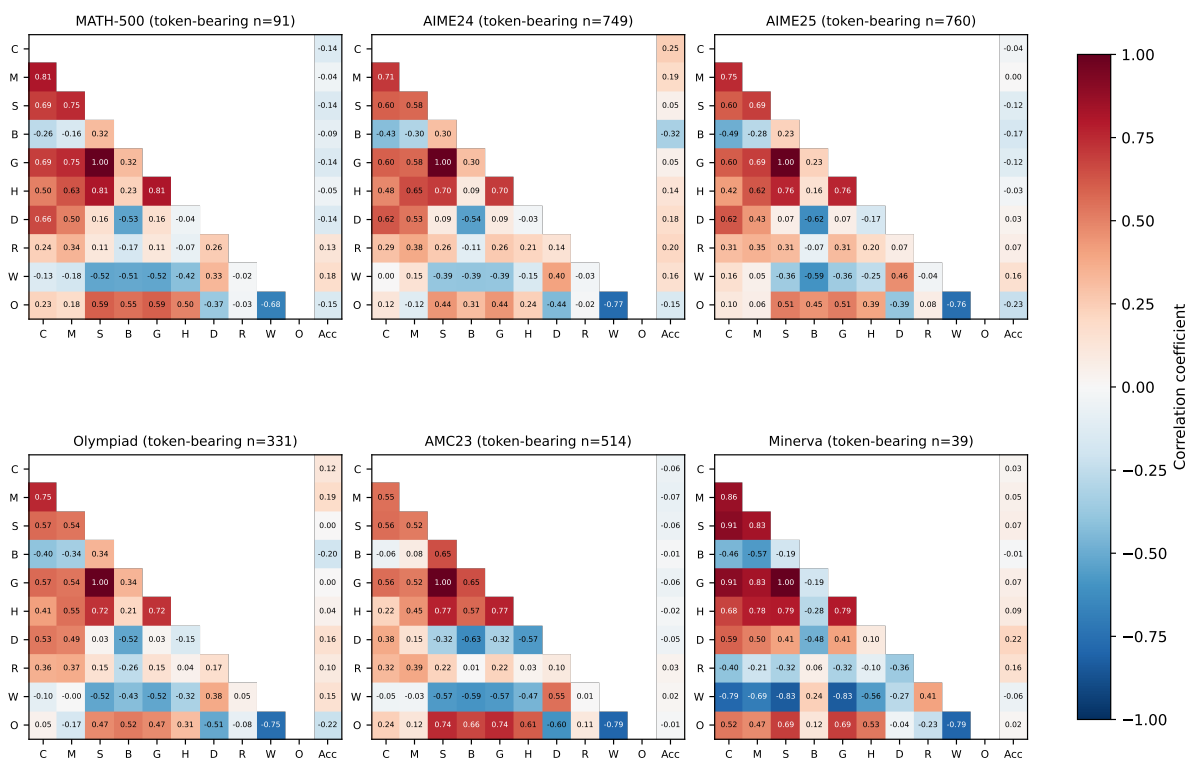


Fig. 71. GPT-OSS-20B / Position: bin 03. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 04

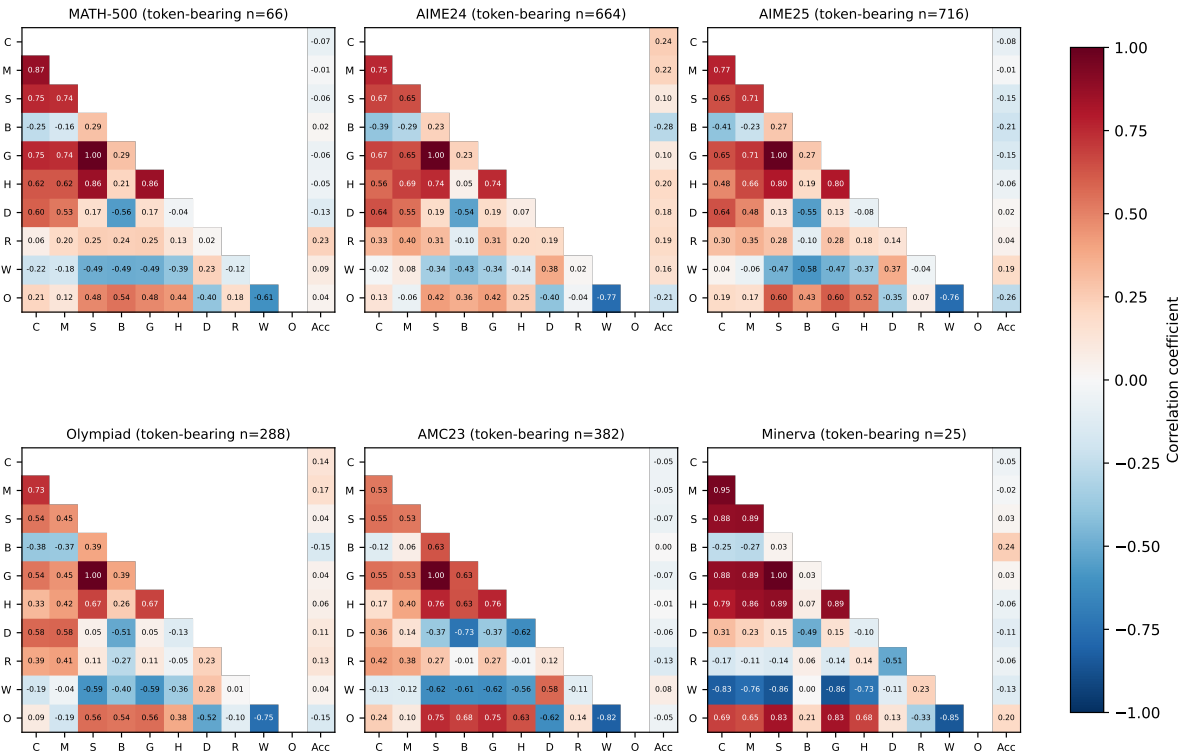


Fig. 72. GPT-OSS-20B / Position: bin 04. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 05

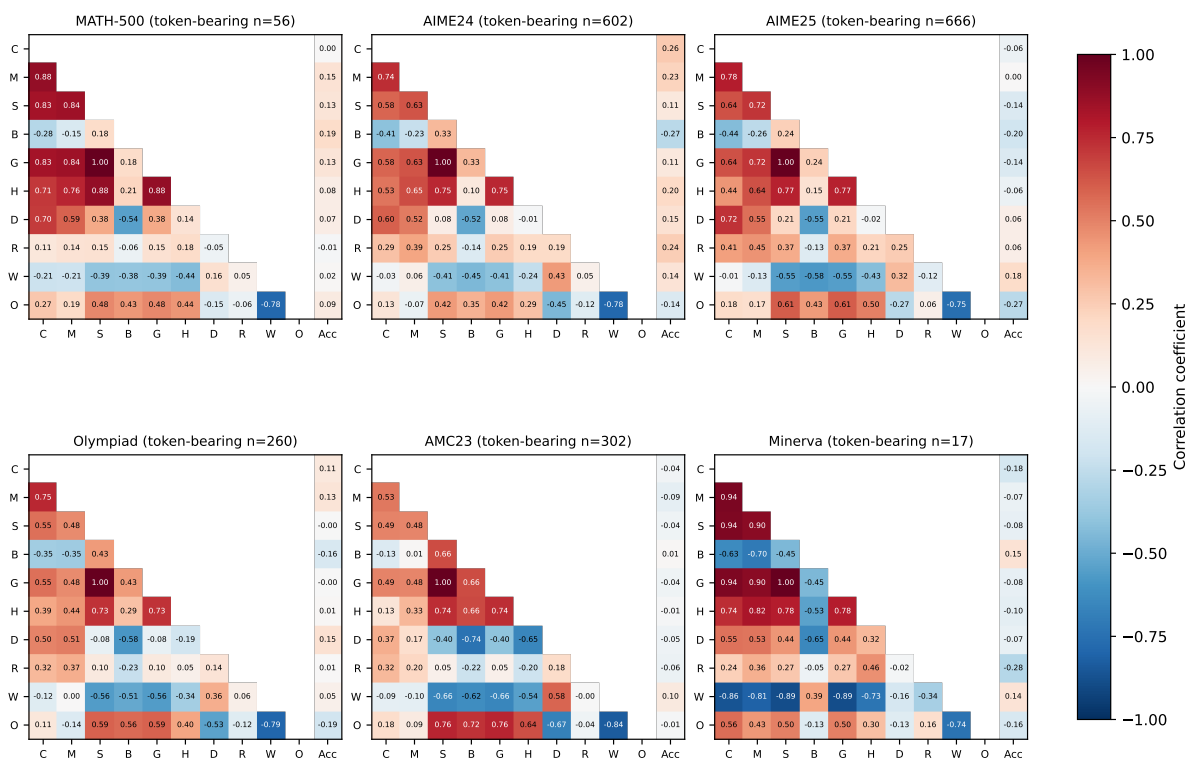


Fig. 73. GPT-OSS-20B / Position: bin 05. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 06

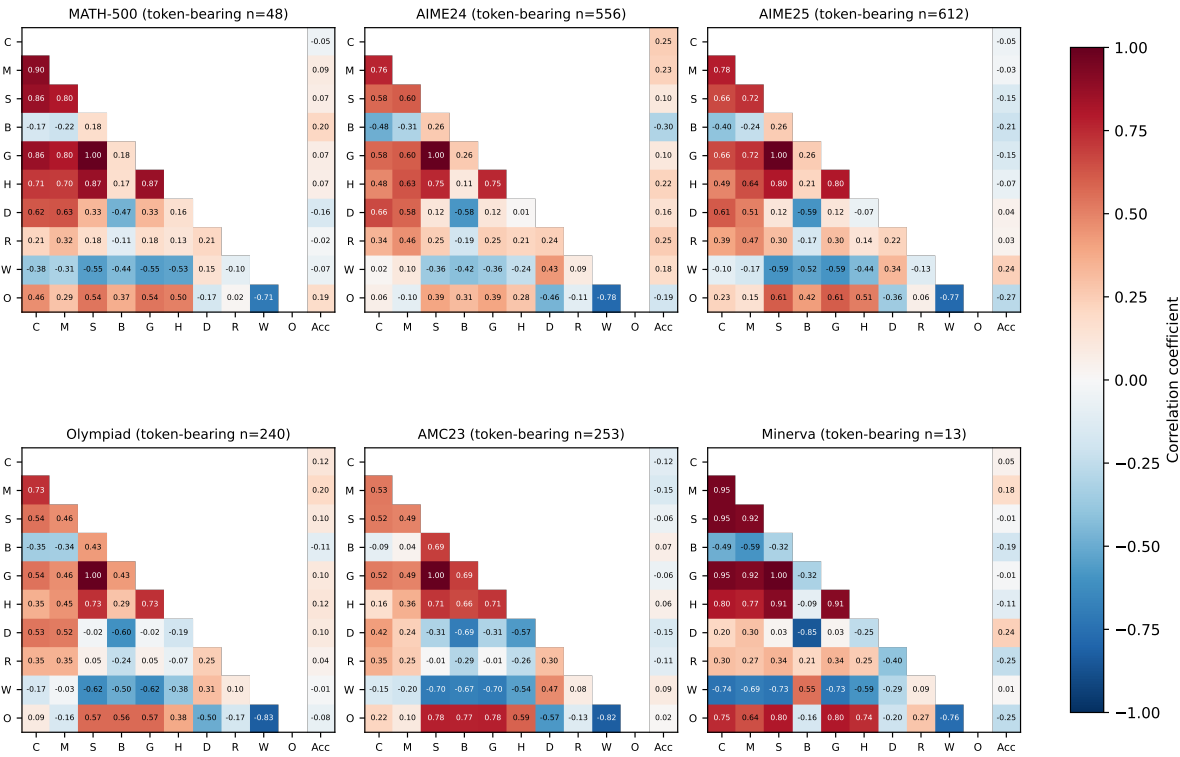


Fig. 74. GPT-OSS-20B / Position: bin 06. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 07

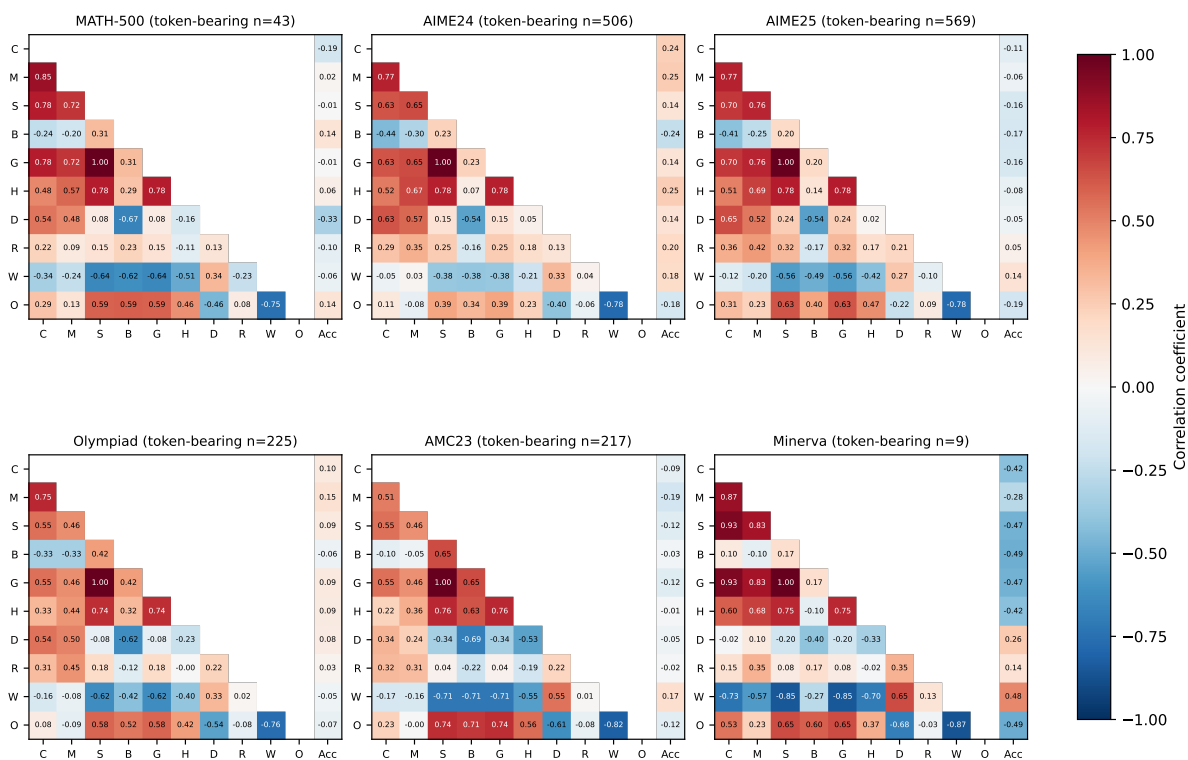


Fig. 75. GPT-OSS-20B / Position: bin 07. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 08

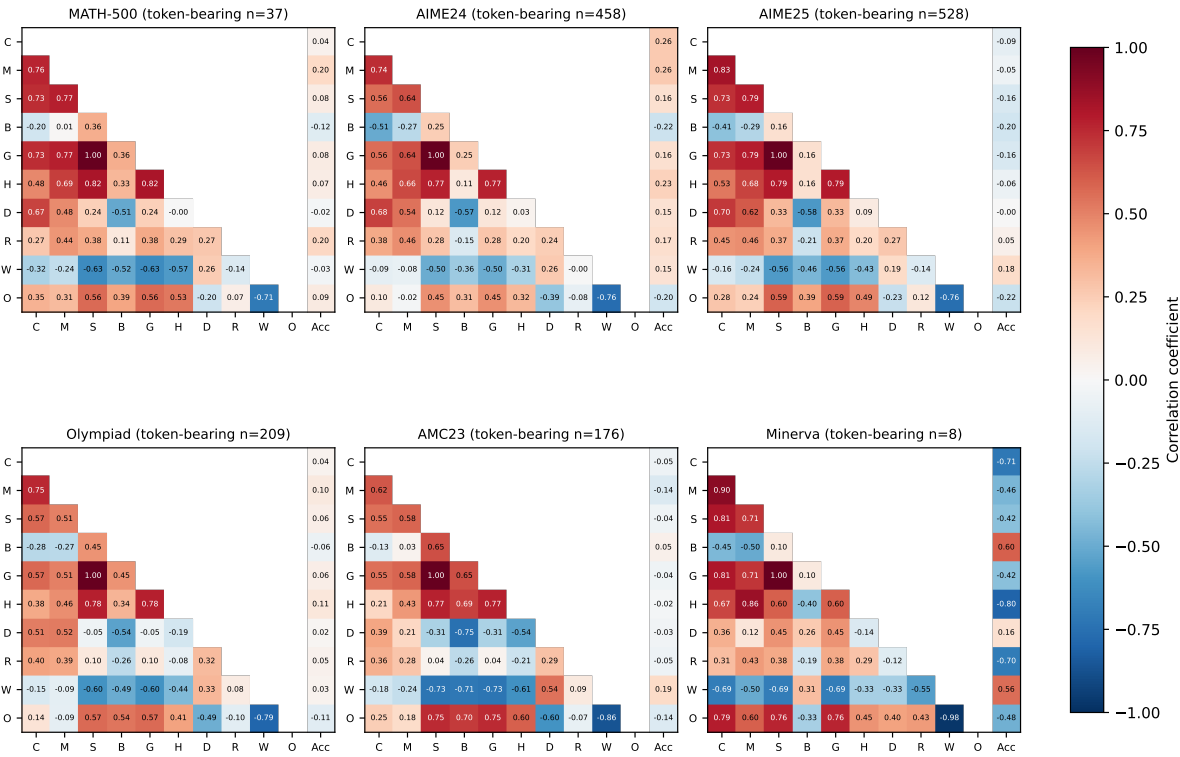


Fig. 76. GPT-OSS-20B / Position: bin 08. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: bin 09

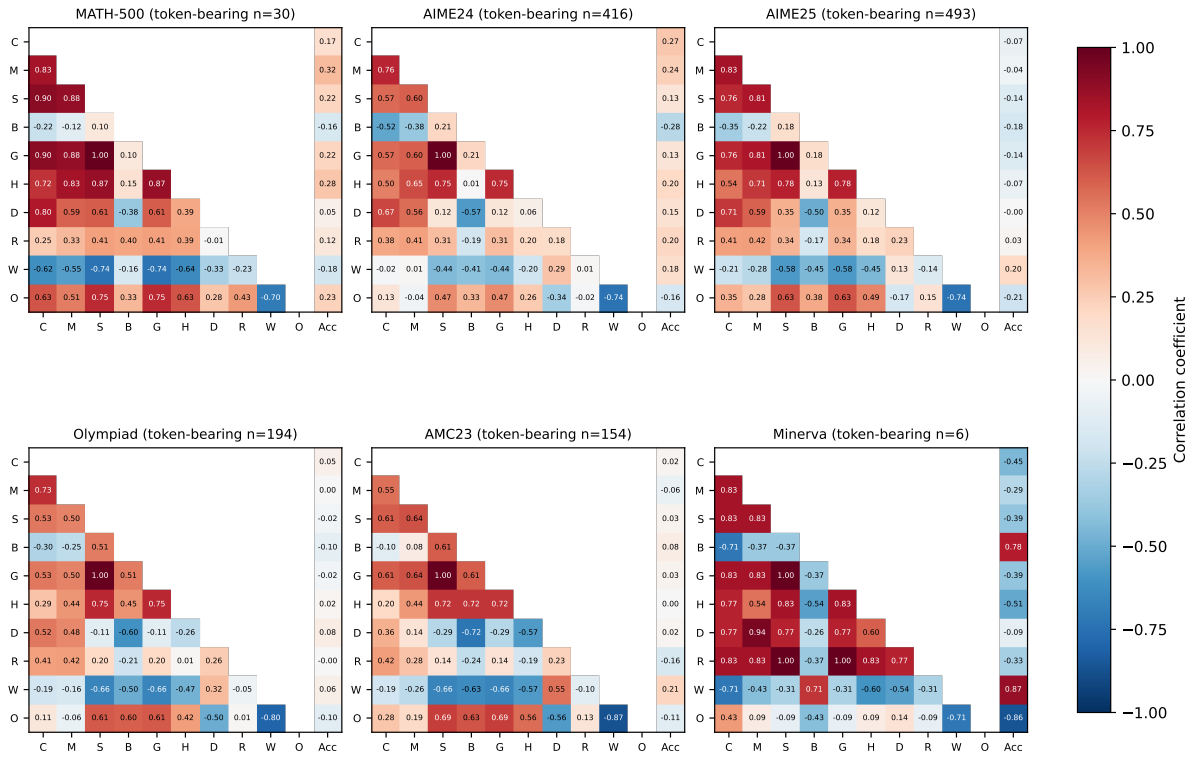


Fig. 77. GPT-OSS-20B / Position: bin 09. Metric correlations and correctness; model-specific cohorts and tokenization.

GPT-OSS-20B: Position: overflow

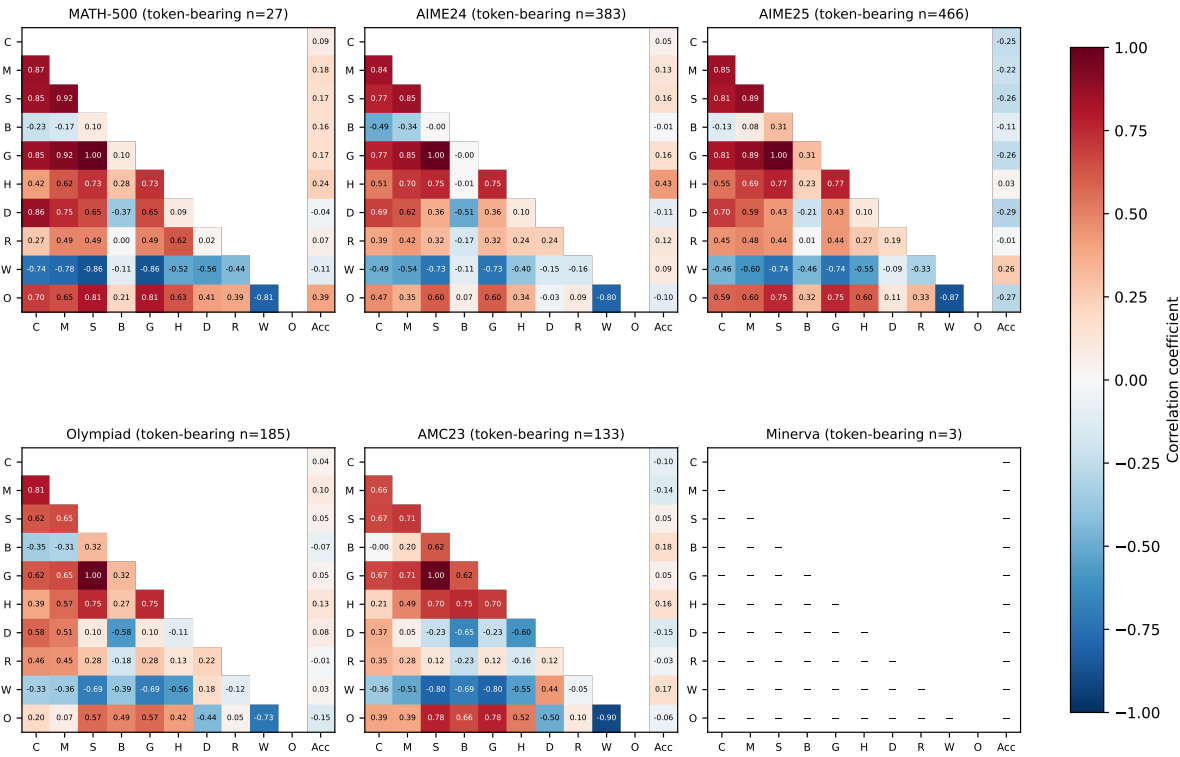


Fig. 78. GPT-OSS-20B / Position: overflow. Metric correlations and correctness; model-specific cohorts and tokenization.

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